

Improved Results for the Minimum Distance Superset Problem

Minh Hai Nguyen^a, Ngoc Anh Nguyen^a, Huong Pham^a, Duc Dong Do^a, Diep Nguyen^b, Minh Hoàng Hà^{a,*}

^aVNU University of Engineering and Technology

^bUniversity of Technology Sydney

Abstract

Given a set D of distances, each is a positive integer, the Minimum Distance Superset Problem (MDSP) requires to retrieve the positions of a set of points on the real line whose pairwise distances make the superset of D . The problem has applications in X-ray crystallography, and recently plays the key role in enzyme restriction, a technique to sequence the DNA structure. We propose new upper and lower bounds, and new techniques to reduce the size of the problem and the number of variables in start-of-the-art Mixed Integer Linear Programming (MILP) formulations. In particular, one of our upper bounds is based on a simple, yet efficient large neighborhood search algorithm that provides very good solutions in short running time. Extensive experiments on randomly generated and real data from the literature show that our new techniques significantly speed up existing exact methods and close 48 open benchmark instances. On real instances based on Luciferase, we can solve to optimality 60 over 76 instances.

Keywords: distance superset, integer programming, enzyme restriction

1. Introduction

The problem of computing all distances between any pair of n points on a real line is rather simple. However, the reverse problem of retrieving the position of n points on the line, given their distances is much more challenging. Formally, the problem can be formulated as following. Given a multiset of k positive integers $D = \{d_1, d_2, \dots, d_k\}$, we need to define the set P such that $D = \Delta(P)$ where $\Delta(P) = \{p_i - p_j \mid p_i, p_j \in P, p_i < p_j\}$. This problem is known as *Partial Digest Problem* (PDP) (12). For example, if $D = \{1, 1, 2\}$, then one solution is $P = \{0, 1, 2\}$ because $\Delta(P) = \{2 - 0, 2 - 1, 1 - 0\} = \{2, 1, 1\}$. This problem has been applied in a number of disciplines.

The first known application of PDP was found in phase retrieval in X-ray crystallography (20; 21), where we have to retrieve coordinate vectors from the distances vectors of atoms. Because atoms lie in physical space, the problem is extended to three dimensions, i.e., $d \in \mathbb{R}^3$. Another appearance of PDP is the later support in DNA sequencing (Skiena et al., 1990 (16); Skiena and Sundaram, 1994 (17)). A DNA strand is composed of nucleotides which have four types A, C, G, T. The DNA sequencing requires determining the order of nucleotides on DNA chain by taking digestion experiment using restriction enzymes. A restriction enzyme could cut DNA at specific sequence patterns of nucleotides, called restriction sites, into fragments. For instance, enzyme BamHI cuts DNA at subsequences GGATCC. By measuring the lengths of the resulting fragments using a standard laboratory technique called electrophoresis, we can retrieve the original locations of the restriction sites by modeling the problem as an instance of the PDP.

In PDP, the set D is exactly equal to the distance set created by P , i.e. $D = \Delta(P)$. However, in reality, we cannot obtain the correct and full $\binom{n}{2}$ distances of D , where $n = |P|$. The erroneous version of Partial Digest Problem, in which the distances in set D are full but exposing the error measurement, has been proposed and studied for a long time. On the other hand, the missing distance PDP variant considering the case where some distances can be missed, i.e. set D is strictly a subset of $\Delta(P)$: $|D| \neq |\Delta(P)|$, has received quite little attention (see Fontoura et al. 2018 (10) for more details). This can be explained by the lack of mathematical background knowledge of the problem and its hardness (10). In this paper, we study a missing distance PDP variant, the *Minimum Distance Superset Problem* (MDSP), in which an

*Corresponding author

arbitrary number of distances is missing and the objective is to find the minimum size P . In (10), three Mixed Integer Programming formulations as well as binary search-based algorithms which can solve the problem to optimality have been proposed for MDSP. However, the results are quite moderate and the solved instances are relatively small. **In this paper, we introduce mathematical results including a graph modeling, via which a number of properties of the problem emerge. We achieve a better result using the Integer Programming model (IP) proposed by (10) with tighter lower bound, upper bound and adding constraints.** We introduce a new heuristic algorithm that is simple but efficient. Before going further, we present some notations and relating mathematical results.

Based on the origin definition of the problem, the elements of D are called the distances and the elements of distance superset P are called points (10). Because D is multiset, which allows the multiple instances of an element. The multiplication of a value d is denoted by $mult(d)$ or alternatively by m_d . In our coming up models, the upper bound and lower bound value for the optimal solution P^* play important role and are denoted by u and l : $l \leq |P^*| \leq u$. These parameters define the number of variables in constructing IP models for problem as in Fontoura et al. (10). The authors of (10) also presented trivial bounds for the MDSP: $1/2 + \sqrt{1/4 + 2k} \leq |P^*| \leq k + 1$. In Subsection 4.2 and 4.3, we introduce the better bounds.

Two solutions are called congruent to each other if one can be reduced to the other via translation operator. The set B is congruent of set A if there exists an integer b such that B is the image of A via translation operator of distance b , i.e., $B = A + b = A + \{b\} = \{a + b \mid a \in A\}$. For example, $P_1 = \{0, 2, 6\}$ and $P_2 = \{3, 5, 9\}$ are congruent because $P_2 = P_1 + 3$. Two congruent solutions are considered the same so we are only interested in congruent solution. Therefore, here and after then we could assume that 0 is the smallest point (the most left point on the line) of solution P .

In general, the points in optimal DS P^* are restricted in $\{0, 1, \dots, B\}$. The bound B helps to narrow the search space in IP algorithms. Fontoura et al. (10) proved that there exist optimal DS P^* such that $0 \leq p \leq B$, $\forall p \in P^*$ where B is sum of all element in D . We will prove later that there also exists a P^* whose the upper bound of element $p \in P^*$ is smaller than half of Fontura 2018's and equals to $(\sum_{d \in D_u} d - d_{min})/2$, where D_u is multiset of u maximal values of D and d_{min} is the smallest element of D . **(Move this part after the MIP)**

We use a directed graph G to visualize the set D and its distance superset P . Each vertex is a point of distance superset P . The arc represents the distance (by weight of the arc) between two vertices with direction points from the smaller to the larger vertices. For example, Figure 1 represents the Distance set $D = \{1, 1, 2, 2, 3, 4, 5, 6, 7, 100\}$ and Distance superset $P = \{0, 1, 3, 5, 6, 7, 100\}$.

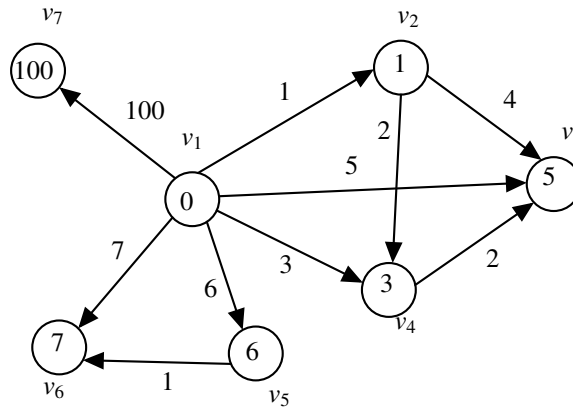


Figure 1: A graph represents Distance Superset

For any path on G , we compute a linear combination of weights along that path with coefficients ± 1 . The weight takes coefficient 1 if the path goes in the direction of arc, and -1 in the opposite direction. If the path is a tour, the linear combination equals to 0, e.g. the tour (v_1, v_5, v_6, v_1) shown in the Figure 1 gives combination: $6 + 1 - 7 = 0$. This combination-idea will help us to prove the new value of B in Subsection 4.5 and reduce size of problem in Subsection 4.1.

Table 1: Related works

Problem	Work	Findings
PDP	Skiena et al. (1990)	Backtracking algorithm
		Problem in c - dimension is NP-hard
		The number of solutions
	Daurat et al. (2002, 2005)	Polynomial Factorization Branch and Bound
		Pyramid Algorithm
	Abbas et al. (2016)	Improved Backtracking algorithm
	Namidi et al. (2011)	Branch and Bound algorithm
Probed PDP	Ahrabian et al. (2013)	Genetic algorithm
	Karp et al. (1995)	Branch and bound algorithm
	Newberg and Naor (1993)	Lower bound for the number of solution
Simplified PDP	Jacek et al. (2011)	Estimation result
		Heuristics algorithm
Erroneous PDP	Skiena et al. (1990)	NP-hard
	Skiena et al. (1994)	Generalization Backtracking algorithm
MDSP	Skiena et al. (1994)	Generalization Backtracking algorithm for small missing fraction
	Cieliebak (2003)	NP-hard
	Thibaut et al. (2018)	Some mathematical results
		IP formulations
		Binary search

The structure of this paper is as following: Section 2 reviews the previous notable works relating the variants of PDP problem. Section 3 briefly reintroduces the IP and Binary search models proposed by Fontoura et al. 2018 (10) with some new proposed constraints. Section 4 presents our main contributions: 1) [Subsection 4.1](#) describes pruning phase that reduces size of D by removing distances that we call the thorn arcs 2) [Subsection 4.2](#) shows how to improve [lower bound](#) by translating the MDSP on $\mathbb{Z}$ into MDSP on cyclic group $\mathbb{Z}/g\mathbb{Z}$ 3) [Subsection 4.3](#) shows how to improve [upper bound](#) improvement based on the multiplications of distances' values 4) [Subsection 4.4](#) introduces a [new heuristics algorithm](#) that is simple but powerful to construct a good solution 5) [Subsection 4.5](#) proves that there exists an [optimal solution](#) whose the maximal point does not exceed $B = (\sum_{d \in D_v} d - d_{\min})/2$. Section 5 presents the experiment results and discusses about these results. Finally, Section 6 summarizes our contributions and discusses some future potential research directions.

2. Literature reviews

Since its birth, the PDP problem has attracted a significant research effort (refer to Fontoura et al. (2018) (10)). In literature, it has different names. The problem is also called Turnpike problem (Dakic, 2000 (7)), or Chords' problem (Daurat et al., 2002 (8), 2005 (9)). If the points lie on a circle, the problem is called Beltway problem (Lemke et al., 2003 (12)). The PDP problem is a part of a broader subject: the Distance Geometry, which is discussed in an annual workshop called Distance Geometry Theory and Application (DGTA) (see Alizadeh et al. 2019 (3)). The variants of and workings related to PDP are summarized in Table 1.

The hardness of PDP is unknown. Until recently, there is no polynomial algorithm to solve PDP nor a proof to show that the problem is NP-hard. We only know that in the case the points distribute in a k -dimensional space, the PDP is proved to be NP-hard (Skiena et al., 1990 (16)). Despite that fact some remarkably efficient algorithms have proposed. Skiena et al. (1990) (16) published an exact algorithm named backtracking algorithm that archives a computational complexity of $O(2^n n \log n)$ in the worst case, which is shown by Zhang 1994 (22), and in average of $O(n^2 \log n)$. However, the Zhang's instance turns out to be solvable in polynomial time via semi-definite programming algorithm due to Dakic 2000 (7). In the 2016 paper, Abbas et al. (1) improved Skiena's algorithm with a number of adding cut - rules for branch and bound strategy and archived better performance on real data. Daurat et al. 2002 (8) proposed two other exact algorithms: one is based on polynomial factorization and another based on solving the

constraints in form of a system of binary quadratic equations, which is called the pyramid algorithm. Another approach is using heuristic algorithm which is fast but obtains only approximation results (see Dakic 2000 (7), Nadimi et al. 2011 (14) and Ahrabian et al. 2013 (2)).

PDP has a number of variants. The first version is called Simplified Partial Digest Problem (SPDP) (see Jacek et al. 2011, (11)), also called Labeled Partial Digest Problem (LPDP), where the information about which distances contains two end points (the leftmost and rightmost points on the line) is given. The distances are classified as interpoint and end distances. An interpoint distance is the distance between two adjacent points while an end distance is the distance between the an interpoint and an end point. The problem was triggered by the fact that two ends of a DNA sequence could be labeled using radioactive-labeling. The main challenge is that the order of points on the sequence is unknown and the indistinctiveness between two ends. The next problem variant mentioned here is called Probe Partial Digest Problem (PPDP) which proposed by Karp et al. (13). Before exposing DNA chain to restriction enzyme, we could mark a location on the chain, e.g. using radical or heat, to get a marked sequence, which is called hybridization probe. The input data of PPDP is now the lengths of fragments that contain the hybridization probe.

Another question relating to PDP and its variants is how many non-congruent solution are there. Skiena et al. 1990 (16) showed that, the number of non-congruent solutions of PDP is expotentially realtively to the number of points: $1/2n^{0.8107144} \leq H(n) \leq 1/2n^{1.2324827}$. Newberg and Naor 1993 (15) proved that the number of order factorizations of the natural number k , e.g. the different factorizations of 12 are $2 \times 6 = 3 \times 4 = 4 \times 3 = 6 \times 2 = 2 \times 2 \times 3 = 2 \times 3 \times 2 = 3 \times 2 \times 2$, is the lower bound for the quantity of non-congruent solutions of PPDP with input of k distances.

The ideal case when $\binom{n}{2}$ distances are given correctly is unrealistic. The source of errors is numerous and can be categorized into four main sources: additional fragments, missing fragments, measurement errors and multiplicity detection (see Cieliebak et al. 2005 (6) for more details). The imperfection of data leads to erroneous variants of PDP. The first variant is PDP with additive error bounds and missing a small fraction of fragments was mentioned by Skiena et al. 1990 (16) and proved to be a strongly NP-hard. Some efficient algorithms to solve erroneous variants of PDP were proposed such as the generalization of backtracking algorithm which addresses small measurement errors, inaccurate multiplicity detection or missing a small fraction of fragments. However only recently, two efficient algorithms based on Mixed Integer programming and Binary search were proposed by Fontoura et al. 2018 (10) which solve the problems with an arbitrarily high number of missing fragments, which the author called *minimum distance superset problem* and briefly written as MDSP. In the paper (10), the authors also provided some mathematical results about the trivial lower bound and upper bound of MDSP and the congruence of solutions. In this paper, we add a new visualization of problem with aid of directed graph and provide some more mathematical results that could improve the proposed algorithms by Fontoura et al. (10).

3. Mixed integer programming model and binary search

In this section, we reintroduce the Integer Programming (IP) formulation and Binary search method proposed by Fontoura et al. (10). The IP model and Binary search model were the best competitors among the models proposed in Fontoura et al. (10). We re-implement these models with the improvements including the **reducing on the distance set D , the upper bound u , the lower bound l for the carnality of the optimal DS, the maximal value B and a number of new constraints to the IP model.**

First, the variables $p_i \in \mathbb{Z}^*$, $0 \leq p_i \leq B$, for $1 \leq i \leq u$, stand for elements of DS P , with u is the upper bound found in the pre-processing phase. To **deal with the symmetry of the search space** we could assume a non decreasing order by constraints: $p_i < p_j$, $1 \leq i < j \leq u$. The optimal DS P^* is a subset of P .

The binary variable x_{ij}^d , where $1 \leq i < j \leq u$, $d \in D$, indicates whether distance between p_i and p_j equals $d \in D$, i.e. $x_{ij}^d = 1$ if and only if $p_j - p_i = d$. We formulate this fact by the following inequality constraints,

$$p_i + x_{ij}^d \times d \leq p_j, \quad 1 \leq i < j \leq u, \quad d \in D, \quad \blacksquare$$

$$p_j \leq p_i + M \times (1 - x_{ij}^d) + d, \quad 1 \leq i < j \leq u, \quad d \in D,$$

where M is a very large number that in experiment we set as $M = B$.

For each $d \in D$, there must be a pair $i < j$ such that $p_j - p_i = d$, therefore,

$$\sum_{i=1}^{u-1} \sum_{j=i+1}^u x_{ij}^d \geq 1, \quad d \in D.$$

The binary variables z_i , for $1 \leq i \leq u$, take value of 1 if $p_i \in P^*$. If $p_i \in P^*$, i.e. $z_i = 1$, the pair of points (p_i, p_j) could be corresponding for at most one distance. This can be done via the following constraint,

$$\sum_{j=i+1}^u x_{ij}^d \leq z_i, \quad 1 \leq i \leq u, \quad d \in D.$$

Because 0 belongs to an optimal solution, we could assume that, $p_1 = 0$ and $z_1 = 1$. We could also assume that $z_i \geq z_j$, $u \geq i > j \geq 1$ to reduce the symmetry of the search space. Realizing that the size of optimal distance superset must be greater than or equals to lower bound l , the first l variables z_i , $1 \leq i \leq l$ must be 1. The following is a new constraint that we contribute to the IP model proposed by Fontoura et al (10),

$$z_i = 1, \quad 1 \leq i \leq l.$$

Finally, we put in IP model an extra constraint for point value restriction,

$$0 \leq p_i \leq B, \quad 1 \leq i \leq u.$$

The objective function is to minimize sum of variable z_i , which equals the cardinality of P^* and equals to $\sum_{i=1}^u z_i$. We have the following IP formulation for MDSP,

$$\begin{aligned} (\mathcal{P}_1) : \min & \quad \sum_{i=1}^u z_i & (1) \\ \text{s.t.} & \quad p_i + x_{ij}^d \times d \leq p_j, \quad 1 \leq i < j \leq u, \quad d \in D, & (2) \\ & \quad p_j \leq p_i + M \times (1 - x_{ij}^d) + d, \quad 1 \leq i < j \leq u, \quad d \in D, & (3) \\ & \quad \sum_{i=1}^{u-1} \sum_{j=i+1}^u x_{ij}^d \geq 1, \quad d \in D, & (4) \\ & \quad \sum_{j \neq i} x_{ij}^d \leq z_i, \quad 1 \leq i \leq u, \quad d \in D, & (5) \\ & \quad x_{ij}^d \in \{0, 1\}, \quad 1 \leq i < j \leq u, \quad d \in D, & (6) \\ & \quad p_i < p_j, \quad 1 \leq i < j \leq u, & (7) \\ & \quad p_i \in Z^*, \quad 1 \leq i \leq u, & (8) \\ & \quad p_1 = 0, & (9) \\ & \quad z_j \leq z_i, \quad 1 \leq i < j \leq u, & (10) \\ & \quad z_i \in \{0, 1\}, \quad 1 \leq i \leq u, & (11) \\ & \quad 0 \leq p_i \leq B, \quad 1 \leq i \leq u, & (12) \\ & \quad z_i = 1, \quad 1 \leq i \leq l. & (13) \end{aligned}$$

We also re-implement the Binary search algorithm (FEAS) proposed by Fontoura et al. 2018 (10) with the improved upper bound u , lower bound l and value upper bound B . The Binary search searches the number of points of the optimal solution, which is denoted by $t \in [l, u]$. A point set of size t is feasible, i.e. exists a point set P of size t such that $D \subset \Delta(P)$, if the following constraints are satisfied,

$$\begin{aligned} (\mathcal{P}_2) : & \quad p_i + x_{ij}^d \times d \leq p_j, \quad 1 \leq i < j \leq t, \quad d \in D, & (14) \\ & \quad p_j \leq p_i + M \times (1 - x_{ij}^d) + d, \quad 1 \leq i < j \leq t, \quad d \in D & (15) \\ & \quad \sum_{i=1}^{t-1} \sum_{j=i+1}^t x_{ij}^d \geq 1, \quad d \in D, & (16) \\ & \quad \sum_{j \neq i} x_{ij}^d = 1, \quad 1 \leq i \leq t, \quad d \in D, & (17) \\ & \quad x_{ij}^d \in \{0, 1\}, \quad 1 \leq i < j \leq t, \quad d \in D, & (18) \\ & \quad p_i < p_j, \quad 1 \leq i < j \leq t, & (19) \\ & \quad 0 \leq p_i \leq B, \quad 1 \leq i \leq t, & (20) \\ & \quad p_i \in Z^*, \quad 1 \leq i \leq t, & (21) \\ & \quad p_1 = 0. & (22) \end{aligned}$$

The binary search schedule could be formulated as described in Algorithm 1.

Algorithm 1 Binary search algorithm

Input: Distance set D , lower bound l , upper bound u

Output: Optimal distance superset P^*

```

1:  $t_{LB} \leftarrow l$ 
2:  $t_{UB} \leftarrow u$ 
3: while  $t_{LB} < t_{UB}$  do
4:    $t_{MID} \leftarrow \lfloor \frac{t_{LB} + t_{UB}}{2} \rfloor$ 
5:   Solve the formulation for  $D$  with  $t = t_{MID}$ 
6:   if a feasible solution of MDSP was not found then
7:      $t_{LB} = t_{MID} + 1$ 
8:   else
9:      $t_{UB} = t_{MID} - 1$ 
10:  end if
11: end while
12: return The latest distance superset

```

4. Main Contributions

4.1. Pruning phase

The aim of the pruning phase is to find the distances that are belonged to a special class that we name the thorn arcs and then move them away from the distances multiset to offload the difficulty of the problem. A thorn arc distance (if there exists) is the distance whose corresponding arc is not included in any simple cycle in any visual graph of the any distance superset. Following Theorem 1, we could safely remove the thorn arc distances to reduce the size of the problem.

For example, with $D = \{1, 2, 3, 4, 5, 6, 7, 100\}$, $d = 100$ is a thorn arc distance. The reason is that 100 is too large comparing to the other elements of D so that every linear combination including 100 and some other elements of D with coefficients ± 1 is nonzero.

We give a formal **definition of thorn arcs**: A distance $d \in D$ is thorn arc if there does not exist elements of $D \setminus \{d\}$ whose linear combination with coefficients ± 1 equals to d . **One property** of the thorn arc is that: if d is the thorn arc of D , $P_{D \setminus \{d\}}^* \cup \{p + d\}$ is an minimum distance super-set of D where $P_{D \setminus \{d\}}^*$ is an minimum distance superset of $D \setminus \{d\}$ and $p \in P_{D \setminus \{d\}}^*$. **We generalize that property in the following theorem which is the main result of this subsection.**

Theorem 1. *Let D_h be the set of thorn arcs of D . If P_h^* is the minimum distance superset of $D \setminus D_h$, we have $P^* = P_h^* \cup (D_h + p)$ is also an optimal solution of D , where $p \in P_h^*$.*

Theorem 1 suggests that we could remove the thorn arcs from the distance set D to reduce the problem size.

An approach to detect the thorn arcs is applying the **Subset sum problem**. Given the integers $w_1, w_2, \dots, w_n$, the Subset sum problem determines if there is any subset of them that sum to preciously another integer W (see Tomohiro and Akihiro, 2016 (18) for more information). If $d \in D$ is a thorn arc, there exists a subset of $(D \setminus \{d\}) \cup -(D \setminus \{d\})$ of which sum of elements equals to d , where $-(D \setminus \{d\})$ is the set of additive inverse value of elements of $(D \setminus \{d\})$. Although Subset sum problem is a NP complete problem (18), the following **dynamic algorithm** could help to solve it efficiently **in case the number and values of integers are not large**.

- We define the boolean-value function $Q(i, x)$, which holds true if $\exists S \subset \{w_1, w_2, \dots, w_i\}$ such that $\sum_{w \in S} w = x$
- We define an array of values of function $Q(i, x)$ according to parameters $1 \leq i \leq n$ and $A \leq x \leq B$, where A, B are the sums of negative integers $-w_1, -w_2, \dots, -w_n$ and positive integers $w_1, w_2, \dots, w_n$ respectively; we limit x between **a and b** because $Q(i, x)$ always holds *false* for $x > B$ or $x < A$

- In the case $i = 1$, $Q(1, x)$ is *true* if $w_1 = x$
- We use the following recursion to compute $Q(n, W)$:

$$Q(i, x) = Q(i - 1, x) \vee Q(i - 1, x - w_i),$$

where $\vee$ is disjunction operator.

In the worst case, we have to compute all value of $Q(i, x)$ for all $1 \leq i \leq n$ and $A \leq x \leq B$. Therefore the complexity of above dynamic algorithm is $O(n \times (B - A))$. A thorn arc test on distance d would require a running time of $O(2|D|-1)2 \sum_{d' \in D, d' \neq d} d' = O(|D| \sum_{d' \in D, d' \neq d} d')$, so the total complexity to check all distance d is $O(|D|^2 \sum_{d \in D} d)$. Base on this complexity, to avoid the situation that this phase could potentially costs a lot of time, we only apply this phase for **instances with less than 1000 distances** and the **maximal distance not exceed 1000**.

4.2. Lower bound based on transforming MDSP on $\mathbb{Z}$ to MDSP on ring $\mathbb{Z}/g\mathbb{Z}$

The improvement on lower bound helps to reduce the symmetry of search space of IP model. More precisely, via constraints (13) we could eliminates l variables $z_1, z_2, \dots, z_l$ by fixing their values to 1. On the other hand, the number of constraints (10) would also decrease by l constraints. In addition, this is also helpful for binary search models. The initial search range of binary search is from l to u , therefore improvement on lower bound l helps to narrow this search range.

To improve lower bound, we transform the MDSP on integer set $\mathbb{Z}$ into MDSP on ring of integers modulo g : $\mathbb{Z}/g\mathbb{Z} = \{0, 1, \dots, g-1\}$. The distance modulo g between points $p, q \in \mathbb{Z}/g\mathbb{Z}$ is defined as following: $d_g(p, q) = \min\{|p - q|, g - |p - q|\}$, e.g. for $g = 5$, $d_5(1, 4) = \min\{|1 - 4|, 5 - |1 - 4|\} = 2$. D is then mapped into the distance multiset modulo g : $D_g = \{\min\{d \bmod g, g - (d \bmod g)\} | d \in D\}$. The modulo g of distance superset is denoted as P_g . The problem Minimum Distance Superset on $\mathbb{Z}/g\mathbb{Z}$ is denoted as MDSP $_g$.

Let consider an example of MDSP $_g$ on distance multiset $D = \{1, 2, 3, 4, 5\}$ and $g = 3$. The distance multiset modulo $g = 3$ is $D_3 = \{1, 1, 0, 1, 1\}$. Back to general case of g , we denote x_i , for $0 \leq i \leq g-1$, as the number of points equal to i in P_g , as a sequence we have $x_0 + \dots + x_{g-1} = |P_g|$. We denote m_j as the number of distances whose value is equal to j in D_g , for $0 \leq j \leq g/2$. It is clear that x_i points equal to i and x_{i+j} points equal to $i+j$ will generate $x_i x_{i+j}$ distances equal to j . Hence, the superset constraints are followings inequalities,

$$\sum_{i=0}^{g-1} x_i x_{(i+j) \bmod g} \geq m_j, \quad 0 < j \leq g/2, \quad (23)$$

$$\sum_{i=0}^{g-1} x_i (x_i - 1) \geq 2m_0. \quad (24)$$

It is clear that the modulo g of the optimal solution P^* of MDSP for D on $\mathbb{Z}$ is also a distance superset of D_g on ring $\mathbb{Z}/g\mathbb{Z}$. As a sequence, we have the following theorem:

Theorem 2. If n_g and n are the size of optimal supersets of MDSP $_g$ and MDSP defined on D_g and D respectively, $n_g \leq n$.

For small value g , we could use brute force to solve MDSP $_g$ for instance of several thousands distances in an acceptable time. Algorithm 2 describes brute force strategy to compute n_g in case $g = 3$. We use WHILE loops to consider every possible value of variable x_i (lines 2, 3, 4). For each value $(x_1, x_2, \dots, x_g)$, we check the validity of constraints (23) - (24) (Line 5). If the constraints are satisfied, we update that best current size n_g (lines 6-8). The complexity of Algorithm 2 is $O(g|D|^g)$. In this paper, we compute n_g for $g \in \{2, 3, 5\}$ and set the largest value among n_2, n_3 and n_5 as the lower bound for MDSP.

4.3. Multiplication based upper bound

As presented in Section 3, the upper bound u directly affects the number of variables and constraints of IP model. Precisely, the numbers of variables p_i, z_{ij} and x_{ij}^d are u, u^2 and $u^2|D|$ respectively, and the number of constraints (5) is $u|D|$. On the other hand, the initial search range of the binary search algorithm is from l to u , therefore improvement on upper bound u helps to narrow this search range. In this section, we define a new algorithm that is fast and effective in detecting an upper bound for the MSDP. That algorithm is called *Arithmetic Sequences Sum*.

Algorithm 2 Brute force lower bound for $g = 3$

Input: Distance set D

Output: Lower bound $l = n_g$

```

1:  $n_g \leftarrow |D_g| + 1$ 
2: while  $x_0 \leq n_g$  do
3:   while  $x_1 \leq n_g - x_0$  do
4:     while  $x_2 \leq n_g - x_0 - x_1$  do
5:       if  $x_0(x_0 - 1) + x_1(x_1 - 1) + x_2(x_2 - 1) \geq 2m_0$  AND  $x_0x_1 + x_1x_2 + x_2x_0 \geq m_1$  then
6:         if  $n_g > x_0 + x_1 + x_2$  then
7:            $n_g = x_0 + x_1 + x_2$ 
8:         end if
9:       end if
10:    end while
11:  end while
12: end while
13: return  $n_g$ 

```

Let D_1 be the set of unique values in D . For example, if $D = \{2, 2, 3, 3, 3, 12, 25, 37, 51, 76, 88, 103, 154, 179, 191, 275\}$ then $D_1 = \{2, 3, 12, 25, 37, 51, 76, 88, 103, 154, 179, 191, 275\}$. Then we extract the subset D_{1X} of D_1 (the details about extraction is described in the bellow paragraph). We compute a distance superset P_{1X} for D_{1X} . Let $D_2 = D \setminus \Delta(P_{1X})$. We then set $|P_{1X}| + |D_2|$ as a new upper bound of problem. The detail algorithm of how to construct D_{1X} and deal with D_{1X} is presented in following paragraphs.

Considering D_1 , we compute $r = \lceil \sqrt{(\max(D_1) + 1)/2} \rceil$ and $s = r + 1$. Let $X = \{ir + js \mid 0 < i \leq s, 0 < j \leq r\}$. It is easily to show that $|X| = rs \approx \max(D_1)/2$. As a sequence, each element d of D_1 has the probability of 0.5 to be represented as $d = ir + js$. If $d \in D_1$ can be formulated as $d = ir + js$ then this formulation is unique, i.e. two representations (if exist) of $d \in D_1$ as $d = ir + js = i'r + j's$ are identical: $i = i', j = j'$. Considering $D_{1X} = D_1 \cap X$, let R and S be the sets of all value i and j appearing in the representation $d = ir + js$ of element d of D_{1X} : $R = \{i \mid \exists d \in D_1 : d = ir + js\}$, $S = \{j \mid \exists d \in D_1 : d = ir + js\}$.

Return to above example, we compute the values $r = \lceil \sqrt{(275 + 1)/2} \rceil = 12$, $s = r + 1 = 13$. The representations of elements of D_1 are as followings, $12 = 1 \times 12 + 0 \times 13$, $25 = 1 \times 12 + 1 \times 13$, $37 = 2 \times 12 + 1 \times 13$, $51 = 1 \times 12 + 3 \times 13$, $76 = 2 \times 12 + 4 \times 13$, $88 = 3 \times 12 + 4 \times 13$, $103 = 1 \times 12 + 7 \times 13$, $154 = 2 \times 12 + 10 \times 13$, $179 = 3 \times 12 + 11 \times 13$, $191 = 4 \times 12 + 11 \times 13$, $275 = 11 \times 12 + 11 \times 13$. Hence, D_{1X} and two sets R and S are: $D_{1X} = \{25, 37, 51, 76, 88, 103, 154, 179, 191, 275\}$, $R = \{1, 2, 3, 4, 11\}$, $S = \{1, 3, 4, 7, 10, 11\}$.

We solve the MDSP on R and S to receive the distance supersets P_R and P_S . One can see that $|R| \ll |D|$, $|S| \ll |D|$ and the values of elements of R and S are smaller than elements of D by a factor of $\sqrt{\max(D)}/2$. Therefore the cost of solving MDSP on R, S is minor comparing the cost of solving MDSP on D . Let $A = \{p_i r + p_j s \mid (p_i, p_j) \in P_R \times P_S\}$. It can be seen that $D_{1X} \subset \Delta(A)$. We then use a greedy algorithm to construct a subset $P_{1X} \subset A$ such that P_{1X} is the superset distance of D_{1X} and $|P_{1X}|$ is as small as possible. This greedy algorithm is shown in Algorithm 3. Here the subtraction operation between a number x and a set P_{1X} is defined as the set of differences between x and all element of the set: $x - P_{1X} = \{x - y \mid y \in P_{1X}\}$. Entering each step of the **WHILE** loop in Algorithm 3, we pop up the element that attributes the most distances from A and add it to result set P_{1X} .

Solving the MDSP problems on sets R and S of the above example, we receive two points sets $P_R = \{0, 1, 2, 3, 4, 11\}$, $P_S = \{0, 1, 4, 11\}$. After that, [Algorithm 3](#) constructs the distance superset of D_{1X} : $P_{1X} = \{0, 12, 37, 88, 191, 275\}$, which has 6 elements, $0 = 0 \times 12 + 0 \times 13$, $12 = 1 \times 12 + 0 \times 13$, $37 = 2 \times 12 + 1 \times 13$, $88 = 3 \times 12 + 4 \times 13$, $191 = 4 \times 12 + 11 \times 13$, $275 = 11 \times 12 + 11 \times 13$. $D_2 = D \setminus \Delta(P_{1X}) = \{2, 2, 3, 3, 3, 3\}$. Finally, the upper bound for this MDSP problem is $|P_{1X}| + |D_2| = 6 + 6 = 12$, significantly smaller than trivial upper bound as $|D| + 1 = 18$.

Algorithm 3 Greedy algorithm to build distance superset for D_{1X}

Input: Distance set D_{1X} , distance superset A

Output: Distance superset P_{1X}

```

1: while  $D_{1X} \neq \emptyset$  do
2:    $a = \max_{x \in A} |(x - P_{1X}) \cap D_{1X}|$ 
3:    $D_{1X} = D_{1X} \setminus ((a - P_{1X}) \cap D_{1X})$ 
4:    $P_{1X} = P_{1X} \cup \{a\}$ 
5:    $A = A \setminus \{a\}$ 
6: end while
7: return  $P_{1X}$ 

```

4.4. Large neighborhood search for MDSP

In this subsection, we introduce another algorithm that also aims to improve upper bound for MDSP. Via construct and destroy process, we receive a distance superset P and then set the cardinality of this superset as an upper bound. The main idea of construction phase is to consider every integer on $[0, B]$ as potential candidate to be added to P . For each integer, we compute the differences between it and all elements of P and count the number of differences which belong to D . The integer whose most number of differences belong to D will be added to P . This construction stops when $\Delta(P)$ fully covers D . After that the destruction is executed by removing a number of elements of P and then the construction is repeated again. The process of construction-destruction is repeated in a certain times. The following step-by-step describes the algorithm.

Step 1. This step aims to construct a distance superset P . Before coming in details, let make clear about some denotation. Let D' denote the set containing uncovered distances by P , i.e. $D' = D \setminus \Delta(P)$. We write the set of distances between integer x and elements of P that belong to D' as $nd(P, D', x) = \{x - a \mid a \in P\} \cap D'$. This step starts with a nonempty set P . At the first time execution Step 1, P is assigned to $\{0\}$ but at the next time, P is assigned to the output of Step 2. We gradually build point set P by consider integers in $[0, B]$. For each integer x , we compute $|nd(P, D', x)|$, then add the integer whose the largest $|nd(P, D', x)|$ to P . If there are more than one integers x whose the largest $|nd(P, D', x)|$, we choose one integer randomly. Then we update P and D' : $D' = D' \setminus nd(P, D', x)$, $P = P \cup \{x\}$.

The Step 1 is executed until D' is empty. Cause of its essence, this step is called *Construction Step*.

Step 2. This step is named *Destruction Step*. The output of this step is a subset $Q \subset P$ which contains b elements of P , where $b = \lfloor 0.1 \times |P| \rfloor$. This step is almost the same as Step 1 except two points: first, we consider only elements of P not the integers between 0 and B ; second, we execute this step until b elements are added to Q . At the beginning, Q is set to $\{0\}$ and D' is set to D . In the processing of Step 2, element $x \in P$ whose the largest $|nd(P, D', x)|$ will be added to Q . After x is added to Q , we update D' , P : $D' = D \setminus nd(P \setminus \{x\}, x, D')$, $P = P \setminus \{x\}$.

We then repeat Step 1. The output Q of Step 2 becomes the input set P of Step 1. We repeat steps 1-2 for t times, which in experiments is set to $t = 100$. The smallest P after t times is the output of this algorithm.

Step 1 takes $O(B(|D|+1)^2)$ in the worst case while Step 2 takes $O(B|D|)$ in the worst case. Therefore, the complexity of this greedy algorithm is $O(tB|D|^2)$. The detail is described in Algorithm 4.

4.5. Reduce maximal value of points

In this Section we propose a better bound for maximal point of P . Fontoura et al. (10) proved that there exists an optimal distance superset of D that the smallest element is 0 and the largest element does not exceed $B' = \sum_{d \in D} d$. They also presented an approach to scale down size of distances in D by dividing all element of D to their greatest common divisor. Relax the distinct points constraint, we will prove that B' could be replaced by a new bound less than half of itself: $B = (\sum_{d \in D_u} d - d_{\min})/2$. The smaller bounded value of B helps to narrow the search domain of integer programming model shown in Subsection 3, hence it speeds up the algorithm.

Considering a distance multiset D and its distance superset P . They are represented using a directed graph G as shown in Section 1. The representation is called Distance set graph (DS graph). In DS graph there is a structure that

Algorithm 4 Large neighborhood search for MDSP

Input: Distance set D

Output: Distance superset P_{best}

```
1:  $t \leftarrow 10$ 
2:  $i \leftarrow 0$ 
3:  $P \leftarrow \{0\}$ 
4:  $P_{best} \leftarrow \phi$ 
5: while  $i < t$  do
6:    $D' \leftarrow D \setminus \Delta(P)$ 
7:   while  $D' \neq \phi$  do
8:      $x \leftarrow \operatorname{argmax}_{y \in P} |\{y - a : a \in P, a \neq y\} \cap D'|$ 
9:      $P \leftarrow P \cup \{x\}$ 
10:     $D' \leftarrow D' \setminus (\{x - a \mid a \in P\} \cap D')$ 
11:   end while
12:   if  $|P| < |P_{best}|$  OR  $P_{best} == \phi$  then
13:      $P_{best} \leftarrow P$ 
14:   end if
15:    $Q \leftarrow \{0\}$ 
16:    $b \leftarrow 0.1 \times |P|$ 
17:    $P \leftarrow P \setminus \{0\}$ 
18:    $D' \leftarrow D$ 
19:   while  $|Q| \leq b$  AND  $D' \neq \phi$  do
20:      $x \leftarrow \operatorname{argmax}_{y \in P} |\{y - a : a \in P, a \neq y\} \cap D'|$ 
21:      $Q \leftarrow Q \cup \{x\}$ 
22:      $P \leftarrow P \setminus \{x\}$ 
23:      $D' \leftarrow D' \setminus (\{x - a \mid a \in P\} \cap D')$ 
24:   end while
25:    $P \leftarrow Q$ 
26:    $i \leftarrow i + 1$ 
27: end while
28: return  $P_{best}$ 
```

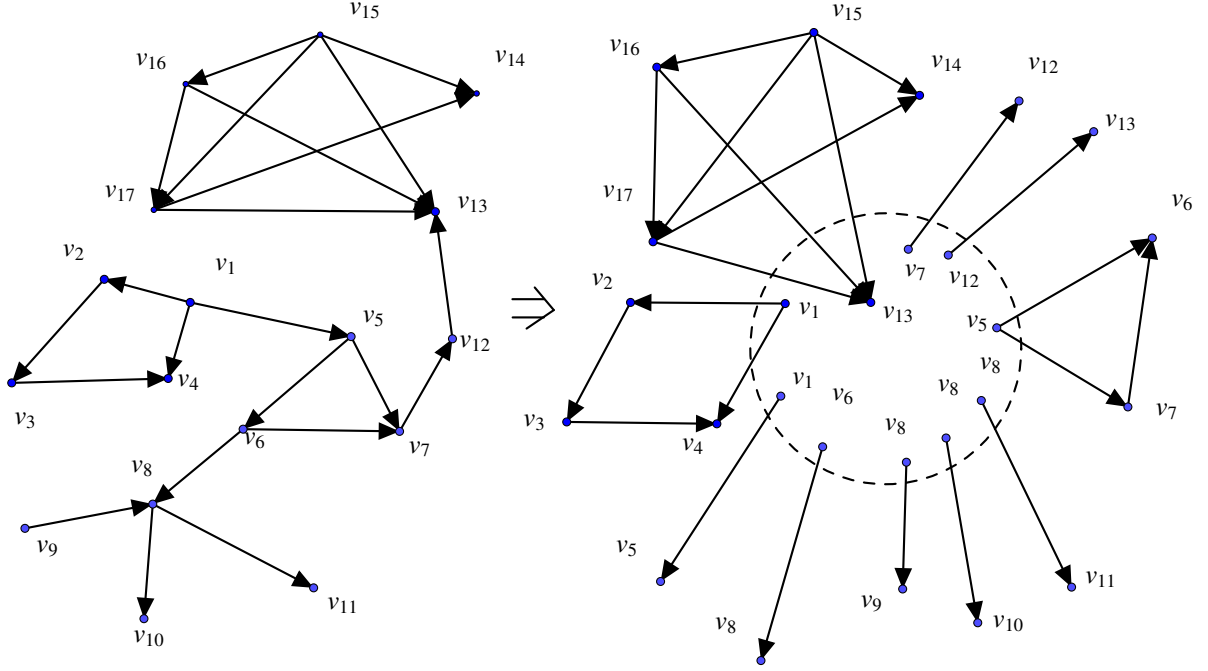


Figure 2: Transform process of distance superset

we name by **cycle component**. A cycle component is an induced subgraph of G constructed of inseparable cycles. To make clear, two cycles C and C' are inseparable if there is a sequence of cycles: $C_0 = C, C_1, \dots, C_m = C'$ such that C_i and C_{i+1} , $0 \leq i < m$, share at least one common arc. It's easily to show that two distinct cycle components share at most one vertex. For example, the graph in Figure 1 has two cycle components (v_1, v_2, v_3, v_4) and (v_1, v_5, v_6) . For the sake of simplicity, an arc that does not contained in any cycle, is also considered as a trivial cycle component. In Figure 1, **arc (v_1, v_7)** is such a trivial cycle component.

Cut-Roll-Paste operation. We first cut the DS graph into cycle components. We paste the cycle components at the minimum point. The received directed graph is called **compact DS graph (CDS)**. The value of vertices in each cycle component of CDS equal the value of vertices in corresponding cycle component of the initial DS graph subtract to the minimum point. **The CDS has the following properties,**

- The vertex values in the same cycle component are distinguished
- The vertex values in different cycle component may be not distinguished.

Given a CDS, we could construct a DS graph by inverse **Cut-Roll-Paste operation**.

An example of Cut-Roll-Paste operation is shown in Figure 2. The common vertex of all cycle components is called **root**. We could turn any cycle component around this root to receive a new CDS that leads to another DS. This is **another symmetric property** of the optimal solutions' space. We will prove that the maximal value of distance superset according to compact form of some optimal distance supersets does not exceed $(\sum_{d \in D_u} d - d_{min})/2$, which will be stated in Theorem 3.

If we relax the distinguish condition for vertex values of DS graph, we have the Distance super-multiset graph (DM graph). The CDS is subclass of DM graph with two restrictions: (i) cycle component shares one common vertex and this vertex is the minimum point of each cycle component; (ii) the vertices belong to the same cycle component are distinguished. We name the common vertex of CDS of cycle components as the root.

We consider another class of DM graph with two restrictions: (i) all cycle component shares one common vertex and this vertex is the minimum point of each cycle component; (ii) every arc that comes in to (out from) a vertex except the root contains is distinguished in value. **This class is named different in out compact DM (DI/OCDM).**

We have the following results

- The CDS is subclass of DI/OCDM (*)
- The smallest DI/OCDM (in size of vertex set) is also a CDS (**).

The Theorem (*) could be proven easily. If a CDS graph contains two arcs (u, v) and (u', v) that come into a non-root vertex, then u and u' belong to the same cycle component but their values are equal to each other which is contrasted to the (ii) condition of CDS. We could prove that there are not any two equal arcs that come from a non-root vertex analogously. Therefore, a CDS satisfies the (i) condition of DI/OCDM definition.

The Theorem (**) could also be proven easily. If the smallest DI/OCDM G^* is not CDS, there are two vertices u and u' in the same cycle component that have the same value. Without losing generality we could assume that u and u' are the smallest equal vertices in the same cycle component of G^* . We could see that the arcs going into u are different to the arcs that go into u' . More precisely, by contrast if there exist arcs (w, u) and (w', u') that share the same value, w and w' are also equal and $w < u$ and $w' < u'$ is contrast to the assumption that u, u' are the smallest equal vertices in a cycle component.

Consider the maximal subgraphs $G_{u'}$ G_u that,

- $G_{u'}$ contains u'
- all arc in $G_{u'}$ that has u' as one end come out of u'
- G_u contains u
- all arc in G_u that has u as one end come out of u
- there exists a bijection $f : G_u \rightarrow G_{u'}$ that $f(u) = u'$ and f pairs each arc of G_u with an arc of the same value of $G_{u'}$.
- G_u and $G_{u'}$ are maximal in the meaning that there does not exist two other subgraphs that satisfy above 5 conditions but are restrictively larger than G_u and $G_{u'}$

We could remove all the arc $a' = (w', v')$ that has an end v' in $V(G_{u'})$ but a' does not belong to $G_{u'}$ and add $a = (w', v)$ with $v = f^{-1}(v') \in V(G_u)$. This is feasible because arc $a = (w', v)$ is not in initial G^* due to the fact that G_u and $G_{u'}$ are maximal. after that, we could cut $G_{u'}$ out and paste it to the root of G^* at the minimum point. The result is a DI/OCDM but smaller than G^* . This is contrast to the fact that G^* of smallest DI/OCDM.

Therefore every smallest DI/OCDM is also a CDS.

Before confronting the theorem about main result of this subsection and its proof, let us give an example, considering the instance *drand-10-110-1* (Fontoura et al. 2018 (10)): $D = \{12, 31, 32, 33, 36, 48, 54, 55, 82, 108\}$ which does not contain any hang distance, using Algorithm Large neighborhood search presented in 4.4, we retrieve the **upper bound**: $u = 8$ then the subset of 8 largest elements of D is $D_u = \{32, 33, 36, 48, 54, 55, 82, 108\}$. Therefore, the value of the bound is $B = (108 + 82 + 55 + 54 + 48 + 36 + 33 + 32 - 12)/2 = 218$ quite smaller than the value $B' = \sum_{d \in D} d = 491$.

Theorem 3. *There exists an optimal solution of MDSP P^* with the maximal value $p_{max} = \max_{p \in P} p$ that does not exceed $\max\{(\sum_{d \in D_u} d - d_{min})/2, d_{max}\}$, where u is an upper bound for $|P^*|$ such that $u \leq |D|$, $D_u \subset D$ is the set of u maximal values of D , d_{min} and d_{max} are minimal and maximal distances in D . In the case, $u = |D|+1$, we have the upper limitation is $\max\{(\sum_{d \in D} d - d_{min})/2, d_{max}\}$.*

Proof. We will prove the theorem in case $u < |D|+1$, the case $u = |D|+1$ could be proved analogously. We present the distance set D and distance superset P using *compact form* of directed graph and name this form as G . The vertex $v_0 = 0$ is the center of this graph. We will prove that the maximal vertex on each *cycle component* does not exceed $\max\{(\sum_{d \in D_u} d - d_{min})/2, d_{max}\}$.

If the cycle component has only one arc (the trivial cycle component) then its maximal vertex equals to the distance weight of this arc and therefore is not large than d_{max} . In other case, if the cycle component has more than one arc, there exists a cycle that starts, ends at v_0 and travels through a number of vertices of the cycle component including

the maximal vertex of this cycle component. We denote that cycle by $C = (v_0, v_1, \dots, v_n)$ and for convenience we consent that $v_{n+1} = v_0$, where $n \geq 3$. The label of vertex v_j according to the point of P^* is also denoted by p_j then $p_0 = 0$. The weight of arc (v_j, v_{j+1}) is the distance between p_j and p_{j+1} and denoted by $d_{jj+1} = |p_j - p_{j+1}| \in D$. Let i be the index of the maximal vertex on C , i.e. $p_i = \max_{0 \leq j \leq n} p_j$. Without losing the generality, we could assume that the arc (v_0, v_1) takes the direction from v_0 to v_1 . We have,

$$p_i = \sigma_{01}d_{01} + \sigma_{12}d_{12} + \dots + \sigma_{i-1,i}d_{i-1,i}, \quad (25)$$

where σ_{jj+1} presents the direction of arc (v_j, v_{j+1}) , for $0 \leq j \leq i-1$, as following: $\sigma_{jj+1} = 1$ if (v_j, v_{j+1}) goes from v_j to v_{j+1} and -1 in other case.

Considering the path from v_i to v_0 through v_n , we also have,

$$-p_i = \sigma_{ii+1}d_{ii+1} + \sigma_{i+1,i+2}d_{i+1,i+2} + \dots + \sigma_{n0}d_{n0}. \quad (26)$$

Combine (25) and (26), we imply that,

$$2p_i \leq \sum_{0 \leq j \leq n} d_{jj+1} = \sum_{d \in D(C)} d, \quad (27)$$

where $D(C)$ is the multiset of distances according to arcs of C .

If C does not contains all distances of D , i.e. $D(C)$ is strictly a subset of D , then as a sequence $|C| < u$, we could imply that,

$$2p_i \leq \sum_{d \in D_u} d - d_{\min}. \quad (28)$$

Consider the left case where $D(C) = D$. We assume that the contrast result where $2p_i > \sum_{d \in D_u} d - d_{\min}$ is true. This happens only if G coincides with C and the arc (v_j, v_{j+1}) goes from j to $j+1$ for every $0 \leq j \leq i-1$ but in oppose direction for the $i \leq j \leq n$. If $i = 1$, then it is easy to see that, $p_i = d_{0i} \leq d_{\max}$. Analogously, The case $n-i = 1$ also leads to following: $p_i = d_{in} \leq d_{\max}$.

If $n \geq 2$ and $n-i \geq 2$, we denote the minimum arc by (v_t, v_{t+1}) , i.e. $d_{tt+1} = \min_{0 \leq j \leq n} d_{jj+1}$. We cut the cycle C into two paths $(v_0, v_1, \dots, v_i)$ and $(v_i, v_{i+1}, \dots, v_{n+1})$. We could swap the location of arcs on path $(v_0, v_1, \dots, v_i)$ or on path $(v_i, v_{i+1}, \dots, v_{n+1})$. We could construct another optimal distance superset with smaller maximal values of vertices by cutting it off the current path the arc (v_t, v_{t+1}) then inserting into in the middle of the other path. For example, considering C with $n = 3, i = 2$ and $t = 1$, and the directions of arcs alongside with weights are as following: $(v_0 \xrightarrow{d_1} v_1), (v_1 \xrightarrow{d_2} v_2), (v_2 \xleftarrow{d_3} v_3), (v_3 \xleftarrow{d_4} v_0)$ we could transform this cycle to $(v_0 \xrightarrow{d_1} v_1), (v_1 \xleftarrow{d_3} v_2), (v_2 \xrightarrow{d_2} v_3), (v_3 \xleftarrow{d_4} v_0)$. The new constructed cycle will have the maximal vertex not exceeds $\sum_{d \in D(C)} d - d_{\min}$.

Therefore, we have $p_{\max} \leq \max\{(\sum_{d \in D(C)} d - d_{\min})/2, d_{\max}\}$. The cycle C does not contains more than u distance from D , hence we could replace $D(C)$ by D_u and retrieve,

$$p_{\max} \leq \max\{(\sum_{d \in D_u} d - d_{\min})/2, d_{\max}\}. \quad (29)$$

□

We believe that, by deeply investigating the structure of directed graph representing distance superset, we could improve B even better in the future. For example, we could replace $d_{\min}$ in formulation $\max\{(\sum_{d \in D_u} d - d_{\min})/2, d_{\max}\}$ in some cases. Considering the reason of inequality (28), the equality occurs only if the following restricted conditions happen: $|C| = u-1$, $d_{\min}$ is the u -th largest element of D (this comes true only if $|D| = u$). We could replace inequality (28) by a better inequality: $2p_i \leq \sum_{d \in D_{u-1}} d$. Considering the reasons that imply (29) for case $D(C) = D$, if $d_{\min}$ could not be a linear combination of distances in D_u with coefficients ± 1 , then we could replace $(\sum_{d \in D_u} d - d_{\min})/2$ by a smaller value, for example a trivial value is $(\sum_{d \in D_u} d - d_{\min} - 1)/2$.

5. Experiments

We implement Integer Linear Programming and Binary Search algorithms with new lower, upper bound, maximal value of distance superset using Java on the computer with configuration: CPU Core i5-7400 3.00GHz, RAM 8.00 GB. The integer linear programming models have been solved by CPLEX 12.8.3. The instances are taken from two sources: one source is Fontoura et al. 2018's data (10), the other is generated from the Luciferase's gene digested by restriction enzymes as Abbas et al. (2016) (1) suggested. For Fontoura et al.'s data, we compare our results with two best algorithm's results of authors. We also give the details about result and time of pre-processing, lower bound, upper bound for each instance.

The data generated by Fontoura et al. 2018 contains 458 instances partitioned into four class: the full instance is the distance multiset of $\binom{n}{2}$ pairwise distances of n randomly generated points; the miss instances received from full instances by randomly removing a certain number of distances; the join instances are the combines of two full instances; the random instances composed of elements randomly chosen on an interval. For each instance, we limit the running time to around 3600 seconds. Table ?? synthesizes the results of instances grouped by type and size. From left to right, the columns show the instance's properties (name, size of distance multiset, number of instances), the gap between resulted lower bound and upper bound and average running time of four experiments: improved Integer programming model, improved Binary search and Fontoura's two Binary search (their best performance algorithms). We could conclude that the new lower bound, upper bound and maximal point values significantly improve the results retrieving smaller gap, smaller average running time and more solved instances (the instances that get zero gap, i.e. lower bound equals upper bound equals optimal solution size).

5.1. Results on randomly generated instances of (10)

Table 4 shows the summarized results of pre-processing, lower bound and upper bound phase of instances gathered by their type and size. We could observe that the lower bound algorithm only works well on drand class. This can be explained as: the full instances' trivial lower bound equals to the optimal solution - therefore lower bound improvement algorithm would not improve lower bound of these instances, the miss instances are gotten from full instances by remove a small number of distances so that their trivial lower bound in almost case equals to the optimal solution, the joint instances encounter the same fact because their optimal solutions tends close to the trivial lower bounds. Table 4 also reveals that on Fontoura et al.'s data, the Large neighborhood search performs much better than multiplication-based upper bound search. It tends to imply that the multiplication-based upper bound search is non-necessary and should be thrown away. However, the Large neighborhood search algorithm is brute force combining greedy strategy with complexity of $O(tB|D|^2)$ could be extremely slow when the values of distance multiset's elements are large. In contrast, the multiplication-based upper bound search algorithm works with the square root of values of distances may be a better choice of large instances.

Tables ??, ??, ??, ?? show the result data of each instance of full, miss, joint, drand class respectively. Table ?? presents the detailed of pre-processing phase, lower bound and upper bound improvement of every instance.

5.2. Results on real instances

We generated the Luciferase data by simulating a partial digestion on the luciferase gense mentioned in (1). The performing restriction enzyme was *TaqI*, which cuts the gene at sub-sequence matching *tcga* pattern. The output distance multiset D containing the distances between the successive *tcga* locations on the luciferase gene, which is $D = \{9, 30, 100, 170, 293, 302, 393, 402, 462, 562, 632, 732, 855, 864, 945, 954, 975, 984, 1025, 1034, 1247, 1277, 1347, 1377, 1809, 1839, 1979, 2009\}$. This full instance has two non-grudient solutions $P_1^* = \{0, 170, 632, 732, 1025, 1034, 1979, 2009\}$ and $P_2^* = \{0, 30, 975, 984, 1277, 1377, 1839, 2009\}$. We then randomly removed R elements from the distances multiset with $R \in \{1, 2, \dots, 14\}$. For each value of R , we extracted 5 instances, numbered from 0 to 4. The instance file is named *luci-R-i.txt*, where R is the number of removed distances, and i is the generated order lies in range from 0 to 4. Table ?? shows the detailed of pre-processing phase, lower bound and upper bound improvement on luciferase instances.

Table 2: Summary on 9 methods (Gap in % and Time in seconds)

Instances	Size	Nb	nIP		nFEAS		nMAX		IP		FEAS		MAX		tIP		tFEAS		tMAX	
			Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
full	10	10	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	1	0.0	0	0.0	1
	15	10	0.0	0	0.0	0	0.0	0	0.0	4	0.0	4	0.0	7	0.0	20	0.0	12	0.0	11
	21	10	0.0	0	0.0	0	0.0	0	6.8	320	0.0	432	0.0	219	0.0	953	0.0	379	0.0	351
	28	10	0.0	0	0.0	0	0.0	0	36.4	2491	17.8	2258	17.8	3092	46.1	3600	28.5	3300	23.3	3559
	36	10	0.0	0	0.0	0	0.0	0	48.2	2522	48.3	3600	37.8	3601	52.6	3600	39.9	3600	47.0	3600
	45	10	0.0	1	0.0	0	0.0	1	48.5	2887	46.4	3600	42.2	3601	62.5	3600	48.5	3600	50.1	3600
Average Solved			0.0	0	0.0	0	0.0	0	23.3	1371	18.8	1649	16.3	1753	26.9	1962	19.5	1815	20.1	1854
				60		60		60		31		36		34		30		33		32
miss	8	10	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0
	13	10	0.0	0	0.0	0	0.0	0	0.0	5	0.0	2	0.0	5	0.0	10	0.0	5	0.0	11
	18	10	0.0	0	0.0	0	0.0	0	12.6	657	0.0	34	0.0	554	3.0	909	3.0	743	3.0	954
	25	10	0.0	0	0.0	0	0.0	0	39.4	2550	22.0	2777	18.9	3274	30.2	3018	27.1	3404	27.9	3288
	32	10	0.0	0	0.0	0	0.0	0	42.2	2834	35.0	3018	29.8	3565	51.8	3600	40.4	3600	37.3	3600
	41	10	0.0	0	0.0	0	0.0	1	53.1	2523	51.4	3600	39.6	3600	60.9	3600	47.9	3600	51.7	3285
Average Solved			0.0	0	0.0	0	0.0	0	24.6	1428	18.1	1572	14.5	1804	24.3	1856	19.7	1892	20.0	1856
				60		60		60		28		36		34		31		30		33
join	20	6	0.0	1670	3.7	1235	9.3	1899	18.5	2013	10.0	1667	17.1	2469	18.4	2418	18.3	1821	15.0	1987
	25	6	3.3	1344	3.3	1116	8.9	2509	17.8	3601	17.6	2295	16.7	3186	31.2	3600	21.1	3470	25.1	3107
	31	6	9.4	3600	11.1	3346	12.7	3601	33.2	3420	28.4	3600	29.9	3600	38.1	3600	33.1	3600	30.2	3600
	38	6	10.4	3604	10.4	3599	10.4	3601	49.9	3059	41.0	3600	34.0	3601	47.7	3600	43.6	3600	40.7	3600
	46	6	9.5	3603	9.5	3600	9.5	3615	48.8	2948	35.1	3600	38.5	3600	66.2	3600	35.9	3600	41.9	3600
	55	6	6.4	3008	6.4	3001	6.4	3015	59.3	2899	57.1	3600	44.1	3601	72.2	3600	52.4	3600	61.1	3600
Average Solved			6.5	2805	7.4	2649	9.5	3040	37.9	2990	31.5	3060	30.1	3343	45.6	3403	34.1	3282	35.7	3249
				14		12		6		3		8		3		3		4		4
drand	5	9	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0
	7	9	0.0	0	0.0	0	0.0	0	0.0	18	0.0	16	0.0	45	0.0	34	0.0	32	0.0	82
	10	9	0.0	455	0.0	241	1.2	674	1.2	653	6.1	512	6.1	910	2.5	985	6.1	612	6.1	1183
	15	9	16.5	2972	15.2	2651	22.0	3135	27.8	3133	27.3	2759	36.7	3478	36.0	3600	31.4	3139	37.3	3405
	20	9	32.9	3600	26.1	3599	32.9	3601	45.6	3604	50.0	3600	50.0	3601	55.1	3600	50.0	3600	45.6	3600
	25	9	32.7	3600	32.7	3599	32.7	3601	46.4	3601	50.4	3600	48.6	3601	56.7	3600	52.9	3600	50.7	3600
Average Solved			13.7	1771	12.3	1682	14.8	1835	20.2	1835	22.3	1748	23.6	1939	25.0	1970	23.4	1831	23.3	1978
				32		30		28		28		29		27		26		28		28

Table 3: Summary on 6 methods (Gap in % and Time in seconds)

Instances	Size	Nb	nIP		nFEAS		nMAX		IP		FEAS		MAX	
			Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
full	55	10	0.8	363	0.0	1	0.0	1	64.4	2735	55.2	3600	52.6	3602
	66	10	0.0	1	0.0	1	0.0	1	72.0	2565	57.2	3600	50.1	3601
	78	10	0.7	362	0.7	362	0.7	362	82.8	798	61.8	3600	53.2	3602
	91	10	2.0	1083	3.2	1442	2.0	1084	84.8	108	65.5	3606	59.8	3603
	105	10	1.2	725	0.6	365	0.6	366	85.8	72	67.8	3607	58.7	3596
	120	10	3.9	1805	4.5	1805	4.9	1806	86.8	43	66.9	3605	60.7	3602
Average Solved			1.5	723	1.5	663	1.4	603	79.4	1053	62.4	3603	55.8	3601
				48		49		50		0		0		0
miss	50	10	0.0	1	0.8	361	0.8	361	61.7	2169	49.0	3600	41.3	3601
	61	10	0.8	362	1.4	361	0.8	363	72.1	1824	55.0	3600	47.6	3570
	72	10	0.7	362	0.7	362	0.7	362	78.6	1132	58.8	3600	51.4	3601
	85	10	2.0	1083	1.3	722	0.7	363	83.7	57	61.7	3601	53.0	3601
	98	10	1.8	724	2.4	1083	4.6	1443	84.8	67	64.9	3603	56.8	3578
	113	10	1.7	726	2.8	1085	5.0	2165	86.0	44	64.2	3603	60.8	3605
	128	10	4.2	1447	4.0	1087	6.1	2165	86.8	24	67.7	3620	59.8	3612
Average Solved			1.6	672	1.9	723	2.7	1032	79.1	760	60.2	3604	53.0	3595
				57		56		50		0		0		0
join	65	6	12.9	3600	12.9	3600	12.9	3602	72.2	1226	54.9	3600	42.7	3599
	76	6	13.3	3600	13.3	3600	13.3	3606	81.3	1274	59.4	3600	49.2	3552
	88	6	11.3	3602	11.3	3600	10.4	3604	84.3	97	57.8	3601	52.3	3601
	101	6	14.2	3600	13.4	3600	13.4	3601	85.3	79	65.0	3606	54.4	3603
	115	6	14.2	3601	12.7	3600	12.7	3602	86.2	50	62.0	3602	53.6	3601
	130	6	12.8	3601	14.1	3601	11.3	3601	87.0	53	72.1	3608	61.5	3604
Average Solved			13.1	3601	13.0	3600	12.3	3603	82.7	463	61.9	3603	52.3	3593
				0		0		0		0		0		0
drand	12	9	0.0	1495	1.2	1318	3.7	2109	9.0	2377	2.5	1672	9.9	2709
	30	9	28.5	3605	27.7	3600	27.7	3601	48.8	3329	52.5	3600	46.4	3603
	35	9	31.8	3616	32.3	3600	31.2	3602	63.0	2954	67.9	3600	54.8	3601
	40	9	32.1	3606	30.9	3600	31.5	3602	58.0	2803	57.6	3600	47.5	3602
	45	9	30.1	3608	29.5	3600	29.5	3615	59.1	3208	58.6	3600	59.2	3602
	50	9	32.4	3604	32.0	3600	32.4	3609	66.5	2796	61.7	3600	51.5	3601
	75	9	31.1	3602	31.5	3600	31.1	3608	76.6	1668	62.5	3601	53.6	3601
	100	9	30.6	3601	30.3	3600	29.8	3402	85.1	163	67.1	3602	61.6	3602
	112	9	29.2	3602	30.8	3601	30.2	3603	85.8	118	68.5	3614	58.5	3604

Continue on next page

Instances	Size	Nb	nIP		nFEAS		nMAX		IP		FEAS		MAX	
			Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time	Gap	Time
Average			27.3	3371	27.4	3346	27.5	3417	61.3	2157	55.4	3388	49.2	3503
Solved		81		9		8		6		4		7		5
luci	13	3	0.0	687	0.0	362	0.0	2442	8.3	2598	0.0	1312	26.7	2453
	14	5	5.0	1192	0.0	813	5.0	2550	16.1	3020	16.0	2109	24.0	2605
	15	3	4.2	2033	0.0	2614	4.2	3048	33.3	3604	30.3	3358	27.7	3601
	16	4	9.4	3109	0.0	2052	3.1	3363	26.1	3602	13.2	2633	22.2	3600
	17	4	6.2	2553	0.0	3045	9.4	3600	33.7	3605	34.0	3600	31.9	3600
	18	3	8.3	3401	0.0	1729	8.3	2919	25.5	3600	34.5	3600	22.2	3601
	19	3	12.5	3605	0.0	3600	8.3	3546	33.0	3608	46.2	3600	38.2	3601
	20	5	7.5	2886	0.0	2277	2.5	2614	45.1	3602	49.3	3600	34.5	3600
	21	3	4.2	1204	0.0	1203	4.2	1203	46.2	3607	50.0	3600	43.3	3600
	22	4	0.0	4	0.0	4	0.0	4	37.0	3600	37.0	3600	23.2	3600
	23	4	0.0	5	0.0	5	0.0	5	42.1	3604	48.5	3600	30.6	3600
	24	3	0.0	6	0.0	6	0.0	6	38.6	3601	50.0	3600	29.5	3600
	25	4	0.0	11	0.0	11	0.0	11	42.6	3601	56.2	3600	19.4	3597
	26	4	0.0	7	0.0	7	0.0	7	49.5	3606	57.3	3600	38.2	3601
	27	4	0.0	12	0.0	12	0.0	11	42.2	3601	54.0	3600	27.8	3600
	28	1	0.0	14	0.0	14	0.0	14	61.9	3601	55.6	3600	55.6	3600
Average			3.7	1311	0.0	1113	2.8	1609	34.7	3439	38.2	3212	29.0	3393
Solved		58		41.0		57.0		45.0		2.0		9.0		4.0

6. Conclusions

In this paper, we show some mathematical results relating graph modelling, and then we prove a value upper bound for some optimal solutions and develop a method to reduce problem size by detecting thorn arc. Our new bounds allow to close the gap for Fontoura et al.'s data, solving more instances for all classes of instances.

For the future work, we recommend to discovery new mathematical properties of MDSP problem. A promising approach is to study the problem in another space V , e.g. the group modulo g as proposed in this paper. The new space V would have the following properties: (i) reserving the difference, i.e. if $x - y = d$ then $\varphi(x) - \varphi(y) = \varphi(d)$, where φ is the map function from the integer set $\mathbb{Z}$ to V (ii) the new MDSP problem on V is easier to solve. We do believe that the key to obtain more fruitful results is Arithmetic Combinatorics, a recent rapidly developing subject of Mathematics. The nature essence of MDSP suggest many relations to Arithmetic Combinatorics, e.g. the sumset operation and arithmetic properties of solution.

The Erroneous MDSP variant (EMDSP) could be an interesting research direction. One simple case is that some distances may flutters around some values, e.g. we could accept a solution that does not exactly cover the distance d but an approximation value in range $[d - \alpha_d, d + \beta_d]$. To deal with this error, then the constraints 2, 3 could be supplant by following inequalities with $\alpha_d = \beta_d = 1$,

$$p_i + x_{ij}^d \times (d - \alpha_d) \leq p_j, \quad 1 \leq i < j \leq u, \quad d \in D_h, \quad (30)$$

$$p_j \leq p_i + M \times (1 - x_{ij}^d) + d_h + \beta_d, \quad 1 \leq i < j \leq u, \quad d \in D_h. \quad (31)$$

This also suggests a new approach to find a better lower bound, i.e., we could reduce size of MDSP problem by dividing all distance to a certain divisor h : $D = \{d_h = \lfloor d/h \rfloor \mid d \in D\}$. We have $\lfloor (x-y)/h \rfloor \leq \lfloor x/h \rfloor - \lfloor y/h \rfloor \leq (x-y)/h + 1$, for every $x \geq y \geq 0$. Therefore, the MDSP is then transformed into the EMDSP with $\alpha_d = 0$ and $\beta_d = 1$. If we take $D = \{d_h = \lceil d/h \rceil \mid d \in D\}$, then analogously we could retrieve an EMDSP with $\alpha_d = 1$ and $\beta_d = 0$. The solution of EMDSP is the lower bound for the MDSP.

A sophisticated strategy to improve the maximal limitation B for points in P , as suggested in the end of Subsection 4.5 is also worthy to research. It may not be a short formulation as $\max\{(\sum_{d \in D_u} d - d_{\min})/2, d_{\max}\}$, but a complicated if-then algorithm.

To conclude, we discuss now the symmetry of the search space. The final phase to solve MSDP is different search algorithms. If we could reduce the symmetry of search space, we bring down the size the search space, hence we hope that the search algorithm would run faster. We did this reduction via proposing the thorn arc pruning phase that removed some distances, the smaller maximal limitation B for value of P , the better lower and upper bound l and u for $|P^*|$. By representing the distance superset and distance set via compact form of directed graph, we discovered the various forms of symmetry of search space: for example sometimes we could swap arcs in a same cycle or turn around the cycle component around the center. This discovers are helpful, e.g. as seen in proof of Theorem 3. Some discovers that we found but were not mentioned in previous parts because we did not retrieve any fruitful results from those. For example, the swapping of arcs on an consecutive similar direction arcs. Or another example is ... On the other hand, reduce the search space may be not always a good way for heuristics algorithms. We may reduce the Anyway, we believe that the compact form that we proposed is a good tool that could impede us further.

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
full-5-30-1	10	5	11	0	0.0	5	0.1	6	13.5	5	12.5
full-5-30-2	10	5	11	0	0.1	5	0.1	10	8.4	5	10.3
full-5-30-3	10	5	11	0	0.0	5	0.1	6	8.3	5	3.2
full-5-30-4	10	5	11	0	0.0	5	0.1	10	3.0	5	2.8
full-5-30-5	10	5	11	0	0.1	5	0.2	9	13.9	5	9.7
full-5-15-1	10	5	11	0	0.1	5	0.1	11	4.5	5	6.8
full-5-15-2	10	5	11	0	0.1	5	0.1	8	14.2	5	7.0

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
full-5-15-3	10	5	11	0	0.1	5	0.1	9	11.1	5	4.0
full-5-15-4	10	5	11	0	0.1	5	0.1	11	6.4	5	6.5
full-5-15-5	10	5	11	0	0.0	5	0.1	10	5.1	5	6.7
full-6-30-1	15	6	16	0	0.1	6	0.1	16	11.4	6	40.6
full-6-30-2	15	6	16	0	2.7	6	0.1	16	14.5	6	11.5
full-6-30-3	15	6	16	0	0.0	6	0.2	13	2.9	6	4.7
full-6-30-4	15	6	16	0	0.1	6	0.1	13	28.0	6	29.6
full-6-30-5	15	6	16	0	0.1	6	0.1	13	18.5	6	57.5
full-6-15-1	15	6	16	0	0.0	6	0.1	12	12.3	6	19.4
full-6-15-2	15	6	16	0	0.1	6	0.2	16	13.4	6	23.2
full-6-15-3	15	6	16	0	0.1	6	0.1	10	7.0	6	6.2
full-6-15-4	15	6	16	0	0.1	6	0.1	11	7.6	6	18.8
full-6-15-5	15	6	16	0	0.1	6	0.1	13	4.2	6	11.4
full-7-30-1	21	7	22	0	0.1	7	0.1	19	42.8	7	90.5
full-7-30-2	21	7	22	0	0.1	7	0.1	16	31.4	7	35.5
full-7-30-3	21	7	22	0	0.1	7	0.1	16	31.4	7	33.0
full-7-30-4	21	7	22	0	0.1	7	0.1	17	51.5	7	43.4
full-7-30-5	21	7	22	0	0.1	7	0.1	13	25.1	7	68.6
full-7-15-1	21	7	22	0	0.1	7	0.1	16	10.2	7	41.2
full-7-15-2	21	7	22	0	0.2	7	0.2	12	14.0	7	49.3
full-7-15-3	21	7	22	0	0.1	7	0.2	14	11.3	7	24.0
full-7-15-4	21	7	22	0	0.1	7	0.1	15	13.0	7	40.8
full-7-15-5	21	7	22	0	0.1	7	0.1	12	7.7	7	45.6
full-8-30-1	28	8	29	0	0.1	8	0.1	16	24.6	8	80.7
full-8-30-2	28	8	29	0	0.3	8	0.1	22	59.3	8	144.4
full-8-30-3	28	8	29	0	0.2	8	0.1	17	31.1	8	110.3
full-8-30-4	28	8	29	0	0.2	8	0.1	20	49.7	8	117.0
full-8-30-5	28	8	29	0	1.6	8	0.1	20	66.2	8	139.7
full-8-15-1	28	8	29	0	0.2	8	0.4	22	28.5	8	78.3
full-8-15-2	28	8	29	0	0.1	8	0.1	14	26.3	8	84.2
full-8-15-3	28	8	29	0	0.1	8	0.1	18	10.8	8	43.2
full-8-15-4	28	8	29	0	0.1	8	0.1	16	11.3	8	56.2
full-8-15-5	28	8	29	0	0.2	8	0.1	16	23.5	8	106.3
full-9-30-1	36	9	37	0	0.5	9	0.1	25	37.7	9	229.9
full-9-30-2	36	9	37	0	0.2	9	0.1	18	66.0	9	166.2
full-9-30-3	36	9	37	0	0.3	9	0.1	23	82.0	9	262.2
full-9-30-4	36	9	37	0	0.4	9	0.1	17	63.0	9	248.1
full-9-30-5	36	9	37	0	0.3	9	0.1	26	73.8	9	248.6
full-9-15-1	36	9	37	0	0.4	9	0.4	17	60.2	9	116.3
full-9-15-2	36	9	37	0	0.1	9	0.1	19	30.1	9	95.7
full-9-15-3	36	9	37	0	0.2	9	0.1	26	32.5	9	114.4
full-9-15-4	36	9	37	0	0.2	9	0.1	20	35.5	9	123.4
full-9-15-5	36	9	37	0	0.1	9	0.1	17	31.7	9	120.7
full-10-30-1	45	10	46	0	0.5	10	0.2	27	87.8	10	374.2
full-10-30-2	45	10	46	0	0.6	10	0.1	27	78.8	10	397.4
full-10-30-3	45	10	46	0	0.3	10	0.1	25	55.6	10	343.8
full-10-30-4	45	10	46	0	0.4	10	0.1	26	134.2	10	569.4

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
full-10-30-5	45	10	46	0	0.8	10	0.1	31	84.2	10	558.7
full-10-15-1	45	10	46	0	0.1	10	0.1	27	71.7	10	324.7
full-10-15-2	45	10	46	0	0.2	10	0.2	29	14.8	10	178.2
full-10-15-3	45	10	46	0	0.2	10	0.1	24	42.6	10	229.0
full-10-15-4	45	10	46	0	0.2	10	0.4	19	141.1	10	1575.3
full-10-15-5	45	10	46	0	0.2	10	0.2	25	35.5	10	269.2
full-11-30-1	55	11	56	0	0.7	11	0.2	34	344.8	11	1316.0
full-11-30-2	55	11	56	0	0.3	11	0.1	30	128.0	11	677.3
full-11-30-3	55	11	56	0	1.2	11	0.1	32	63.8	11	611.4
full-11-30-4	55	11	56	0	0.5	11	0.2	37	291.3	11	1009.7
full-11-30-5	55	11	56	0	0.8	11	0.1	33	351.0	11	907.9
full-11-15-1	55	11	56	0	0.3	11	0.1	27	63.3	12	468.5
full-11-15-2	55	11	56	0	0.4	11	0.2	27	54.3	11	485.8
full-11-15-3	55	11	56	0	0.3	11	0.1	29	39.9	11	335.0
full-11-15-4	55	11	56	0	0.4	11	0.1	32	36.8	11	302.5
full-11-15-5	55	11	56	0	0.3	11	0.2	28	58.4	11	409.7
full-12-30-1	66	12	67	0	1.3	12	0.2	33	252.0	12	1691.0
full-12-30-2	66	12	67	0	0.8	12	0.1	31	494.8	12	1666.4
full-12-30-3	66	12	67	0	0.5	12	0.2	38	78.7	12	1239.4
full-12-30-4	66	12	67	0	0.9	12	0.2	39	273.0	12	1772.0
full-12-30-5	66	12	67	0	1.0	12	0.1	41	304.2	12	1590.3
full-12-15-1	66	12	67	0	0.4	12	0.2	32	39.0	12	749.8
full-12-15-2	66	12	67	0	0.7	12	0.2	24	77.2	12	696.9
full-12-15-3	66	12	67	0	0.4	12	0.1	27	82.9	12	691.5
full-12-15-4	66	12	67	0	0.3	12	0.1	30	70.4	12	849.4
full-12-15-5	66	12	67	0	0.5	12	0.2	36	66.7	12	1018.3
full-13-30-1	78	13	79	0	1.2	13	0.2	39	484.8	13	2749.5
full-13-30-2	78	13	79	0	14.3	13	0.2	36	491.4	13	2751.0
full-13-30-3	78	13	79	0	0.8	13	0.2	36	82.0	13	1550.0
full-13-30-4	78	13	79	0	0.8	13	0.1	36	291.4	13	1823.9
full-13-30-5	78	13	79	0	1.6	13	0.2	36	380.3	13	2562.2
full-13-15-1	78	13	79	0	0.4	13	0.2	41	99.8	13	1262.7
full-13-15-2	78	13	79	0	1.0	13	0.2	40	105.2	13	1257.6
full-13-15-3	78	13	79	0	0.5	13	0.2	35	123.6	14	1246.0
full-13-15-4	78	13	79	0	0.7	13	0.1	34	84.5	13	1337.2
full-13-15-5	78	13	79	0	0.5	13	0.2	39	81.7	13	1259.6
full-14-30-1	91	14	92	0	6.4	14	0.1	38	566.9	14	4693.6
full-14-30-2	91	14	92	0	1.2	14	0.1	42	574.7	15	4817.1
full-14-30-3	91	14	92	0	2.1	14	0.2	38	584.3	14	4647.0
full-14-30-4	91	14	92	0	1.1	14	0.2	35	184.5	14	3013.3
full-14-30-5	91	14	92	0	1.1	14	0.2	37	305.9	14	3593.3
full-14-15-1	91	14	92	0	0.6	14	0.1	34	129.9	14	2330.6
full-14-15-2	91	14	92	0	0.9	14	0.2	38	95.5	14	2202.9
full-14-15-3	91	14	92	0	0.6	14	0.2	32	136.4	15	2121.9
full-14-15-4	91	14	92	0	2.5	14	0.2	32	100.7	15	2443.5
full-14-15-5	91	14	92	0	0.5	14	0.1	38	107.9	14	2309.2
full-15-30-1	105	15	106	0	1.6	15	0.2	50	569.3	15	7441.6

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
full-15-30-2	105	15	106	0	2.9	15	0.2	53	587.0	15	6907.2
full-15-30-3	105	15	106	0	2.7	15	0.2	52	571.3	15	6918.3
full-15-30-4	105	15	106	0	2.5	15	0.2	50	576.5	15	5863.8
full-15-30-5	105	15	106	0	3.7	15	0.2	47	445.4	15	7316.5
full-15-15-1	105	15	106	0	0.8	15	0.2	39	135.4	16	3365.3
full-15-15-2	105	15	106	0	0.9	15	0.2	54	127.4	15	3416.9
full-15-15-3	105	15	106	0	1.1	15	0.2	47	134.6	15	3598.0
full-15-15-4	105	15	106	0	0.8	15	0.2	37	147.2	15	3116.6
full-15-15-5	105	15	106	0	2.2	15	0.1	39	126.8	16	3562.0
full-16-30-1	120	16	121	0	1.3	16	0.2	58	559.9	17	8899.7
full-16-30-2	120	16	121	0	6.8	16	0.2	44	460.0	16	10641.4
full-16-30-3	120	16	121	0	3.4	16	0.2	49	1349.3	16	12092.7
full-16-30-4	120	16	121	0	3.0	16	0.2	44	606.0	16	8692.5
full-16-30-5	120	16	121	0	3.0	16	0.2	52	570.8	17	10123.9
full-16-15-1	120	16	121	0	0.9	16	0.2	56	102.6	19	4023.2
full-16-15-2	120	16	121	0	1.0	16	0.2	55	110.6	17	3931.3
full-16-15-3	120	16	121	0	2.6	16	0.2	49	350.8	16	5321.5
full-16-15-4	120	16	121	0	2.9	16	0.2	50	159.6	16	6267.5
full-16-15-5	120	16	121	0	0.9	16	0.2	53	145.0	17	5026.3
miss-5-15-2-4	8	5	9	0	0.0	5	0.1	7	4.1	5	4.1
miss-5-15-2-5	8	5	9	0	0.3	5	0.2	6	7.3	5	3.3
miss-5-30-2-1	8	5	9	0	0.0	5	0.1	9	6.9	5	4.0
miss-5-30-2-2	8	5	9	0	0.0	5	0.1	8	3.2	5	4.1
miss-5-30-2-3	8	5	9	0	0.1	5	0.1	9	5.9	5	4.8
miss-5-30-2-4	8	5	9	0	0.0	5	0.1	9	13.6	5	12.1
miss-5-30-2-5	8	5	9	0	0.0	5	0.1	9	14.9	5	6.8
miss-5-15-2-1	8	5	9	0	0.0	5	0.1	6	5.4	5	1.7
miss-5-15-2-2	8	5	9	0	0.0	5	0.1	9	1.4	5	3.9
miss-5-15-2-3	8	5	9	0	0.0	5	0.2	9	18.7	5	4.8
miss-6-15-2-4	13	6	14	0	0.0	6	0.1	13	6.9	6	11.2
miss-6-15-2-5	13	6	14	0	0.1	6	0.2	8	10.8	6	11.6
miss-6-30-2-1	13	6	14	0	0.0	6	0.1	11	10.8	6	11.9
miss-6-30-2-2	13	6	14	0	0.0	6	0.1	12	9.0	6	7.8
miss-6-30-2-3	13	6	14	0	0.1	6	0.1	14	6.1	6	14.4
miss-6-30-2-4	13	6	14	0	0.1	6	0.1	11	10.9	6	38.5
miss-6-30-2-5	13	6	14	0	0.2	6	0.1	14	57.4	6	13.4
miss-6-15-2-1	13	6	14	0	0.0	6	0.1	11	9.5	6	9.1
miss-6-15-2-2	13	6	14	0	0.0	6	0.1	8	6.6	6	10.8
miss-6-15-2-3	13	6	14	0	0.1	6	0.1	9	5.7	6	9.4
miss-7-15-3-4	18	7	19	0	0.0	7	0.1	13	6.0	7	23.1
miss-7-15-3-5	18	7	19	0	0.5	7	0.2	16	22.4	7	24.4
miss-7-30-3-1	18	7	19	0	0.1	7	0.1	19	49.9	7	42.0
miss-7-30-3-2	18	7	19	0	0.2	7	0.1	16	38.0	7	41.0
miss-7-30-3-3	18	7	19	0	0.1	7	0.1	13	27.1	7	40.8
miss-7-30-3-4	18	7	19	0	0.1	7	0.2	19	31.3	7	59.9
miss-7-30-3-5	18	7	19	0	0.1	7	0.1	18	10.3	7	28.4
miss-7-15-3-1	18	7	19	0	0.0	7	0.1	12	12.0	7	11.7

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
miss-7-15-3-2	18	7	19	0	0.0	7	0.2	11	7.7	7	21.7
miss-7-15-3-3	18	7	19	0	0.0	7	0.1	15	9.8	7	34.1
miss-8-15-3-4	25	8	26	0	0.1	8	0.1	15	14.8	8	61.4
miss-8-15-3-5	25	8	26	0	0.1	8	0.2	17	12.8	8	34.0
miss-8-30-3-1	25	8	26	0	0.2	8	0.1	19	54.7	8	84.2
miss-8-30-3-2	25	8	26	0	0.5	8	0.1	19	121.3	8	107.6
miss-8-30-3-3	25	8	26	0	0.1	8	0.1	18	34.3	8	78.1
miss-8-30-3-4	25	8	26	0	0.1	8	0.1	20	44.7	8	107.9
miss-8-30-3-5	25	8	26	0	0.2	8	0.1	19	55.7	8	81.6
miss-8-15-3-1	25	8	26	0	0.1	8	0.2	19	16.7	8	48.8
miss-8-15-3-2	25	8	26	0	0.1	8	0.1	17	32.3	8	88.5
miss-8-15-3-3	25	8	26	0	0.1	8	0.2	18	10.8	8	69.5
miss-9-15-4-4	32	9	33	0	0.1	9	0.1	22	25.0	9	118.4
miss-9-15-4-5	32	9	33	0	1.3	9	0.2	28	19.4	9	87.0
miss-9-30-4-1	32	9	33	0	0.3	9	0.1	23	102.4	9	164.8
miss-9-30-4-2	32	9	33	0	0.2	9	0.1	19	96.8	9	174.9
miss-9-30-4-3	32	9	33	0	0.3	9	0.1	26	52.6	9	186.3
miss-9-30-4-4	32	9	33	0	0.1	9	0.1	24	57.4	9	203.4
miss-9-30-4-5	32	9	33	0	0.3	9	0.1	24	137.6	9	193.3
miss-9-15-4-1	32	9	33	0	0.5	9	0.1	19	18.5	9	86.7
miss-9-15-4-2	32	9	33	0	0.2	9	0.1	22	33.2	9	128.8
miss-9-15-4-3	32	9	33	0	320.7	9	0.1	27	63.6	9	144.8
miss-10-15-4-4	41	10	42	0	0.3	10	0.1	27	21.4	10	185.1
miss-10-15-4-5	41	10	42	0	0.2	10	0.1	24	24.1	10	149.8
miss-10-30-4-1	41	10	42	0	0.5	10	0.1	26	95.9	10	336.8
miss-10-30-4-2	41	10	42	0	0.5	10	0.1	29	142.1	10	408.2
miss-10-30-4-3	41	10	42	0	0.6	10	0.1	30	253.2	10	568.0
miss-10-30-4-4	41	10	42	0	0.3	10	0.1	26	89.3	10	450.2
miss-10-30-4-5	41	10	42	0	1.0	10	0.2	24	252.1	10	558.0
miss-10-15-4-1	41	10	42	0	0.2	10	0.2	20	41.1	10	249.0
miss-10-15-4-2	41	10	42	0	0.2	10	0.1	23	69.5	10	243.4
miss-10-15-4-3	41	10	42	0	0.3	10	0.1	26	43.1	10	204.6
miss-11-15-5-4	50	11	51	0	0.2	11	0.1	23	66.5	11	354.1
miss-11-15-5-5	50	11	51	0	0.3	11	0.1	33	51.6	11	270.5
miss-11-30-5-1	50	11	51	0	0.5	11	0.1	29	349.7	11	747.3
miss-11-30-5-2	50	11	51	0	0.5	11	0.1	35	146.9	11	1116.4
miss-11-30-5-3	50	11	51	0	0.4	11	0.1	29	58.1	11	588.4
miss-11-30-5-4	50	11	51	0	1.5	11	0.1	32	351.6	11	749.6
miss-11-30-5-5	50	11	51	0	0.3	11	0.1	26	87.2	11	447.7
miss-11-15-5-1	50	11	51	0	0.3	11	0.1	28	83.0	11	325.5
miss-11-15-5-2	50	11	51	0	0.1	11	0.1	24	39.0	11	280.9
miss-11-15-5-3	50	11	51	0	0.4	11	0.1	27	37.3	11	362.4
miss-12-15-5-4	61	12	62	0	0.4	12	0.1	27	82.1	12	844.7
miss-12-15-5-5	61	12	62	0	0.3	12	0.2	31	33.8	12	368.6
miss-12-30-5-1	61	12	62	0	0.8	12	0.1	36	316.3	12	1425.6
miss-12-30-5-2	61	12	62	0	1.0	12	0.1	31	350.5	12	1520.3
miss-12-30-5-3	61	12	62	0	0.5	12	0.1	27	360.6	12	1211.6

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
miss-12-30-5-4	61	12	62	0	1.9	12	0.1	30	194.3	12	1078.3
miss-12-30-5-5	61	12	62	0	1.3	12	0.1	39	227.2	12	1074.8
miss-12-15-5-1	61	12	62	0	0.7	12	0.2	30	61.8	12	641.3
miss-12-15-5-2	61	12	62	0	0.2	12	0.1	30	69.9	12	634.4
miss-12-15-5-3	61	12	62	0	0.3	12	0.1	29	73.2	13	557.3
miss-13-15-6-4	72	13	73	0	3.9	13	0.1	36	37.1	13	747.5
miss-13-15-6-5	72	13	73	0	0.3	13	0.1	34	25.4	14	529.2
miss-13-30-6-1	72	13	73	0	1.2	13	0.1	37	520.7	13	2321.3
miss-13-30-6-2	72	13	73	0	2.0	13	0.1	43	278.6	13	1909.8
miss-13-30-6-3	72	13	73	0	1.2	13	0.1	47	490.1	13	2548.8
miss-13-30-6-4	72	13	73	0	1.3	13	0.1	32	530.7	13	2145.7
miss-13-30-6-5	72	13	73	0	2.9	13	0.2	40	348.8	13	1712.0
miss-13-15-6-1	72	13	73	0	0.5	13	0.1	34	108.8	13	1152.5
miss-13-15-6-2	72	13	73	0	0.4	13	0.1	32	56.4	13	858.6
miss-13-15-6-3	72	13	73	0	0.5	13	0.2	34	99.9	13	1267.7
miss-14-15-6-4	85	14	86	0	0.8	14	0.2	41	50.8	14	1969.9
miss-14-15-6-5	85	14	86	0	0.5	14	0.1	41	88.3	14	1852.9
miss-14-30-6-1	85	14	86	0	1.0	14	0.1	45	486.5	14	2657.4
miss-14-30-6-2	85	14	86	0	1.8	14	0.1	45	440.7	14	3433.5
miss-14-30-6-3	85	14	86	0	0.9	14	0.1	39	342.4	14	2631.8
miss-14-30-6-4	85	14	86	0	1.3	14	0.2	38	411.4	15	3735.3
miss-14-30-6-5	85	14	86	0	2.1	14	0.2	40	634.6	14	4396.9
miss-14-15-6-1	85	14	86	0	0.6	14	0.1	36	155.0	15	2065.2
miss-14-15-6-2	85	14	86	0	0.7	14	0.2	41	165.6	15	2240.0
miss-14-15-6-3	85	14	86	0	1.6	14	0.2	46	129.7	14	2203.6
miss-15-15-7-4	98	15	99	0	1.1	15	0.2	44	154.1	15	3575.9
miss-15-15-7-5	98	15	99	0	1.0	15	0.2	41	134.0	15	3300.2
miss-15-30-7-1	98	15	99	0	1.0	15	0.2	43	494.9	15	4519.1
miss-15-30-7-2	98	15	99	0	28.1	15	0.2	40	436.7	15	5433.0
miss-15-30-7-3	98	15	99	0	5.8	15	0.2	38	610.1	15	6142.9
miss-15-30-7-4	98	15	99	0	1.4	15	0.2	50	285.2	15	4916.3
miss-15-30-7-5	98	15	99	0	1.9	15	0.2	38	427.6	15	5558.5
miss-15-15-7-1	98	15	99	0	0.7	15	0.2	48	109.2	17	2530.5
miss-15-15-7-2	98	15	99	0	0.8	15	0.2	44	137.8	16	3489.3
miss-15-15-7-3	98	15	99	0	0.6	15	0.1	41	79.4	15	2268.5
miss-16-15-7-4	113	16	114	0	0.8	16	0.2	52	101.9	16	3503.2
miss-16-15-7-5	113	16	114	0	0.8	16	0.2	38	143.5	18	4080.8
miss-16-30-7-1	113	16	114	0	1.7	16	0.2	50	355.4	16	5982.4
miss-16-30-7-2	113	16	114	0	1.8	16	0.2	49	1020.3	16	8242.4
miss-16-30-7-3	113	16	114	0	1.9	16	0.2	46	567.6	16	8334.6
miss-16-30-7-4	113	16	114	0	1.2	16	0.1	45	613.2	16	7540.5
miss-16-30-7-5	113	16	114	0	2.5	16	0.1	47	432.1	16	9517.4
miss-16-15-7-1	113	16	114	0	1.2	16	0.2	45	347.5	16	4343.2
miss-16-15-7-2	113	16	114	0	1.2	16	0.2	51	197.4	17	4842.9
miss-16-15-7-3	113	16	114	0	0.9	16	0.2	36	123.1	16	4531.2
miss-17-15-8-4	128	17	129	0	0.9	17	0.1	58	157.7	19	6417.2
miss-17-15-8-5	128	17	129	0	14.6	17	24.9	54	273.4	17	5880.5

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
miss-17-30-8-1	128	17	129	0	3.8	17	0.2	53	1144.8	17	11280.9
miss-17-30-8-2	128	17	129	0	5.5	17	0.2	63	1144.2	17	12514.6
miss-17-30-8-3	128	17	129	0	4.0	17	0.2	55	576.9	17	12437.7
miss-17-30-8-4	128	17	129	0	6.0	17	0.1	50	573.3	18	11897.7
miss-17-30-8-5	128	17	129	0	2.5	17	0.2	50	542.2	17	10542.3
miss-17-15-8-1	128	17	129	0	5.3	17	0.2	47	310.5	17	5735.6
miss-17-15-8-2	128	17	129	0	1.2	17	0.1	43	107.5	20	5231.8
miss-17-15-8-3	128	17	129	0	1.4	17	0.2	48	345.4	19	6811.0
joint-5-30-5-30-1	20	7	21	0	0.1	7	0.1	17	26.8	9	96.5
joint-5-30-5-30-2	20	7	21	0	0.1	7	0.1	16	10.7	9	72.1
joint-5-30-5-30-3	20	7	21	0	0.2	8	0.2	17	10.3	9	60.3
joint-5-15-5-15-1	20	7	21	0	0.0	7	0.1	17	4.3	8	24.7
joint-5-15-5-15-2	20	7	21	0	0.1	7	0.2	16	18.1	8	52.6
joint-5-15-5-15-3	20	7	21	0	0.1	7	0.1	19	9.4	8	27.9
joint-6-15-5-15-1	25	8	26	0	0.1	8	0.1	22	15.7	9	61.8
joint-6-15-5-15-2	25	8	26	0	0.1	8	0.2	20	7.6	9	40.6
joint-6-15-5-15-3	25	8	26	0	0.1	8	0.2	24	9.2	9	62.8
joint-6-30-5-30-1	25	8	26	0	0.2	8	0.2	24	15.1	10	107.7
joint-6-30-5-30-2	25	8	26	0	0.1	9	0.1	14	14.2	9	66.5
joint-6-30-5-30-3	25	8	26	0	0.1	8	0.1	20	23.3	9	80.7
joint-7-15-5-15-2	31	9	32	0	0.1	9	0.1	24	15.7	10	101.8
joint-7-15-5-15-3	31	9	32	0	0.1	9	0.1	22	12.7	10	70.9
joint-7-30-5-30-1	31	9	32	0	0.1	9	0.1	19	34.6	10	148.5
joint-7-30-5-30-2	31	9	32	0	0.2	9	0.1	18	35.7	11	144.9
joint-7-30-5-30-3	31	9	32	0	0.2	9	0.1	22	35.5	11	156.0
joint-7-15-5-15-1	31	9	32	0	0.1	9	0.1	24	15.7	10	98.0
joint-8-30-5-30-3	38	10	39	0	0.1	10	0.1	23	43.7	11	226.8
joint-8-15-5-15-1	38	10	39	0	0.1	10	0.1	22	13.4	11	116.6
joint-8-15-5-15-2	38	10	39	0	0.2	10	0.1	22	36.8	12	175.9
joint-8-15-5-15-3	38	10	39	0	0.1	10	0.1	21	19.1	11	114.6
joint-8-30-5-30-1	38	10	39	0	0.3	10	0.2	22	74.7	11	302.9
joint-8-30-5-30-2	38	10	39	0	0.3	10	0.1	22	112.0	11	389.3
joint-9-15-5-15-1	46	11	47	0	0.2	11	0.1	27	64.1	12	256.5
joint-9-15-5-15-2	46	11	47	0	0.3	11	0.2	27	19.5	12	253.9
joint-9-15-5-15-3	46	11	47	0	0.2	11	0.2	25	40.9	12	199.1
joint-9-30-5-30-1	46	11	47	0	0.4	11	0.1	27	79.7	12	490.7
joint-9-30-5-30-2	46	11	47	0	0.5	11	0.2	32	114.8	13	602.8
joint-9-30-5-30-3	46	11	47	0	0.2	11	0.2	27	23.9	12	208.3
joint-10-30-5-30-3	55	11	56	0	0.6	12	0.1	28	124.7	13	764.4
joint-10-15-5-15-1	55	11	56	0	0.7	12	0.1	32	41.7	12	331.8
joint-10-15-5-15-2	55	11	56	0	0.3	12	0.2	26	63.5	13	418.3
joint-10-15-5-15-3	55	11	56	0	0.2	12	0.1	30	44.8	13	274.6
joint-10-30-5-30-1	55	11	56	0	0.4	12	0.1	29	79.8	13	664.2
joint-10-30-5-30-2	55	11	56	0	0.5	12	0.1	39	383.8	13	866.7
joint-11-15-5-15-1	65	12	66	0	0.3	13	0.2	33	25.1	14	527.2
joint-11-15-5-15-2	65	12	66	0	0.5	12	0.2	30	69.5	14	743.4
joint-11-15-5-15-3	65	12	66	0	0.3	12	0.2	33	81.2	14	705.1

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
joint-11-30-5-30-1	65	12	66	0	0.7	12	0.1	37	356.9	14	1178.7
joint-11-30-5-30-2	65	12	66	0	0.5	13	0.1	33	151.2	15	1126.5
joint-11-30-5-30-3	65	12	66	0	0.5	12	0.2	28	115.5	14	811.1
joint-12-30-5-30-1	76	13	77	0	0.7	13	0.2	36	351.1	15	2355.1
joint-12-30-5-30-2	76	13	77	0	3.2	13	0.2	34	276.2	15	2300.2
joint-12-30-5-30-3	76	13	77	0	1.3	13	0.2	35	351.6	15	1779.9
joint-12-15-5-15-1	76	13	77	0	0.5	13	0.2	36	41.7	15	900.4
joint-12-15-5-15-2	76	13	77	0	0.3	13	0.1	30	84.3	15	1024.2
joint-12-15-5-15-3	76	13	77	0	0.5	13	0.1	33	80.5	15	1075.3
joint-13-15-5-15-1	88	14	89	0	0.5	14	0.2	42	66.1	16	1465.7
joint-13-15-5-15-2	88	14	89	0	0.5	14	0.2	44	73.4	16	1320.4
joint-13-15-5-15-3	88	14	89	0	0.4	14	0.1	43	82.4	16	1420.2
joint-13-30-5-30-1	88	14	89	0	2.2	14	0.2	35	508.8	17	3789.2
joint-13-30-5-30-2	88	14	89	0	1.1	15	0.2	41	338.1	16	2987.9
joint-13-30-5-30-3	88	14	89	0	1.5	15	0.1	47	766.4	16	4814.4
joint-14-15-5-15-2	101	15	102	0	1.1	15	0.2	37	87.2	18	1811.0
joint-14-15-5-15-3	101	15	102	0	0.6	15	0.2	43	77.7	18	2169.6
joint-14-30-5-30-1	101	15	102	0	2.8	15	0.2	47	491.4	17	5042.2
joint-14-30-5-30-2	101	15	102	0	224.7	15	4.0	46	910.0	18	4853.5
joint-14-30-5-30-3	101	15	102	0	144.3	15	0.2	54	451.3	17	5850.4
joint-14-15-5-15-1	101	15	102	0	1.0	15	0.2	41	195.3	17	3393.8
joint-15-30-5-30-3	115	16	116	0	2.3	16	0.2	48	581.9	18	8931.6
joint-15-15-5-15-1	115	16	116	0	12.4	16	0.2	41	133.5	20	4651.4
joint-15-15-5-15-2	115	16	116	0	11.1	16	0.2	46	107.0	19	3055.5
joint-15-15-5-15-3	115	16	116	0	1.8	16	0.2	57	348.1	18	4240.8
joint-15-30-5-30-1	115	16	116	0	10.6	16	0.2	53	508.8	18	7776.5
joint-15-30-5-30-2	115	16	116	0	7.4	16	0.2	41	498.8	19	6622.7
joint-16-15-5-15-1	130	17	131	0	4.9	17	0.2	60	316.8	19	6242.8
joint-16-15-5-15-2	130	17	131	0	1.5	17	0.2	59	191.5	20	6125.9
joint-16-15-5-15-3	130	17	131	0	1.9	17	0.3	56	195.4	19	6911.4
joint-16-30-5-30-1	130	17	131	0	2.3	17	0.2	44	527.0	20	8826.2
joint-16-30-5-30-2	130	17	131	0	2.3	17	0.2	48	463.7	20	12630.3
joint-16-30-5-30-3	130	17	131	0	5.1	17	0.2	64	1191.0	19	14380.8
drand-5-110-2	5	4	6	5	3.5	-	-	-	-	-	-
drand-5-110-3	5	4	6	5	2.6	-	-	-	-	-	-
drand-5-200-1	5	4	6	5	3.4	-	-	-	-	-	-
drand-5-200-2	5	4	6	5	0.1	-	-	-	-	-	-
drand-5-200-3	5	4	6	2	0.1	5	0.2	6	1.4	5	3.3
drand-5-75-1	5	4	6	5	0.1	-	-	-	-	-	-
drand-5-75-2	5	4	6	0	0.0	4	0.2	6	5.0	5	5.8
drand-5-75-3	5	4	6	5	0.1	-	-	-	-	-	-
drand-5-110-1	5	4	6	2	0.0	5	0.1	5	3.6	5	2.3
drand-7-200-1	7	5	8	7	0.1	-	-	-	-	-	-
drand-7-200-2	7	5	8	7	0.1	-	-	-	-	-	-
drand-7-200-3	7	5	8	1	0.0	5	0.1	8	6.7	7	22.5
drand-7-75-1	7	5	8	2	0.1	6	0.1	8	4.3	7	7.8
drand-7-75-2	7	5	8	0	0.1	5	0.2	8	1.8	6	9.2

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
drand-7-75-3	7	5	8	0	0.1	5	0.5	8	50.8	6	18.8
drand-7-110-1	7	5	8	1	0.5	5	0.1	8	3.6	7	5.5
drand-7-110-2	7	5	8	1	0.1	5	0.2	8	2.6	7	18.0
drand-7-110-3	7	5	8	1	0.0	6	0.8	8	38.9	7	243.8
drand-10-110-1	10	5	11	0	2.0	6	11.7	11	156.2	8	124.8
drand-10-110-2	10	5	11	0	3.0	6	23.6	11	141.6	8	178.7
drand-10-110-3	10	5	11	0	6.2	6	93.7	11	300.5	8	102.4
drand-10-200-1	10	5	11	0	3.3	5	33.7	11	220.7	8	178.2
drand-10-200-2	10	5	11	0	4.1	5	37.9	11	277.2	9	173.4
drand-10-200-3	10	5	11	0	2.6	6	12.1	11	139.2	8	116.2
drand-10-75-1	10	5	11	0	1.6	6	20.8	11	110.8	7	86.3
drand-10-75-2	10	5	11	0	2.7	5	40.9	11	329.4	8	190.7
drand-10-75-3	10	5	11	0	2.0	5	12.3	11	151.8	7	92.5
drand-12-200-1	12	6	13	0	2.1	6	0.1	13	6.5	9	81.9
drand-12-200-2	12	6	13	0	1.0	6	0.1	13	49.6	9	92.3
drand-12-200-3	12	6	13	0	0.2	6	0.2	13	20.4	8	81.4
drand-12-75-1	12	6	13	0	0.0	6	0.2	11	10.9	7	33.5
drand-12-75-2	12	6	13	0	0.9	6	0.2	13	20.2	8	43.8
drand-12-75-3	12	6	13	0	0.0	6	0.1	11	6.3	8	32.7
drand-12-110-1	12	6	13	0	0.1	6	0.2	13	70.4	8	69.9
drand-12-110-2	12	6	13	0	0.1	6	0.2	13	22.8	8	67.9
drand-12-110-3	12	6	13	0	0.2	6	0.2	13	46.9	9	69.1
drand-15-200-2	15	6	16	0	0.2	7	0.2	16	37.1	10	152.1
drand-15-200-3	15	6	16	0	0.2	7	0.2	16	60.5	10	101.2
drand-15-75-1	15	6	16	0	0.2	6	0.2	14	23.8	9	29.7
drand-15-75-2	15	6	16	0	0.7	7	0.2	16	10.6	9	49.2
drand-15-75-3	15	6	16	0	0.3	7	0.1	13	41.8	8	56.3
drand-15-110-1	15	6	16	0	0.2	7	0.2	16	40.6	10	75.0
drand-15-110-2	15	6	16	0	0.2	6	0.1	16	30.4	10	86.6
drand-15-110-3	15	6	16	0	0.1	6	0.1	16	6.1	9	83.1
drand-15-200-1	15	6	16	0	1.8	6	0.1	15	26.9	11	191.4
drand-20-200-3	20	7	21	0	0.4	7	0.2	21	76.9	11	189.8
drand-20-75-1	20	7	21	0	0.4	7	0.1	20	36.8	11	97.5
drand-20-75-2	20	7	21	0	0.1	7	0.2	17	27.8	10	70.5
drand-20-75-3	20	7	21	0	0.2	7	0.1	18	26.6	10	67.8
drand-20-110-1	20	7	21	0	0.1	8	0.2	19	40.3	11	153.5
drand-20-110-2	20	7	21	0	0.3	7	0.2	18	37.4	11	138.9
drand-20-110-3	20	7	21	0	0.2	7	0.1	16	63.3	10	100.0
drand-20-200-1	20	7	21	0	2.9	8	0.1	20	386.7	12	256.0
drand-20-200-2	20	7	21	0	0.3	7	0.2	15	66.1	11	222.8
drand-25-75-1	25	8	26	0	0.3	8	0.2	15	43.9	11	111.0
drand-25-75-2	25	8	26	0	0.1	8	0.2	18	18.2	11	95.5
drand-25-75-3	25	8	26	0	0.4	9	0.2	22	17.7	12	155.7
drand-25-110-1	25	8	26	0	0.5	8	0.1	20	84.9	12	175.1
drand-25-110-2	25	8	26	0	0.1	8	0.2	20	79.1	12	188.8
drand-25-110-3	25	8	26	0	0.5	8	0.2	20	75.4	12	161.2
drand-25-200-1	25	8	26	0	1.9	8	0.2	23	61.0	12	248.9

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
drand-25-200-2	25	8	26	0	0.6	8	0.1	25	130.6	14	584.9
drand-25-200-3	25	8	26	0	0.5	8	0.2	24	240.8	13	369.8
drand-30-75-2	30	9	31	0	0.1	9	0.2	22	21.8	12	148.3
drand-30-75-3	30	9	31	0	0.1	9	0.2	19	39.0	12	175.5
drand-30-110-1	30	9	31	0	0.3	9	0.2	25	110.0	12	193.0
drand-30-110-2	30	9	31	0	0.3	9	0.2	24	45.7	13	294.7
drand-30-110-3	30	9	31	0	0.1	9	0.2	21	51.5	11	202.7
drand-30-200-1	30	9	31	0	3.4	9	0.2	28	164.0	14	490.8
drand-30-200-2	30	9	31	0	0.6	9	0.2	28	230.2	14	390.6
drand-30-200-3	30	9	31	0	0.3	9	0.2	25	370.3	14	594.9
drand-30-75-1	30	9	31	0	4.4	9	22.6	22	305.9	12	232.2
drand-35-75-3	35	9	36	0	0.7	10	0.2	21	44.7	13	244.5
drand-35-110-1	35	9	36	0	0.3	10	0.2	23	99.2	14	453.4
drand-35-110-2	35	9	36	0	0.6	10	0.2	25	109.3	15	521.1
drand-35-110-3	35	9	36	0	0.3	10	3.5	24	166.9	14	745.6
drand-35-200-1	35	9	36	0	0.6	9	0.2	32	271.0	15	730.0
drand-35-200-2	35	9	36	0	9.2	9	44.1	30	599.7	14	716.4
drand-35-200-3	35	9	36	0	2.3	9	1.7	28	233.0	14	799.6
drand-35-75-1	35	9	36	0	0.2	9	0.2	24	42.8	13	218.0
drand-35-75-2	35	9	36	0	1.5	9	0.2	23	42.5	13	276.5
drand-40-110-1	40	10	41	0	0.4	10	0.1	24	127.7	14	619.1
drand-40-110-2	40	10	41	0	0.4	10	0.2	23	106.7	15	524.4
drand-40-110-3	40	10	41	0	1.2	10	8.6	25	302.5	14	631.3
drand-40-200-1	40	10	41	0	5.9	10	0.9	30	155.6	16	975.8
drand-40-200-2	40	10	41	0	0.7	10	0.1	27	503.3	16	1249.0
drand-40-200-3	40	10	41	0	2.8	10	0.2	31	279.1	16	954.1
drand-40-75-1	40	10	41	0	0.3	10	0.3	24	57.2	14	266.8
drand-40-75-2	40	10	41	0	0.3	10	0.3	26	142.9	14	1173.9
drand-40-75-3	40	10	41	0	0.2	10	0.3	24	118.4	14	317.6
drand-45-110-1	45	10	46	0	1.8	11	0.2	31	130.9	15	766.4
drand-45-110-2	45	10	46	0	1.8	10	0.2	26	103.3	15	645.5
drand-45-110-3	45	10	46	0	0.6	10	0.2	27	127.9	15	741.3
drand-45-200-1	45	10	46	0	365.0	11	4.1	33	507.4	17	1978.1
drand-45-200-2	45	10	46	0	0.7	10	0.2	33	439.0	16	1312.4
drand-45-200-3	45	10	46	0	35.9	11	0.2	34	494.5	17	1519.0
drand-45-75-1	45	10	46	0	0.3	11	0.2	27	58.8	14	368.1
drand-45-75-2	45	10	46	0	0.4	11	0.2	25	61.9	15	404.6
drand-45-75-3	45	10	46	0	0.6	11	0.2	24	71.8	14	406.8
drand-50-110-3	50	11	51	0	3.8	11	0.3	29	187.5	16	1176.1
drand-50-200-1	50	11	51	0	58.1	11	3.5	33	338.8	18	2130.1
drand-50-200-2	50	11	51	0	374.8	11	4.1	37	489.7	17	1836.6
drand-50-200-3	50	11	51	0	17.1	11	0.2	40	348.2	18	2392.6
drand-50-75-1	50	11	51	0	0.6	11	0.1	29	65.7	15	533.0
drand-50-75-2	50	11	51	0	2.0	11	0.1	27	57.5	16	567.7
drand-50-75-3	50	11	51	0	20.9	11	0.2	32	81.9	15	487.2
drand-50-110-1	50	11	51	0	41.6	11	0.1	35	83.1	16	1030.3
drand-50-110-2	50	11	51	0	0.3	11	0.1	31	111.1	16	879.3

Continue on next page

Table 4: Results preprocessing, lower bound and upper bound phases (Time in ms)

Instances	Size	Trivial		HD		LB		UB		DUB	
		tLB	tUB	HD	Time	nLB	Time	nUB	Time	nDUB	Time
drand-75-200-2	75	13	76	0	1.2	13	0.2	45	492.0	21	4985.7
drand-75-200-3	75	13	76	0	10.4	13	0.2	45	499.5	21	5517.3
drand-75-75-1	75	13	76	0	108.1	13	0.1	33	54.4	17	1369.1
drand-75-75-2	75	13	76	0	1.2	13	0.2	30	79.5	19	1757.7
drand-75-75-3	75	13	76	0	0.4	13	0.2	39	70.6	18	1409.9
drand-75-110-1	75	13	76	0	1.5	13	0.2	35	135.2	19	2202.0
drand-75-110-2	75	13	76	0	378.7	14	0.2	43	113.5	18	1942.9
drand-75-110-3	75	13	76	0	2.1	13	0.1	40	143.1	19	2597.4
drand-75-200-1	75	13	76	0	22.3	13	0.2	40	722.8	20	4335.6
drand-100-110-2	100	15	101	0	1.1	15	0.7	47	143.1	22	4194.8
drand-100-110-3	100	15	101	0	7.6	15	0.7	45	150.6	21	4121.1
drand-100-200-1	100	15	101	0	68.0	15	0.7	51	600.0	23	8222.1
drand-100-200-2	100	15	101	0	4.8	15	0.7	51	508.1	25	10308.8
drand-100-200-3	100	15	101	0	15.9	15	0.7	58	625.1	24	9893.0
drand-100-75-1	100	15	101	0	11.8	15	0.6	41	59.1	20	2689.9
drand-100-75-2	100	15	101	0	4.8	15	0.7	39	103.8	20	2500.1
drand-100-75-3	100	15	101	0	6.1	16	0.9	44	88.9	21	2764.4
drand-100-110-1	100	15	101	0	8.7	15	12.6	49	369.1	21	4746.2
drand-112-110-3	112	16	113	0	3.8	16	0.3	53	135.9	22	5673.2
drand-112-200-1	112	16	113	0	30.2	16	0.3	54	580.8	24	12928.6
drand-112-200-2	112	16	113	0	18.5	16	0.3	50	506.7	24	12392.6
drand-112-200-3	112	16	113	0	79.9	16	0.4	58	564.8	25	14894.1
drand-112-75-1	112	16	113	0	86.8	16	73.0	57	375.4	21	4158.7
drand-112-75-2	112	16	113	0	0.7	16	0.3	42	84.3	22	4123.5
drand-112-75-3	112	16	113	0	1.6	16	0.3	41	78.5	21	3201.9
drand-112-110-1	112	16	113	0	6.4	16	3.2	45	143.2	22	6233.4
drand-112-110-2	112	16	113	0	84.2	16	0.3	55	134.9	23	6351.1

Table 5: Results of 9 methods on full instances

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time
full-5-30-1	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	2	5	5	0.0	1	5	5	0.0	2
full-5-30-2	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	1	5	5	0.0	0	5	5	0.0	1	5	5	0.0	0	5	5	0.0	1
full-5-30-3	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
full-5-30-4	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	1
full-5-30-5	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	1	5	5	0.0	0	5	5	0.0	0
full-5-15-1	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	1	5	5	0.0	0	5	5	0.0	0
full-5-15-2	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
full-5-15-3	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	1	5	5	0.0	0	5	5	0.0	0
full-5-15-4	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
full-5-15-5	10	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
full-6-30-1	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	4	6	6	0.0	11	6	6	0.0	7	6	6	0.0	77	6	6	0.0	46	6	6	0.0	12
full-6-30-2	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	5	6	6	0.0	2	6	6	0.0	5	6	6	0.0	8	6	6	0.0	5	6	6	0.0	5
full-6-30-3	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	3	6	6	0.0	0	6	6	0.0	0	6	6	0.0	5	6	6	0.0	2	6	6	0.0	2
full-6-30-4	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	3	6	6	0.0	16	6	6	0.0	37	6	6	0.0	59	6	6	0.0	19	6	6	0.0	29
full-6-30-5	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	6	6	6	0.0	1	6	6	0.0	4	6	6	0.0	23	6	6	0.0	28	6	6	0.0	31
full-6-15-1	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	3	6	6	0.0	0	6	6	0.0	4	6	6	0.0	7	6	6	0.0	2	6	6	0.0	11
full-6-15-2	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	4	6	6	0.0	2	6	6	0.0	6	6	6	0.0	4	6	6	0.0	11	6	6	0.0	8
full-6-15-3	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	4	6	6	0.0	1	6	6	0.0	1	6	6	0.0	5	6	6	0.0	3	6	6	0.0	4
full-6-15-4	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	4	6	6	0.0	12	6	6	0.0	3	6	6	0.0	6	6	6	0.0	5	6	6	0.0	8
full-6-15-5	15	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	3	6	6	0.0	0	6	6	0.0	1	6	6	0.0	6	6	6	0.0	3	6	6	0.0	4
full-7-30-1	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	22	68.2	451	7	7	0.0	229	7	7	0.0	954	7	7	0.0	659	7	7	0.0	1855	7	7	0.0	695
full-7-30-2	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	22	7	7	0.0	217	7	7	0.0	68	7	7	0.0	2392	7	7	0.0	65	7	7	0.0	908
full-7-30-3	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	99	7	7	0.0	110	7	7	0.0	16	7	7	0.0	1157	7	7	0.0	613	7	7	0.0	15
full-7-30-4	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	1529	7	7	0.0	315	7	7	0.0	507	7	7	0.0	2841	7	7	0.0	34	7	7	0.0	136
full-7-30-5	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	150	7	7	0.0	47	7	7	0.0	115	7	7	0.0	465	7	7	0.0	29	7	7	0.0	770
full-7-15-1	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	22	7	7	0.0	3	7	7	0.0	8	7	7	0.0	58	7	7	0.0	6	7	7	0.0	8
full-7-15-2	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	59	7	7	0.0	12	7	7	0.0	306	7	7	0.0	385	7	7	0.0	22	7	7	0.0	296
full-7-15-3	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	99	7	7	0.0	20	7	7	0.0	39	7	7	0.0	192	7	7	0.0	186	7	7	0.0	155
full-7-15-4	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	65	7	7	0.0	3362	7	7	0.0	100	7	7	0.0	1182	7	7	0.0	875	7	7	0.0	492
full-7-15-5	21	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	705	7	7	0.0	2	7	7	0.0	77	7	7	0.0	202	7	7	0.0	102	7	7	0.0	38
full-8-30-1	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	29	72.4	0	8	8	0.0	2771	8	8	0.0	3040	8	14	42.9	3600	8	13	38.5	3600	8	8	0.0	3233
full-8-30-2	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	13	38.5	3601	8	18	55.6	3600	8	12	33.3	3600	8	25	68.0	3600	8	18	55.6	3600	8	10	20.0	3600
full-8-30-3	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	29	72.4	371	8	12	33.3	3600	8	12	33.3	3600	8	16	50.0	3600	8	13	38.5	3600	8	10	20.0	3600
full-8-30-4	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	29	72.4	2527	8	8	0.0	525	8	9	11.1	3572	8	23	65.2	3600	8	13	38.5	3600	8	13	38.5	3600
full-8-30-5	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	14	42.9	3601	8	18	55.6	3600	8	8	0.0	3591	8	20	60.0	3600	8	18	55.6	3600	8	18	55.6	3600
full-8-15-1	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	8	0.0	3283	8	8	0.0	587	8	8	0.0	2628	8	14	42.9	3600	8	8	0.0	3082	8	10	20.0	3600
full-8-15-2	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	11	27.3	3601	8	8	0.0	928	8	12	33.3	3600	8	13	38.5	3600	8	13	38.5	3600	8	10	20.0	3600
full-8-15-3	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	8	0.0	723	8	8	0.0	1093	8	12	33.3	3601	8	11	27.3	3600	8	8	0.0	1816	8	8	0.0	3554
full-8-15-4	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	11	27.3	3601	8	12	33.3	3600	8	12	33.3	3600	8	12	33.3	3600	8	8	0.0	2905	8	10	20.0	3600
full-8-15-5	28	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	9	11.1	3601	8	8	0.0	2272	8	8	0.0	89	8	12	33.3	3600	8	10	20.0	3600	8	13	38.5	3600
full-9-30-1	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	37	75.7	0	9	23	60.9	3600	9	15	40.0	3601	9	22	59.1	3600	9	16	43.8	3600	9	23	60.9	3600
full-9-30-2	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	16	43.7	3602	9	23	60.9	3600	9	15	40.0	3601	9	18	50.0	3600	9	16	43.8	3600	9	16	43.8	3600

Continue on next page

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time				
full-9-30-3	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	37	75.7	0	9	15	40.0	3600	9	11	18.2	3600	9	36	75.0	3600	9	16	43.8	3600	9	23	60.9	3600
full-9-30-4	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	37	75.7	0	9	23	60.9	3600	9	15	40.0	3600	9	24	62.5	3600	9	16	43.8	3600	9	23	60.9	3600
full-9-30-5	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	17	47.1	3603	9	23	60.9	3600	9	15	40.0	3601	9	21	57.1	3600	9	23	60.9	3600	9	16	43.8	3600
full-9-15-1	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	14	35.7	3602	9	15	40.0	3600	9	15	40.0	3601	9	14	35.7	3600	9	16	43.8	3600	9	16	43.8	3600
full-9-15-2	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	13	30.8	3602	9	15	40.0	3600	9	15	40.0	3602	9	15	40.0	3600	9	12	25.0	3600	9	16	43.8	3600
full-9-15-3	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	13	30.8	3602	9	15	40.0	3600	9	15	40.0	3601	9	23	60.9	3600	9	16	43.8	3600	9	16	43.8	3600
full-9-15-4	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	14	35.7	3602	9	15	40.0	3600	9	15	40.0	3600	9	20	55.0	3600	9	12	25.0	3600	9	16	43.8	3600
full-9-15-5	36	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	13	30.8	3602	9	15	40.0	3600	9	15	40.0	3603	9	13	30.8	3600	9	12	25.0	3600	9	12	25.0	3600
full-10-30-1	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	1	10	19	47.4	3607	10	18	44.4	3600	10	18	44.4	3601	10	26	61.5	3600	10	28	64.3	3600	10	19	47.4	3600
full-10-30-2	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	17	41.2	3607	10	18	44.4	3600	10	13	23.1	3600	10	26	61.5	3600	10	28	64.3	3600	10	19	47.4	3600
full-10-30-3	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	22	54.5	3607	10	18	44.4	3600	10	18	44.4	3600	10	27	63.0	3600	10	19	47.4	3600	10	28	64.3	3600
full-10-30-4	45	10	10	0.0	1	10	10	0.0	1	10	10	0.0	1	10	16	37.5	3611	10	18	44.4	3602	10	18	44.4	3601	10	46	78.3	3600	10	28	64.3	3600	10	46	78.3	3600
full-10-30-5	45	10	10	0.0	1	10	10	0.0	1	10	10	0.0	1	10	20	50.0	3612	10	28	64.3	3600	10	18	44.4	3600	10	45	77.8	3600	10	28	64.3	3600	10	19	47.4	3600
full-10-15-1	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	46	78.3	1	10	18	44.4	3600	10	28	64.3	3601	10	42	76.2	3600	10	19	47.4	3600	10	28	64.3	3600
full-10-15-2	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	13	23.1	3606	10	18	44.4	3600	10	18	44.4	3601	10	17	41.2	3600	10	14	28.6	3600	10	14	28.6	3600
full-10-15-3	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	16	37.5	3606	10	18	44.4	3600	10	18	44.4	3601	10	20	50.0	3600	10	14	28.6	3600	10	19	47.4	3600
full-10-15-4	45	10	10	0.0	2	10	10	0.0	0	10	10	0.0	3	10	46	78.3	0	10	18	44.4	3600	10	18	44.4	3601	10	27	63.0	3600	10	14	28.6	3600	10	19	47.4	3600
full-10-15-5	45	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	16	37.5	3608	10	18	44.4	3600	10	13	23.1	3600	10	21	52.4	3600	10	19	47.4	3600	10	14	28.6	3600
full-11-30-1	55	11	11	0.0	2	11	11	0.0	2	11	11	0.0	2	11	29	62.1	3624	11	33	66.7	3600	11	56	80.4	3611	-	-	-	-	-	-	-	-	-	-	-	-
full-11-30-2	55	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	26	57.7	3618	11	21	47.6	3600	11	21	47.6	3602	-	-	-	-	-	-	-	-	-	-	-	-
full-11-30-3	55	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	30	63.3	3617	11	21	47.6	3600	11	21	47.6	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-11-30-4	55	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	44	75.0	3630	11	33	66.7	3600	11	33	66.7	3602	-	-	-	-	-	-	-	-	-	-	-	-
full-11-30-5	55	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	39	71.8	3628	11	33	66.7	3600	11	33	66.7	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-11-15-1	55	11	12	8.3	3619	11	11	0.0	1	11	11	0.0	1	11	56	80.4	1	11	21	47.6	3600	11	21	47.6	3600	-	-	-	-	-	-	-	-	-	-	-	-
full-11-15-2	55	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	19	42.1	3627	11	33	66.7	3600	11	21	47.6	3600	-	-	-	-	-	-	-	-	-	-	-	-
full-11-15-3	55	11	11	0.0	0	11	11	0.0	0	11	11	0.0	0	11	56	80.4	1979	11	21	47.6	3600	11	21	47.6	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-11-15-4	55	11	11	0.0	0	11	11	0.0	0	11	11	0.0	0	11	56	80.4	0	11	21	47.6	3600	11	15	26.7	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-11-15-5	55	11	11	0.0	0	11	11	0.0	0	11	11	0.0	1	11	16	31.2	3624	11	21	47.6	3600	11	21	47.6	3602	-	-	-	-	-	-	-	-	-	-	-	-
full-12-30-1	66	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	67	82.1	33	12	25	52.0	3600	12	25	52.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-12-30-2	66	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	58	79.3	3657	12	39	69.2	3600	12	25	52.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
full-12-30-3	66	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	38	68.4	3652	12	25	52.0	3600	12	25	52.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-12-30-4	66	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	60	80.0	3674	12	39	69.2	3601	12	25	52.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-12-30-5	66	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	43	72.1	3663	12	25	52.0	3600	12	25	52.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
full-12-15-1	66	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	33	63.6	3665	12	25	52.0	3600	12	25	52.0	3602	-	-	-	-	-	-	-	-	-	-	-	-
full-12-15-2	66	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	18	33.3	3647	12	25	52.0	3600	12	18	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
full-12-15-3	66	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	67	82.1	1	12	25	52.0	3600	12	25	52.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
full-12-15-4	66	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	67	82.1	0	12	25	52.0	3600	12	25	52.0	3601	-	-	-	-	-</							

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time				
full-13-15-2	78	13	13	0.0	1	13	13	0.0	1	13	13	0.0	1	13	79	83.5	99	13	29	55.2	3600	13	29	55.2	3601	-	-	-	-	-	-	-	-				
full-13-15-3	78	13	14	7.1	3601	13	13	0.0	1	13	13	0.0	1	13	79	83.5	0	13	29	55.2	3600	13	29	55.2	3601	-	-	-	-	-	-	-	-				
full-13-15-4	78	13	13	0.0	1	13	13	0.0	1	13	13	0.0	1	13	79	83.5	0	13	29	55.2	3600	13	29	55.2	3600	-	-	-	-	-	-	-	-				
full-13-15-5	78	13	13	0.0	1	13	13	0.0	1	13	13	0.0	1	13	78	83.3	3736	13	46	71.7	3601	13	29	55.2	3605	-	-	-	-	-	-	-	-				
full-14-30-1	91	14	14	0.0	5	14	14	0.0	5	14	14	0.0	5	14	92	84.8	23	14	92	84.8	3616	14	33	57.6	3601	-	-	-	-	-	-	-	-				
full-14-30-2	91	14	15	6.7	3600	14	15	6.7	3599	14	14	0.0	5	14	92	84.8	50	14	92	84.8	3616	14	92	84.8	3614	-	-	-	-	-	-	-	-				
full-14-30-3	91	14	14	0.0	5	14	14	0.0	5	14	14	0.0	5	14	92	84.8	56	14	33	57.6	3600	14	53	73.6	3602	-	-	-	-	-	-	-	-				
full-14-30-4	91	14	14	0.0	3	14	14	0.0	3	14	14	0.0	3	14	92	84.8	97	14	53	73.6	3603	14	53	73.6	3603	-	-	-	-	-	-	-	-				
full-14-30-5	91	14	14	0.0	4	14	14	0.0	4	14	14	0.0	4	14	92	84.8	110	14	92	84.8	3620	14	33	57.6	3601	-	-	-	-	-	-	-	-				
full-14-15-1	91	14	14	0.0	2	14	16	12.5	3600	14	14	0.0	3	14	92	84.8	236	14	33	57.6	3601	14	23	39.1	3600	-	-	-	-	-	-	-	-				
full-14-15-2	91	14	14	0.0	2	14	15	6.7	3600	14	15	6.7	3609	14	92	84.8	62	14	23	39.1	3600	14	33	57.6	3602	-	-	-	-	-	-	-	-				
full-14-15-3	91	14	15	6.7	3603	14	14	0.0	2	14	15	6.7	3600	14	92	84.8	117	14	33	57.6	3601	14	33	57.6	3601	-	-	-	-	-	-	-	-				
full-14-15-4	91	14	15	6.7	3606	14	15	6.7	3600	14	15	6.7	3606	14	92	84.8	228	14	33	57.6	3601	14	33	57.6	3601	-	-	-	-	-	-	-	-				
full-14-15-5	91	14	14	0.0	2	14	14	0.0	2	14	14	0.0	2	14	92	84.8	105	14	33	57.6	3600	14	23	39.1	3602	-	-	-	-	-	-	-	-				
full-15-30-1	105	15	15	0.0	8	15	15	0.0	8	15	15	0.0	8	15	106	85.8	13	15	106	85.8	3632	15	37	59.5	3601	-	-	-	-	-	-	-	-				
full-15-30-2	105	15	15	0.0	7	15	15	0.0	7	15	15	0.0	7	15	106	85.8	30	15	37	59.5	3601	15	37	59.5	3601	-	-	-	-	-	-	-	-				
full-15-30-3	105	15	15	0.0	7	15	15	0.0	8	15	15	0.0	8	15	106	85.8	46	15	106	85.8	3628	15	37	59.5	3601	-	-	-	-	-	-	-	-				
full-15-30-4	105	15	15	0.0	6	15	15	0.0	6	15	15	0.0	6	15	106	85.8	62	15	60	75.0	3605	15	60	75.0	3606	-	-	-	-	-	-	-	-				
full-15-30-5	105	15	15	0.0	8	15	15	0.0	8	15	15	0.0	8	15	106	85.8	58	15	60	75.0	3605	15	60	75.0	3605	-	-	-	-	-	-	-	-				
full-15-15-1	105	15	16	6.2	3600	15	16	6.2	3600	15	15	0.0	3	15	106	85.8	122	15	37	59.5	3601	15	25	40.0	3600	-	-	-	-	-	-	-	-				
full-15-15-2	105	15	15	0.0	4	15	15	0.0	4	15	15	0.0	4	15	106	85.8	49	15	37	59.5	3600	15	37	59.5	3603	-	-	-	-	-	-	-	-				
full-15-15-3	105	15	15	0.0	4	15	15	0.0	4	15	16	6.2	3607	15	106	85.8	82	15	37	59.5	3601	15	37	59.5	3601	-	-	-	-	-	-	-	-				
full-15-15-4	105	15	15	0.0	3	15	15	0.0	3	15	15	0.0	3	15	106	85.8	125	15	37	59.5	3600	15	25	40.0	3600	-	-	-	-	-	-	-	-				
full-15-15-5	105	15	16	6.2	3600	15	15	0.0	4	15	15	0.0	4	15	106	85.8	128	15	37	59.5	3601	15	37	59.5	3543	-	-	-	-	-	-	-	-				
full-16-30-1	120	16	17	5.9	3602	16	18	11.1	3600	16	17	5.9	3604	16	121	86.8	7	16	41	61.0	3601	16	41	61.0	3602	-	-	-	-	-	-	-	-				
full-16-30-2	120	16	16	0.0	11	16	16	0.0	11	16	16	0.0	11	16	121	86.8	19	16	68	76.5	3609	16	41	61.0	3602	-	-	-	-	-	-	-	-				
full-16-30-3	120	16	16	0.0	13	16	16	0.0	13	16	16	0.0	13	16	121	86.8	21	16	68	76.5	3607	16	68	76.5	3611	-	-	-	-	-	-	-	-				
full-16-30-4	120	16	16	0.0	9	16	16	0.0	9	16	16	0.0	9	16	121	86.8	24	16	68	76.5	3610	16	41	61.0	3603	-	-	-	-	-	-	-	-				
full-16-30-5	120	16	17	5.9	3601	16	16	0.0	11	16	16	0.0	11	16	121	86.8	32	16	68	76.5	3610	16	41	61.0	3603	-	-	-	-	-	-	-	-				
full-16-15-1	120	16	19	15.8	3600	16	19	15.8	3600	16	20	20.0	3604	16	121	86.8	83	16	41	61.0	3601	16	41	61.0	3601	-	-	-	-	-	-	-	-				
full-16-15-2	120	16	17	5.9	3600	16	17	5.9	3600	16	18	11.1	3600	16	121	86.8	33	16	41	61.0	3601	16	41	61.0	3602	-	-	-	-	-	-	-	-				
full-16-15-3	120	16	16	0.0	6	16	17	5.9	3601	16	17	5.9	3604	16	121	86.8	43	16	68	76.5	3606	16	28	42.9	3600	-	-	-	-	-	-	-	-				
full-16-15-4	120	16	16	0.0	6	16	17	5.9	3600	16	16	0.0	6	16	121	86.8	71	16	41	61.0	3601	16	41	61.0	3601	-	-	-	-	-	-	-	-				
full-16-15-5	120	16	17	5.9	3600	16	16	0.0	5	16	17	5.9	3600	16	121	86.8	94	16	28	42.9	3601	16	41	61.0	3601	-	-	-	-	-	-	-	-				

Table 6: Results of 9 methods on miss instances

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time
miss-5-15-2-4	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-15-2-5	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-30-2-1	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-30-2-2	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-30-2-3	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-30-2-4	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-30-2-5	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-15-2-1	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-15-2-2	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-5-15-2-3	8	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0
miss-6-15-2-4	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	2	6	6	0.0	1	6	6	0.0	2	6	6	0.0	9	6	6	0.0	2	6	6	0.0	3
miss-6-15-2-5	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	2	6	6	0.0	1	6	6	0.0	5	6	6	0.0	2	6	6	0.0	2	6	6	0.0	11
miss-6-30-2-1	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	2	6	6	0.0	3	6	6	0.0	9	6	6	0.0	5	6	6	0.0	4	6	6	0.0	39
miss-6-30-2-2	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	14	6	6	0.0	3	6	6	0.0	5	6	6	0.0	23	6	6	0.0	15	6	6	0.0	7
miss-6-30-2-3	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	11	6	6	0.0	1	6	6	0.0	4	6	6	0.0	9	6	6	0.0	2	6	6	0.0	2
miss-6-30-2-4	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	8	6	6	0.0	3	6	6	0.0	2	6	6	0.0	1	6	6	0.0	2	6	6	0.0	4
miss-6-30-2-5	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	3	6	6	0.0	1	6	6	0.0	10	6	6	0.0	29	6	6	0.0	8	6	6	0.0	20
miss-6-15-2-1	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	8	6	6	0.0	2	6	6	0.0	16	6	6	0.0	24	6	6	0.0	8	6	6	0.0	14
miss-6-15-2-2	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	1	6	6	0.0	0	6	6	0.0	1	6	6	0.0	1	6	6	0.0	1	6	6	0.0	2
miss-6-15-2-3	13	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	1	6	6	0.0	1	6	6	0.0	1	6	6	0.0	2	6	6	0.0	3	6	6	0.0	7
miss-7-15-3-4	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	63	7	7	0.0	7	7	7	0.0	32	7	7	0.0	188	7	7	0.0	82	7	7	0.0	30
miss-7-15-3-5	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	19	63.2	1	7	7	0.0	21	7	7	0.0	8	7	7	0.0	72	7	7	0.0	100	7	7	0.0	218
miss-7-30-3-1	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	108	7	7	0.0	8	7	7	0.0	191	7	7	0.0	722	7	7	0.0	242	7	7	0.0	706
miss-7-30-3-2	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	1506	7	7	0.0	165	7	7	0.0	2639	7	7	0.0	3451	7	10	30.0	3600	7	10	30.0	3600
miss-7-30-3-3	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	1284	7	7	0.0	36	7	7	0.0	1079	7	7	0.0	244	7	7	0.0	1455	7	7	0.0	1183
miss-7-30-3-4	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	3384	7	7	0.0	49	7	7	0.0	350	7	10	30.0	3600	7	7	0.0	1207	7	7	0.0	167
miss-7-30-3-5	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	126	7	7	0.0	41	7	7	0.0	1112	7	7	0.0	245	7	7	0.0	680	7	7	0.0	2600
miss-7-15-3-1	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	28	7	7	0.0	4	7	7	0.0	21	7	7	0.0	55	7	7	0.0	14	7	7	0.0	47
miss-7-15-3-2	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	69	7	7	0.0	6	7	7	0.0	60	7	7	0.0	297	7	7	0.0	14	7	7	0.0	119
miss-7-15-3-3	18	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	19	63.2	0	7	7	0.0	5	7	7	0.0	47	7	7	0.0	214	7	7	0.0	36	7	7	0.0	869
miss-8-15-3-4	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	10	20.0	3600	8	8	0.0	297	8	8	0.0	3269	8	10	20.0	3600	8	9	11.1	3600	8	8	0.0	3541
miss-8-15-3-5	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	26	69.2	0	8	8	0.0	2041	8	12	33.3	3600	8	11	27.3	3600	8	8	0.0	1640	8	8	0.0	2126
miss-8-30-3-1	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	13	38.5	3601	8	12	33.3	3600	8	12	33.3	3601	8	13	38.5	3600	8	12	33.3	3600	8	12	33.3	3600
miss-8-30-3-2	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	13	38.5	3600	8	12	33.3	3600	8	12	33.3	3600	8	16	50.0	3600	8	12	33.3	3600	8	17	52.9	3600
miss-8-30-3-3	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	10	20.0	3601	8	12	33.3	3600	8	9	11.1	3600	8	13	38.5	3600	8	12	33.3	3600	8	17	52.9	3600
miss-8-30-3-4	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	10	20.0	3601	8	12	33.3	3600	8	8	0.0	2265	8	8	0.0	509	8	10	20.0	3600	8	12	33.3	3600
miss-8-30-3-5	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	13	38.5	3601	8	17	52.9	3600	8	12	33.3	3600	8	15	46.7	3600	8	12	33.3	3600	8	12	33.3	3600
miss-8-15-3-1	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	26	69.2	298	8	8	0.0	3026	8	12	33.3	3600	8	13	38.5	3600	8	12	33.3	3600	8	8	0.0	2011
miss-8-15-3-2	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	9	11.1	3600	8	12	33.3	3600	8	9	11.1	3600	8	14	42.9	3600	8	17	52.9	3600	8	17	52.9	3600
miss-8-15-3-3	25	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	26	69.2	0	8	8	0.0	810	8	8	0.0	2000	8	8	0.0	870	8	10	20.0	3600	8	10	20.0	3600
miss-9-15-4-4	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	11	18.2	3601	9	14	35.7	3600	9	17	47.1	3600	9	15	40.0	3600	9	12	25.0	3600				
miss-9-15-4-5	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	33	72.7	0	9	9	0.0	296	9	9	0.0	3274	9	12	25.0	3600	9	12	25.0	3600	9	10	10.0	3600

Continue on next page

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time				
miss-9-30-4-1	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	15	40.0	3601	9	21	57.1	3600	9	14	35.7	3601	9	26	65.4	3600	9	21	57.1	3600	9	15	40.0	3600
miss-9-30-4-2	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	33	72.7	3127	9	14	35.7	3600	9	14	35.7	3601	9	25	64.0	3600	9	15	40.0	3600	9	15	40.0	3600
miss-9-30-4-3	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	13	30.8	3602	9	14	35.7	3600	9	14	35.7	3600	9	24	62.5	3600	9	21	57.1	3600	9	15	40.0	3600
miss-9-30-4-4	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	12	25.0	3601	9	14	35.7	3600	9	14	35.7	3601	9	28	67.9	3600	9	15	40.0	3600	9	15	40.0	3600
miss-9-30-4-5	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	20	55.0	3601	9	14	35.7	3600	9	14	35.7	3601	9	26	65.4	3600	9	15	40.0	3600	9	33	72.7	3600
miss-9-15-4-1	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	33	72.7	0	9	21	57.1	3600	9	14	35.7	3611	9	11	18.2	3600	9	15	40.0	3600	9	15	40.0	3600
miss-9-15-4-2	32	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	9	10	10.0	3601	9	21	57.1	3600	9	11	18.2	3600	9	20	55.0	3600	9	12	25.0	3600	9	12	25.0	3600
miss-9-15-4-3	32	9	9	0.0	1	9	9	0.0	1	9	9	0.0	1	9	12	25.0	3601	9	9	0.0	1082	9	14	35.7	3601	9	17	47.1	3600	9	15	40.0	3600	9	15	40.0	3600
miss-10-15-4-4	41	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	15	33.3	3603	10	17	41.2	3600	10	13	23.1	3601	10	17	41.2	3600	10	14	28.6	3600	10	18	44.4	3600
miss-10-15-4-5	41	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	16	37.5	3604	10	17	41.2	3600	10	13	23.1	3600	10	15	33.3	3600	10	18	44.4	3600	10	10	0.0	452
miss-10-30-4-1	41	10	10	0.0	0	10	10	0.0	0	10	10	0.0	1	10	23	56.5	3604	10	26	61.5	3600	10	17	41.2	3600	10	32	68.8	3600	10	26	61.5	3600	10	26	61.5	3600
miss-10-30-4-2	41	10	10	0.0	1	10	10	0.0	0	10	10	0.0	1	10	24	58.3	3605	10	17	41.2	3600	10	17	41.2	3600	10	42	76.2	3600	10	18	44.4	3600	10	18	44.4	3600
miss-10-30-4-3	41	10	10	0.0	1	10	10	0.0	1	10	10	0.0	1	10	42	76.2	0	10	26	61.5	3600	10	17	41.2	3600	10	42	76.2	3600	10	26	61.5	3600	10	18	44.4	3600
miss-10-30-4-4	41	10	10	0.0	1	10	10	0.0	1	10	10	0.0	1	10	42	76.2	0	10	26	61.5	3600	10	17	41.2	3600	10	40	75.0	3600	10	26	61.5	3600	10	42	76.2	3600
miss-10-30-4-5	41	10	10	0.0	1	10	10	0.0	1	10	10	0.0	1	10	17	41.2	3605	10	26	61.5	3600	10	26	61.5	3602	10	42	76.2	3600	10	18	44.4	3600	10	26	61.5	3600
miss-10-15-4-1	41	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	21	52.4	3605	10	17	41.2	3600	10	17	41.2	3600	10	28	64.3	3600	10	18	44.4	3600	10	26	61.5	3600
miss-10-15-4-2	41	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	42	76.2	0	10	26	61.5	3600	10	17	41.2	3601	10	23	56.5	3600	10	18	44.4	3600	10	26	61.5	3600
miss-10-15-4-3	41	10	10	0.0	0	10	10	0.0	0	10	10	0.0	0	10	13	23.1	3604	10	17	41.2	3600	10	17	41.2	3600	10	17	41.2	3600	10	18	44.4	3600	10	26	61.5	3600
miss-11-15-5-4	50	11	11	0.0	0	11	11	0.0	0	11	11	0.0	0	11	19	42.1	3611	11	20	45.0	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-15-5-5	50	11	11	0.0	0	11	11	0.0	0	11	11	0.0	0	11	51	78.4	8	11	20	45.0	3600	11	20	45.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-30-5-1	50	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	19	42.1	3613	11	31	64.5	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-30-5-2	50	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	28	60.7	3618	11	31	64.5	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-30-5-3	50	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	51	78.4	0	11	31	64.5	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-30-5-4	50	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	51	78.4	0	11	20	45.0	3600	11	20	45.0	3602	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-30-5-5	50	11	11	0.0	1	11	11	0.0	1	11	11	0.0	1	11	24	54.2	3613	11	20	45.0	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-15-5-1	50	11	11	0.0	0	11	11	0.0	0	11	11	0.0	0	11	24	54.2	3611	11	20	45.0	3600	11	15	26.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-15-5-2	50	11	11	0.0	0	11	12	8.3	3601	11	12	8.3	3600	11	51	78.4	0	11	15	26.7	3600	11	15	26.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-11-15-5-3	50	11	11	0.0	0	11	11	0.0	0	11	11	0.0	0	11	22	50.0	3615	11	20	45.0	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-15-5-4	61	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	22	45.5	3635	12	37	67.6	3600	12	24	50.0	3296	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-15-5-5	61	12	12	0.0	0	12	12	0.0	0	12	12	0.0	0	12	62	80.6	5	12	17	29.4	3600	12	24	50.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-30-5-1	61	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	38	68.4	3639	12	37	67.6	3600	12	24	50.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-30-5-2	61	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	35	65.7	3653	12	24	50.0	3600	12	37	67.6	3601	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-30-5-3	61	12	12	0.0	2	12	12	0.0	2	12	12	0.0	2	12	34	64.7	3644	12	37	67.6	3600	12	24	50.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-30-5-4	61	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	62	80.6	0	12	24	50.0	3600	12	24	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-30-5-5	61	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	46	73.9	3637	12	24	50.0	3600	12	24	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-15-5-1	61	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	62	80.6	13	12	37	67.6	3600	12	24	50.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
miss-12-15-5-2	61	12	12	0.0	1	12	12	0.0	1	12	12	0.0	1	12	62																						

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time				
miss-13-30-6-5	72	13	13	0.0	2	13	13	0.0	2	13	13	0.0	2	13	73	82.2	70	13	43	69.8	3601	13	27	51.9	3600	-	-	-	-	-	-	-	-	-	-		
miss-13-15-6-1	72	13	13	0.0	1	13	13	0.0	1	13	13	0.0	1	13	73	82.2	10	13	43	69.8	3600	13	27	51.9	3600	-	-	-	-	-	-	-	-	-	-		
miss-13-15-6-2	72	13	13	0.0	1	13	14	7.1	3600	13	13	0.0	1	13	28	53.6	3687	13	27	51.9	3600	13	27	51.9	3601	-	-	-	-	-	-	-	-	-	-		
miss-13-15-6-3	72	13	13	0.0	1	13	13	0.0	1	13	13	0.0	1	13	73	82.2	9	13	27	51.9	3600	13	19	31.6	3600	-	-	-	-	-	-	-	-	-	-		
miss-14-15-6-4	85	14	14	0.0	2	14	14	0.0	2	14	14	0.0	2	14	86	83.7	32	14	31	54.8	3600	14	31	54.8	3600	-	-	-	-	-	-	-	-	-	-		
miss-14-15-6-5	85	14	14	0.0	2	14	14	0.0	2	14	14	0.0	2	14	86	83.7	2	14	31	54.8	3600	14	31	54.8	3602	-	-	-	-	-	-	-	-	-	-		
miss-14-30-6-1	85	14	14	0.0	3	14	14	0.0	3	14	14	0.0	3	14	86	83.7	70	14	31	54.8	3600	14	31	54.8	3601	-	-	-	-	-	-	-	-	-	-		
miss-14-30-6-2	85	14	14	0.0	4	14	14	0.0	4	14	14	0.0	4	14	86	83.7	50	14	50	72.0	3601	14	31	54.8	3600	-	-	-	-	-	-	-	-	-	-		
miss-14-30-6-3	85	14	14	0.0	3	14	14	0.0	3	14	14	0.0	3	14	86	83.7	192	14	31	54.8	3600	14	31	54.8	3600	-	-	-	-	-	-	-	-	-	-		
miss-14-30-6-4	85	14	15	6.7	3602	14	14	0.0	4	14	14	0.0	4	14	86	83.7	0	14	50	72.0	3601	14	31	54.8	3602	-	-	-	-	-	-	-	-	-	-		
miss-14-30-6-5	85	14	14	0.0	5	14	14	0.0	5	14	14	0.0	5	14	86	83.7	46	14	50	72.0	3602	14	31	54.8	3601	-	-	-	-	-	-	-	-	-	-		
miss-14-15-6-1	85	14	15	6.7	3604	14	15	6.7	3599	14	15	6.7	3607	14	86	83.7	6	14	50	72.0	3601	14	31	54.8	3600	-	-	-	-	-	-	-	-	-	-		
miss-14-15-6-2	85	14	15	6.7	3600	14	15	6.7	3600	14	14	0.0	2	14	86	83.7	167	14	31	54.8	3600	14	22	36.4	3600	-	-	-	-	-	-	-	-	-	-		
miss-14-15-6-3	85	14	14	0.0	2	14	14	0.0	2	14	14	0.0	2	14	86	83.7	6	14	31	54.8	3600	14	31	54.8	3600	-	-	-	-	-	-	-	-	-	-		
miss-15-15-7-4	98	15	15	0.0	4	15	17	11.8	3601	15	15	0.0	4	15	99	84.8	25	15	35	57.1	3601	15	35	57.1	3601	-	-	-	-	-	-	-	-	-	-		
miss-15-15-7-5	98	15	15	0.0	3	15	15	0.0	3	15	15	0.0	3	15	99	84.8	1	15	57	73.7	3603	15	35	57.1	3602	-	-	-	-	-	-	-	-	-	-		
miss-15-30-7-1	98	15	15	0.0	5	15	15	0.0	5	15	15	0.0	5	15	99	84.8	63	15	99	84.8	3618	15	35	57.1	3601	-	-	-	-	-	-	-	-	-	-		
miss-15-30-7-2	98	15	15	0.0	6	15	15	0.0	6	15	15	0.0	6	15	99	84.8	39	15	57	73.7	3605	15	35	57.1	3602	-	-	-	-	-	-	-	-	-	-		
miss-15-30-7-3	98	15	15	0.0	7	15	16	6.2	3600	15	15	0.0	7	15	99	84.8	150	15	35	57.1	3601	15	35	57.1	3601	-	-	-	-	-	-	-	-	-	-		
miss-15-30-7-4	98	15	15	0.0	5	15	15	0.0	5	15	16	6.2	3601	15	99	84.8	255	15	35	57.1	3601	15	57	73.7	3606	-	-	-	-	-	-	-	-	-	-		
miss-15-30-7-5	98	15	15	0.0	6	15	15	0.0	6	15	16	6.2	3604	15	99	84.8	28	15	57	73.7	3603	15	35	57.1	3601	-	-	-	-	-	-	-	-	-	-		
miss-15-15-7-1	98	15	17	11.8	3600	15	16	6.2	3600	15	18	16.7	3600	15	99	84.8	4	15	35	57.1	3600	15	35	57.1	3601	-	-	-	-	-	-	-	-	-	-		
miss-15-15-7-2	98	15	16	6.2	3600	15	15	0.0	4	15	15	0.0	4	15	99	84.8	104	15	35	57.1	3601	15	35	57.1	3364	-	-	-	-	-	-	-	-	-	-		
miss-15-15-7-3	98	15	15	0.0	2	15	15	0.0	2	15	18	16.7	3600	15	99	84.8	3	15	35	57.1	3601	15	24	37.5	3601	-	-	-	-	-	-	-	-	-	-		
miss-16-15-7-4	113	16	16	0.0	4	16	16	0.0	4	16	17	5.9	3600	16	114	86.0	23	16	40	60.0	3601	16	40	60.0	3601	-	-	-	-	-	-	-	-	-	-		
miss-16-15-7-5	113	16	18	11.1	3600	16	19	15.8	3601	16	19	15.8	3601	16	114	86.0	0	16	27	40.7	3600	16	27	40.7	3600	-	-	-	-	-	-	-	-	-	-		
miss-16-30-7-1	113	16	16	0.0	6	16	17	5.9	3600	16	17	5.9	3602	16	114	86.0	46	16	65	75.4	3605	16	40	60.0	3601	-	-	-	-	-	-	-	-	-	-		
miss-16-30-7-2	113	16	16	0.0	9	16	16	0.0	9	16	16	0.0	9	16	114	86.0	22	16	65	75.4	3609	16	65	75.4	3610	-	-	-	-	-	-	-	-	-	-		
miss-16-30-7-3	113	16	16	0.0	9	16	16	0.0	9	16	16	0.0	9	16	114	86.0	122	16	40	60.0	3602	16	65	75.4	3615	-	-	-	-	-	-	-	-	-	-		
miss-16-30-7-4	113	16	16	0.0	8	16	16	0.0	8	16	16	0.0	8	16	114	86.0	131	16	65	75.4	3606	16	65	75.4	3608	-	-	-	-	-	-	-	-	-	-		
miss-16-30-7-5	113	16	16	0.0	10	16	16	0.0	10	16	16	0.0	10	16	114	86.0	18	16	65	75.4	3606	16	40	60.0	3612	-	-	-	-	-	-	-	-	-	-		
miss-16-15-7-1	113	16	16	0.0	5	16	16	0.0	5	16	18	11.1	3604	16	114	86.0	2	16	40	60.0	3601	16	27	40.7	3600	-	-	-	-	-	-	-	-	-	-		
miss-16-15-7-2	113	16	17	5.9	3600	16	17	5.9	3600	16	17	5.9	3600	16	114	86.0	78	16	40	60.0	3601	16	40	60.0	3601	-	-	-	-	-	-	-	-	-	-		
miss-16-15-7-3	113	16	16	0.0	5	16	16	0.0	5	16	17	5.9	3603	16	114	86.0	1	16	40	60.0	3601	16	40	60.0	3601	-	-	-	-	-	-	-	-	-	-		
miss-17-15-8-4	128	17	19	10.5	3602	17	19	10.5	3600	17	18	5.6	3604	17	129	86.8	10	17	44	61.4	3601	17	44	61.4	3604	-	-	-	-	-	-	-	-	-	-		
miss-17-15-8-5	128	17	17	0.0	6	17	17	0.0	6	17	19	10.5	3603	17	129	86.8	0	17	44	61.4	3601	17	30	43.3	3601	-	-	-	-	-	-	-	-	-	-		
miss-17-30-8-1	128	17	17	0.0	12	17	17	0.0	12	17	17	0.0	12	17	129	86.8	29	17	129	86.8	3663	17	129	86.8	3676	-	-	-	-	-	-	-	-	-	-		
miss-17-30-8-2	128	17	17	0.0	14	17	17	0.0	13	17	18	5.6	3600	17	129	86.8	14	17	73	76.7	3612	17	73	76.7	3612	-	-	-	-	-	-	-	-	-	-		
miss-17-30-8-3	128	17	17	0.0	13	17	17	0.0	13	17	17	0.0	13	17	129	86.8	68	17	129	86.8	3704	17	44	61.4	3609	-	-	-	-	-	-	-	-	-	-		
miss-17-30-8-4	128	17	18	5.6	3601	17	17	0.0	12	17	18	5.6	3600	17	129	86.8	64	17																			

Table 7: Results of 9 methods on join instances

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time				
joint-5-30-5-30-1	20	9	9	0.0	3599	7	9	22.2	3599	7	9	22.2	3600	7	9	22.2	3600	7	10	30.0	3600	7	10	30.0	3600	7	13	46.2	3600	7	10	30.0	3600	7	10	30.0	3600
joint-5-30-5-30-2	20	9	9	0.0	3599	9	9	0.0	2668	7	9	22.2	3599	7	21	66.7	1429	7	10	30.0	3600	7	9	22.2	3600	7	10	30.0	3600	7	10	30.0	3600	7	10	30.0	3600
joint-5-30-5-30-3	20	9	9	0.0	2729	9	9	0.0	1080	8	9	11.1	3599	7	9	22.2	3600	9	9	0.0	2368	7	10	30.0	3600	7	12	41.7	3600	7	14	50.0	3600	7	10	30.0	3600
joint-5-15-5-15-1	20	8	8	0.0	58	8	8	0.0	34	8	8	0.0	283	8	8	0.0	3335	8	8	0.0	181	8	8	0.0	331	8	8	0.0	1189	8	8	0.0	33	8	8	0.0	443
joint-5-15-5-15-2	20	8	8	0.0	25	8	8	0.0	16	8	8	0.0	252	8	8	0.0	75	8	8	0.0	240	7	8	12.5	3600	8	8	0.0	2362	8	8	0.0	64	8	8	0.0	342
joint-5-15-5-15-3	20	8	8	0.0	10	8	8	0.0	11	8	8	0.0	61	8	8	0.0	38	8	8	0.0	13	8	8	0.0	84	8	8	0.0	155	8	8	0.0	31	8	8	0.0	339
joint-6-15-5-15-1	25	9	9	0.0	1613	9	9	0.0	504	8	9	11.1	3600	8	9	11.1	3600	9	9	0.0	3354	8	9	11.1	3601	8	10	20.0	3600	9	9	0.0	2817	8	12	33.3	3600
joint-6-15-5-15-2	25	9	9	0.0	1145	9	9	0.0	275	9	9	0.0	654	8	9	11.1	3600	9	9	0.0	220	9	9	0.0	1112	8	9	11.1	3600	8	10	20.0	3600	9	9	0.0	642
joint-6-15-5-15-3	25	9	9	0.0	754	9	9	0.0	1618	8	9	11.1	3600	8	9	11.1	3600	9	9	0.0	2090	8	9	11.1	3601	8	12	33.3	3600	8	10	20.0	3600	8	9	11.1	3600
joint-6-30-5-30-1	25	8	10	20.0	3599	8	10	20.0	3599	8	10	20.0	3601	8	12	33.3	3603	8	17	52.9	3600	8	12	33.3	3600	8	15	46.7	3600	8	12	33.3	3600	8	17	52.9	3600
joint-6-30-5-30-2	25	9	9	0.0	0	9	9	0.0	0	9	9	0.0	0	8	10	20.0	3600	8	17	52.9	3600	8	12	33.3	3601	8	14	42.9	3600	8	12	33.3	3600	8	10	20.0	3600
joint-6-30-5-30-3	25	9	9	0.0	952	9	9	0.0	703	8	9	11.1	3600	8	10	20.0	3600	9	9	0.0	903	8	9	11.1	3600	8	12	33.3	3600	8	10	20.0	3600	8	12	33.3	3600
joint-7-15-5-15-2	31	10	10	0.0	3599	9	10	10.0	3599	9	10	10.0	3601	9	13	30.8	3602	9	11	18.2	3600	9	11	18.2	3600	9	11	18.2	3600	9	11	18.2	3600	9	14	35.7	3600
joint-7-15-5-15-3	31	10	10	0.0	3599	10	10	0.0	2077	9	10	10.0	3600	9	10	10.0	3601	10	11	9.1	3600	9	14	35.7	3600	9	12	25.0	3600	9	11	18.2	3600	9	11	18.2	3600
joint-7-30-5-30-1	31	9	10	10.0	3600	9	10	10.0	3599	9	10	10.0	3602	9	13	30.8	3601	9	14	35.7	3600	9	14	35.7	3600	9	16	43.8	3600	9	14	35.7	3600	9	11	18.2	3600
joint-7-30-5-30-2	31	9	11	18.2	3600	9	11	18.2	3599	9	11	18.2	3601	9	13	30.8	3601	9	14	35.7	3600	9	14	35.7	3601	9	17	47.1	3600	9	14	35.7	3600	9	20	55.0	3600
joint-7-30-5-30-3	31	9	11	18.2	3599	9	11	18.2	3599	9	11	18.2	3602	9	32	71.9	2511	9	14	35.7	3600	9	14	35.7	3601	9	25	64.0	3600	9	20	55.0	3600	9	14	35.7	3600
joint-7-15-5-15-1	31	9	10	10.0	3600	9	10	10.0	3599	9	10	10.0	3600	9	12	25.0	3601	9	14	35.7	3600	9	11	18.2	3601	9	13	30.8	3600	9	14	35.7	3600	9	11	18.2	3600
joint-8-30-5-30-3	38	10	11	9.1	3600	10	11	9.1	3599	10	11	9.1	3600	10	18	44.4	3603	10	16	37.5	3600	10	16	37.5	3600	10	22	54.5	3600	10	17	41.2	3600	10	17	41.2	3600
joint-8-15-5-15-1	38	10	11	9.1	3599	10	11	9.1	3599	10	11	9.1	3600	10	39	74.4	2384	10	12	16.7	3600	10	12	16.7	3601	10	15	33.3	3600	10	13	23.1	3600	10	13	23.1	3600
joint-8-15-5-15-2	38	10	12	16.7	3625	10	12	16.7	3599	10	12	16.7	3600	10	13	23.1	3603	10	16	37.5	3600	10	16	37.5	3601	10	18	44.4	3600	10	24	58.3	3600	10	17	41.2	3600
joint-8-15-5-15-3	38	10	11	9.1	3601	10	11	9.1	3599	10	11	9.1	3601	10	14	28.6	3602	10	16	37.5	3600	10	12	16.7	3600	10	14	28.6	3600	10	13	23.1	3600	10	13	23.1	3600
joint-8-30-5-30-1	38	10	11	9.1	3600	10	11	9.1	3600	10	11	9.1	3600	10	39	74.4	1558	10	24	58.3	3600	10	16	37.5	3606	10	32	68.8	3600	10	17	41.2	3600	10	17	41.2	3600
joint-8-30-5-30-2	38	10	11	9.1	3600	10	11	9.1	3600	10	11	9.1	3603	10	22	54.5	3604	10	24	58.3	3600	10	24	58.3	3600	10	23	56.5	3600	10	39	74.4	3600	10	39	74.4	3600
joint-9-15-5-15-1	46	11	12	8.3	3621	11	12	8.3	3599	11	12	8.3	3600	11	15	26.7	3607	11	14	21.4	3600	11	14	21.4	3600	11	26	57.7	3600	11	15	26.7	3600	11	20	45.0	3600
joint-9-15-5-15-2	46	11	12	8.3	3600	11	12	8.3	3600	11	12	8.3	3660	11	19	42.1	3607	11	29	62.1	3600	11	14	21.4	3600	11	27	59.3	3600	11	20	45.0	3600	11	20	45.0	3600
joint-9-15-5-15-3	46	11	12	8.3	3600	11	12	8.3	3599	11	12	8.3	3615	11	17	35.3	3610	11	14	21.4	3600	11	19	42.1	3600	11	24	54.2	3600	11	15	26.7	3600	11	15	26.7	3600
joint-9-30-5-30-1	46	11	12	8.3	3600	11	12	8.3	3600	11	12	8.3	3602	11	47	76.6	0	11	19	42.1	3600	11	19	42.1	3600	11	47	76.6	3600	11	20	45.0	3600	11	20	45.0	3600
joint-9-30-5-30-2	46	11	13	15.4	3600	11	13	15.4	3601	11	13	15.4	3601	11	47	76.6	3259	11	19	42.1	3600	11	29	62.1	3600	11	46	76.1	3600	11	20	45.0	3600	11	20	45.0	3600
joint-9-30-5-30-3	46	11	12	8.3	3600	11	12	8.3	3599	11	12	8.3	3609	11	17	35.3	3607	11	14	21.4	3600	11	19	42.1	3602	11	41	73.2	3600	11	15	26.7	3600	11	20	45.0	3600
joint-10-30-5-30-3	55	12	13	7.7	3615	12	13	7.7	3602	12	13	7.7	3602	11	56	80.4	0	11	33	66.7	3600	11	21	47.6	3600	11	56	80.4	3600	11	33	66.7	3600	11	33	66.7	3600
joint-10-15-5-15-1	55	12	12	0.0	0	12	12	0.0	0	12	12	0.0	0	11	17	35.3	3616	11	21	47.6	3600	11	21	47.6	3600	11	44	75.0	3600	11	22	50.0	3600	11	33	66.7	3600
joint-10-15-5-15-2	55	12	13	7.7	3616	12	13	7.7	3601	12	13	7.7	3640	11	56	80.4	2919	11	21	47.6	3600	11	21	47.6	3602	11	36	69.4	3600	11	22	50.0	3600	11	22	50.0	3600
joint-10-15-5-15-3	55	12	13	7.7	3602	12	13	7.7	3599	12	13	7.7	3643	11	19	42.1	3616	11	21	47.6	3600	11	15	26.7	3602	11	21	47									

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX				
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	
joint-12-30-5-30-1	76	13	15	13.3	3600	13	15	13.3	3600	13	15	13.3	3601	13	59	78.0	3745	13	45	71.1	3601	13	77	83.1	3606	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-12-30-5-30-2	76	13	15	13.3	3600	13	15	13.3	3600	13	15	13.3	3603	13	77	83.1	12	13	28	53.6	3601	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-12-30-5-30-3	76	13	15	13.3	3602	13	15	13.3	3599	13	15	13.3	3605	13	58	77.6	3737	13	45	71.1	3601	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-12-15-5-15-1	76	13	15	13.3	3600	13	15	13.3	3600	13	15	13.3	3609	13	77	83.1	39	13	28	53.6	3600	13	20	35.0	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-12-15-5-15-2	76	13	15	13.3	3601	13	15	13.3	3600	13	15	13.3	3600	13	77	83.1	52	13	28	53.6	3600	13	20	35.0	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-12-15-5-15-3	76	13	15	13.3	3600	13	15	13.3	3600	13	15	13.3	3618	13	77	83.1	56	13	28	53.6	3600	13	20	35.0	3303	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-13-15-5-15-1	88	14	16	12.5	3609	14	16	12.5	3600	14	16	12.5	3601	14	89	84.3	168	14	22	36.4	3600	14	22	36.4	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-13-15-5-15-2	88	14	16	12.5	3602	14	16	12.5	3600	14	16	12.5	3605	14	89	84.3	20	14	22	36.4	3600	14	22	36.4	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-13-15-5-15-3	88	14	16	12.5	3600	14	16	12.5	3600	14	16	12.5	3601	14	89	84.3	60	14	32	56.2	3600	14	32	56.2	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-13-30-5-30-1	88	14	17	17.6	3602	14	17	17.6	3599	14	17	17.6	3606	14	89	84.3	261	14	51	72.5	3602	14	32	56.2	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-13-30-5-30-2	88	15	16	6.2	3600	15	16	6.2	3600	15	16	6.2	3602	14	89	84.3	31	14	51	72.5	3601	14	51	72.5	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-13-30-5-30-3	88	15	16	6.2	3600	15	16	6.2	3600	15	16	6.2	3607	14	89	84.3	45	14	51	72.5	3602	14	32	56.2	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-14-15-5-15-2	101	15	18	16.7	3601	15	18	16.7	3600	15	18	16.7	3600	15	102	85.3	90	15	25	40.0	3600	15	25	40.0	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-14-15-5-15-3	101	15	18	16.7	3600	15	18	16.7	3600	15	18	16.7	3600	15	102	85.3	9	15	36	58.3	3600	15	25	40.0	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-14-30-5-30-1	101	15	17	11.8	3600	15	17	11.8	3600	15	17	11.8	3600	15	102	85.3	184	15	102	85.3	3625	15	58	74.1	3606	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-14-30-5-30-2	101	15	18	16.7	3601	15	17	11.8	3600	15	17	11.8	3603	15	102	85.3	131	15	58	74.1	3605	15	36	58.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-14-30-5-30-3	101	15	17	11.8	3600	15	17	11.8	3600	15	17	11.8	3600	15	102	85.3	14	15	58	74.1	3604	15	58	74.1	3608	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-14-15-5-15-1	101	15	17	11.8	3600	15	17	11.8	3600	15	17	11.8	3603	15	102	85.3	49	15	36	58.3	3601	15	25	40.0	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-15-30-5-30-3	115	16	18	11.1	3600	16	18	11.1	3600	16	18	11.1	3602	16	116	86.2	67	16	66	75.8	3606	16	40	60.0	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-15-15-5-15-1	115	16	20	20.0	3601	16	18	11.1	3601	16	19	15.8	3601	16	116	86.2	44	16	40	60.0	3601	16	27	40.7	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-15-15-5-15-2	115	16	19	15.8	3602	16	19	15.8	3600	16	18	11.1	3601	16	116	86.2	130	16	27	40.7	3600	16	27	40.7	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-15-15-5-15-3	115	16	18	11.1	3601	16	18	11.1	3600	16	18	11.1	3603	16	116	86.2	16	16	66	75.8	3604	16	40	60.0	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-15-30-5-30-1	115	16	18	11.1	3600	16	18	11.1	3601	16	18	11.1	3604	16	116	86.2	24	16	40	60.0	3601	16	40	60.0	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-15-30-5-30-2	115	16	19	15.8	3600	16	19	15.8	3600	16	19	15.8	3601	16	116	86.2	21	16	40	60.0	3601	16	40	60.0	3603	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-16-15-5-15-1	130	17	19	10.5	3601	17	19	10.5	3600	17	19	10.5	3600	17	131	87.0	3	17	74	77.0	3608	17	45	62.2	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-16-15-5-15-2	130	17	20	15.0	3601	17	19	10.5	3601	17	19	10.5	3601	17	131	87.0	121	17	45	62.2	3601	17	45	62.2	3603	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-16-15-5-15-3	130	17	19	10.5	3601	17	21	19.0	3601	17	19	10.5	3601	17	131	87.0	98	17	45	62.2	3603	17	45	62.2	3603	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-16-30-5-30-1	130	17	20	15.0	3602	17	21	19.0	3601	17	20	15.0	3601	17	131	87.0	4	17	74	77.0	3613	17	30	43.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-16-30-5-30-2	130	17	20	15.0	3601	17	19	10.5	3601	17	19	10.5	3601	17	131	87.0	26	17	74	77.0	3611	17	45	62.2	3604	-	-	-	-	-	-	-	-	-	-	-	-	-
joint-16-30-5-30-3	130	17	19	10.5	3602	17	20	15.0	3600	17	19	10.5	3601	17	131	87.0	65	17	74	77.0	3611	17	74	77.0	3612	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 8: Results of 9 methods on miss instances

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX				
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time					
drand-5-110-2	5	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0					
drand-5-110-3	5	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0					
drand-5-200-1	5	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0					
drand-5-200-2	5	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0					
drand-5-200-3	5	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0					
drand-5-75-1	5	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0					
drand-5-75-2	5	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0					
drand-5-75-3	5	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0					
drand-5-110-1	5	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0	5	5	0.0	0					
drand-7-200-1	7	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	8	0.0	94	8	8	0.0	76	8	8	0.0	250	8	8	0.0	136	8	8	0.0	154	8	8	0.0	382	
drand-7-200-2	7	8	8	0.0	0	8	8	0.0	0	8	8	0.0	0	8	8	0.0	54	8	8	0.0	50	8	8	0.0	120	8	8	0.0	135	8	8	0.0	112	8	8	0.0	298	
drand-7-200-3	7	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	3	7	7	0.0	3	7	7	0.0	4	7	7	0.0	8	7	7	0.0	4	7	7	0.0	10	
drand-7-75-1	7	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	3	7	7	0.0	3	7	7	0.0	5	7	7	0.0	6	7	7	0.0	5	7	7	0.0	7	
drand-7-75-2	7	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	0	6	6	0.0	1	6	6	0.0	1	6	6	0.0	0	6	6	0.0	2	
drand-7-75-3	7	6	6	0.0	1	6	6	0.0	0	6	6	0.0	1	6	6	0.0	1	6	6	0.0	0	6	6	0.0	2	6	6	0.0	2	6	6	0.0	1	6	6	0.0	3	
drand-7-110-1	7	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	3	7	7	0.0	2	7	7	0.0	6	7	7	0.0	7	7	7	0.0	5	7	7	0.0	12	
drand-7-110-2	7	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	3	7	7	0.0	3	7	7	0.0	5	7	7	0.0	6	7	7	0.0	5	7	7	0.0	10	
drand-7-110-3	7	7	7	0.0	0	7	7	0.0	0	7	7	0.0	0	7	7	0.0	3	7	7	0.0	3	7	7	0.0	8	7	7	0.0	6	7	7	0.0	6	7	7	0.0	12	
drand-10-110-1	10	8	8	0.0	138	8	8	0.0	97	8	8	0.0	113	8	8	0.0	426	8	8	0.0	165	8	8	0.0	507	8	8	0.0	567	8	8	0.0	369	8	8	0.0	1076	
drand-10-110-2	10	8	8	0.0	242	8	8	0.0	135	8	8	0.0	322	8	8	0.0	485	8	8	0.0	240	8	8	0.0	831	8	8	0.0	1216	8	8	0.0	423	8	8	0.0	893	
drand-10-110-3	10	8	8	0.0	134	8	8	0.0	107	8	8	0.0	220	8	8	0.0	270	8	8	0.0	184	8	8	0.0	780	8	8	0.0	677	8	8	0.0	342	8	8	0.0	778	
drand-10-200-1	10	8	8	0.0	233	8	8	0.0	124	8	8	0.0	193	8	8	0.0	584	8	8	0.0	254	8	8	0.0	780	8	8	0.0	1920	8	8	0.0	457	8	8	0.0	1420	
drand-10-200-2	10	9	9	0.0	3175	9	9	0.0	1611	8	9	11.1	3599	8	9	11.1	3600	5	11	54.5	3600	5	11	54.5	3600	7	9	22.2	3600	5	11	54.5	3600	5	11	54.5	3600	
drand-10-200-3	10	8	8	0.0	79	8	8	0.0	41	8	8	0.0	1231	8	8	0.0	326	8	8	0.0	60	8	8	0.0	1122	8	8	0.0	302	8	8	0.0	113	8	8	0.0	2494	
drand-10-75-1	10	7	7	0.0	4	7	7	0.0	2	7	7	0.0	10	7	7	0.0	7	7	0.0	6	7	7	0.0	10	7	7	0.0	9	7	7	0.0	12	7	7	0.0	11		
drand-10-75-2	10	8	8	0.0	77	8	8	0.0	45	8	8	0.0	350	8	8	0.0	165	8	8	0.0	84	8	8	0.0	488	8	8	0.0	536	8	8	0.0	175	8	8	0.0	293	
drand-10-75-3	10	7	7	0.0	10	7	7	0.0	5	7	7	0.0	27	7	7	0.0	13	7	7	0.0	20	7	7	0.0	73	7	7	0.0	41	7	7	0.0	16	7	7	0.0	78	
drand-12-200-1	12	9	9	0.0	2400	9	9	0.0	1078	8	9	11.1	3599	8	9	11.1	3600	9	9	0.0	3023	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-200-2	12	9	9	0.0	3600	8	9	11.1	3599	8	9	11.1	3599	7	9	22.2	3601	8	9	11.1	3600	6	9	33.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-200-3	12	8	8	0.0	620	8	8	0.0	1264	8	8	0.0	2494	8	8	0.0	1939	8	8	0.0	2125	6	9	33.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-75-1	12	7	7	0.0	116	7	7	0.0	8	7	7	0.0	39	7	7	0.0	332	7	7	0.0	189	7	7	0.0	106	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-75-2	12	8	8	0.0	437	8	8	0.0	240	8	8	0.0	485	8	8	0.0	789	8	8	0.0	364	8	8	0.0	922	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-75-3	12	8	8	0.0	408	8	8	0.0	316	8	8	0.0	822	8	8	0.0	329	8	8	0.0	434	8	8	0.0	2808	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-110-1	12	8	8	0.0	1480	8	8	0.0	1082	8	8	0.0	1697	7	8	12.5	3600	8	8	0.0	534	8	8	0.0	3189	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-110-2	12	8	8	0.0	792	8	8	0.0	1161	8	8	0.0	2642	7	8	12.5	3600	8	8	0.0	1184	8	8	0.0	2952	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-12-110-3	12	9	9	0.0	3599	9	9	0.0	3110	8	9	11.1	3599	7	9	22.2	3600	8	9	11.1	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-15-200-2	15	7	10	30.0	3600	7	10	30.0	3599	7	10	30.0	3602	6	12	46.8	3600	6	11	45.5	3600	6	11	45.5	3601	6	14	57.1	3600	6	11	45.5	3600	6	11	45.5	3600	
drand-15-200-3	15	10	10	0.0	3599	7	10	30.0	3599	7	10	30.0	3601	7	11	36.4	3600	6	11	45.5	3600	6	11	45.5	3602	6	11	45.5	3600	6	11	45.5	3600	6	11	45.5	3600	
drand-15-75-1	15	9	9	0.0	3507	9	9	0.0	1368	8	9	11.1	3599	7	9	22.2	3600	9	9	0.0	2306	6	11	45.5	3601	7	10	30.0	3600	9	10	10.0	3600	6	11	45.5	3600	
drand-15-75-2	15	9	9	0.0	1293	9	9	0.0	159	9	9	0.0	2355	9	9	0.0	1272	9	9	0.0	267	9	9	0.0	2492	8	10	20.0	3600	9	9	0.0	1470	9	9	0.0	3295	
drand-15-75-3	15	8	8	0.0	353	8	8	0.0.																														

Continue on next page

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time				
drand-15-110-2	15	6	10	40.0	3600	9	10	10.0	3599	6	10	40.0	3600	7	11	36.4	3600	6	11	45.5	3600	6	11	45.5	3602	7	11	36.4	3600	6	11	45.5	3600	6	11	45.5	3600
drand-15-110-3	15	6	9	33.3	3600	8	9	11.1	3599	8	9	11.1	3599	7	11	36.4	3604	6	11	45.5	3600	6	11	45.5	3601	7	9	22.2	3600	6	11	45.5	3600	6	11	45.5	3600
drand-15-200-1	15	6	11	45.5	3600	6	11	45.5	3599	6	11	45.5	3602	7	12	41.7	3600	6	11	45.5	3600	6	11	45.5	3601	6	13	53.8	3600	6	11	45.5	3600	6	16	62.5	3600
drand-20-200-3	20	7	11	36.4	3600	7	11	36.4	3599	7	11	36.4	3601	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3601	7	17	58.8	3600	7	14	50.0	3600	7	14	50.0	3600
drand-20-75-1	20	7	11	36.4	3599	10	11	9.1	3599	7	11	36.4	3599	8	12	33.3	3633	7	14	50.0	3600	7	14	50.0	3601	7	16	56.3	3600	7	14	50.0	3600	7	14	50.0	3600
drand-20-75-2	20	7	10	30.0	3599	9	10	10.0	3599	7	10	30.0	3601	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3600	7	17	58.8	3600	7	14	50.0	3600	7	10	30.0	3600
drand-20-75-3	20	7	10	30.0	3599	9	10	10.0	3599	7	10	30.0	3601	7	12	41.7	3600	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3600	7	10	30.0	3600
drand-20-110-1	20	8	11	27.3	3600	8	11	27.3	3599	8	11	27.3	3600	7	13	46.2	3600	7	14	50.0	3600	7	14	50.0	3601	7	15	53.3	3600	7	14	50.0	3600	7	14	50.0	3600
drand-20-110-2	20	7	11	36.4	3599	7	11	36.4	3599	7	11	36.4	3601	7	13	46.2	3600	7	14	50.0	3600	7	14	50.0	3602	7	13	46.2	3600	7	14	50.0	3600	7	14	50.0	3600
drand-20-110-3	20	7	10	30.0	3599	7	11	36.4	3599	7	10	30.0	3601	7	11	36.4	3600	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3600	7	14	50.0	3600
drand-20-200-1	20	8	12	33.3	3600	8	12	33.3	3600	8	12	33.3	3602	7	15	53.3	3601	7	14	50.0	3600	7	14	50.0	3602	7	18	61.1	3600	7	14	50.0	3600	7	14	50.0	3600
drand-20-200-2	20	7	11	36.4	3600	7	11	36.4	3599	7	11	36.4	3601	7	15	53.3	3600	7	14	50.0	3600	7	14	50.0	3601	7	18	61.1	3600	7	14	50.0	3600	7	14	50.0	3600
drand-25-75-1	25	8	11	27.3	3599	8	11	27.3	3599	8	11	27.3	3601	8	12	33.3	3601	8	12	33.3	3600	8	12	33.3	3601	8	16	50.0	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-75-2	25	8	11	27.3	3599	8	11	27.3	3599	8	11	27.3	3600	8	13	38.5	3601	8	12	33.3	3600	8	17	52.9	3600	8	15	46.7	3600	8	17	52.9	3600	8	12	33.3	3600
drand-25-75-3	25	9	12	25.0	3599	9	12	25.0	3599	9	12	25.0	3601	8	15	46.7	3600	8	17	52.9	3600	8	17	52.9	3601	8	16	50.0	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-110-1	25	8	12	33.3	3600	8	12	33.3	3599	8	12	33.3	3602	8	14	42.9	3601	8	17	52.9	3600	8	17	52.9	3601	8	20	60.0	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-110-2	25	8	12	33.3	3599	8	12	33.3	3599	8	12	33.3	3601	8	14	42.9	3601	8	17	52.9	3600	8	17	52.9	3601	8	20	60.0	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-110-3	25	8	12	33.3	3599	8	12	33.3	3599	8	12	33.3	3601	8	16	50.0	3601	8	17	52.9	3600	8	12	33.3	3600	8	17	52.9	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-200-1	25	8	12	33.3	3600	8	12	33.3	3599	8	12	33.3	3601	8	18	55.6	3600	8	17	52.9	3600	8	17	52.9	3601	8	21	61.9	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-200-2	25	8	14	42.9	3600	8	14	42.9	3600	8	14	42.9	3602	8	19	57.9	3600	8	26	69.2	3600	8	17	52.9	3602	8	24	66.7	3600	8	17	52.9	3600	8	17	52.9	3600
drand-25-200-3	25	8	13	38.5	3600	8	13	38.5	3600	8	13	38.5	3602	8	16	50.0	3601	8	17	52.9	3600	8	17	52.9	3600	8	21	61.9	3600	8	17	52.9	3600	8	17	52.9	3600
drand-30-75-2	30	9	12	25.0	3599	9	12	25.0	3599	9	12	25.0	3601	9	15	40.0	3601	9	14	35.7	3600	9	14	35.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-75-3	30	9	12	25.0	3599	9	12	25.0	3599	9	12	25.0	3601	9	16	43.7	3602	9	20	55.0	3600	9	14	35.7	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-110-1	30	9	12	25.0	3599	9	12	25.0	3599	9	12	25.0	3601	9	13	30.8	3601	9	20	55.0	3600	9	20	55.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-110-2	30	9	13	30.8	3633	9	13	30.8	3599	9	13	30.8	3601	9	17	47.1	3601	9	20	55.0	3600	9	20	55.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-110-3	30	9	11	18.2	3599	9	11	18.2	3599	9	11	18.2	3601	9	31	71.0	1146	9	20	55.0	3600	9	14	35.7	3619	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-200-1	30	9	14	35.7	3600	9	14	35.7	3600	9	14	35.7	3601	9	19	52.6	3606	9	20	55.0	3600	9	20	55.0	3602	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-200-2	30	9	14	35.7	3619	9	14	35.7	3600	9	14	35.7	3601	9	21	57.1	3601	9	20	55.0	3600	9	20	55.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-200-3	30	9	14	35.7	3601	9	14	35.7	3600	9	14	35.7	3601	9	21	57.1	3601	9	31	71.0	3600	9	20	55.0	3602	-	-	-	-	-	-	-	-	-	-	-	-
drand-30-75-1	30	9	12	25.0	3600	9	11	18.2	3599	9	11	18.2	3601	9	15	40.0	3601	9	14	35.7	3600	9	14	35.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
drand-35-75-3	35	10	13	23.1	3599	10	13	23.1	3599	10	13	23.1	3601	9	22	59.1	3602	9	22	59.1	3600	9	22	59.1	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-35-110-1	35	10	14	28.6	3600	10	14	28.6	3600	10	13	23.1	3601	9	19	52.6	3602	9	22	59.1	3600	9	22	59.1	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-35-110-2	35	10	15	33.3	3626	10	15	33.3	3600	10	15	33.3	3602	9	23	60.9	3602	9	36	75.0	3600	9	22	59.1	3602	-	-	-	-	-	-	-	-	-	-	-	-
drand-35-110-3	35	10	14	28.6	3618	10	14	28.6	3600	10	14	28.6	3601	9	24	62.5	3602	9	36	75.0	3600	9	22	59.1	3602	-	-	-	-	-	-	-	-	-	-	-	-
drand-35-200-1	35	9	15	40.0	3613	9	15	40.0	3599	9	15	40.0	3603	9	36	75.0	3	9	36	75.0	3600	9	22	59.1	3600	-	-	-	-	-	-	-	-	-	-	-	-
drand-																																					

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX				
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	
drand-40-200-3	40	10	16	37.5	3601	10	16	37.5	3599	10	16	37.5	3600	10	26	61.5	3604	10	25	60.0	3600	10	17	41.2	3600	-	-	-	-	-	-	-	-	-	-	-	-	
drand-40-75-1	40	10	14	28.6	3622	10	14	28.6	3599	10	14	28.6	3601	10	19	47.4	3603	10	25	60.0	3600	10	17	41.2	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-40-75-2	40	10	14	28.6	3617	10	13	23.1	3599	10	14	28.6	3601	10	18	44.4	3603	10	25	60.0	3600	10	17	41.2	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-40-75-3	40	10	14	28.6	3600	10	14	28.6	3599	10	14	28.6	3601	10	41	75.6	2	10	17	41.2	3600	10	17	41.2	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-110-1	45	11	15	26.7	3625	11	15	26.7	3601	11	15	26.7	3615	10	24	58.3	3608	10	28	64.3	3600	10	28	64.3	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-110-2	45	10	15	33.3	3610	10	15	33.3	3600	10	15	33.3	3601	10	19	47.4	3608	10	18	44.4	3600	10	28	64.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-110-3	45	10	15	33.3	3623	10	15	33.3	3600	10	15	33.3	3646	10	24	58.3	3607	10	28	64.3	3600	10	28	64.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-200-1	45	11	17	35.3	3600	11	17	35.3	3600	11	17	35.3	3601	10	35	71.4	3610	10	46	78.3	3600	10	46	78.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-200-2	45	10	16	37.5	3600	10	16	37.5	3601	10	16	37.5	3604	10	37	73.0	3614	10	28	64.3	3600	10	28	64.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-200-3	45	11	17	35.3	3599	11	17	35.3	3599	11	17	35.3	3610	10	46	78.3	1	10	46	78.3	3600	10	28	64.3	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-75-1	45	11	14	21.4	3616	11	14	21.4	3600	11	14	21.4	3600	10	25	60.0	3607	10	18	44.4	3600	10	18	44.4	3603	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-75-2	45	11	15	26.7	3600	11	14	21.4	3600	11	14	21.4	3652	10	17	41.2	3606	10	18	44.4	3600	10	18	44.4	3604	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-45-75-3	45	11	14	21.4	3600	11	14	21.4	3600	11	14	21.4	3602	10	18	44.4	3607	10	18	44.4	3600	10	18	44.4	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-110-3	50	11	16	31.2	3600	11	16	31.2	3600	11	16	31.2	3602	11	25	56.0	3612	11	20	45.0	3600	11	20	45.0	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-200-1	50	11	18	38.9	3600	11	17	35.3	3599	11	18	38.9	3600	11	28	60.7	3615	11	31	64.5	3600	11	31	64.5	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-200-2	50	11	17	35.3	3600	11	17	35.3	3599	11	17	35.3	3601	11	32	65.6	3614	11	51	78.4	3601	11	31	64.5	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-200-3	50	11	18	38.9	3605	11	18	38.9	3600	11	18	38.9	3602	11	36	69.4	3618	11	31	64.5	3600	11	31	64.5	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-75-1	50	11	15	26.7	3600	11	15	26.7	3600	11	15	26.7	3612	11	27	59.3	3616	11	20	45.0	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-75-2	50	11	16	31.2	3617	11	16	31.2	3600	11	15	26.7	3602	11	51	78.4	1	11	31	64.5	3600	11	20	45.0	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-75-3	50	11	15	26.7	3600	11	15	26.7	3600	11	16	31.2	3648	11	23	52.2	3610	11	31	64.5	3600	11	20	45.0	3605	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-110-1	50	11	16	31.2	3599	11	16	31.2	3600	11	16	31.2	3613	11	51	78.4	3480	11	31	64.5	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-50-110-2	50	11	16	31.2	3615	11	16	31.2	3600	11	16	31.2	3601	11	51	78.4	0	11	31	64.5	3600	11	20	45.0	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-200-2	75	13	21	38.1	3600	13	21	38.1	3600	13	21	38.1	3611	13	76	82.9	46	13	44	70.5	3601	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-200-3	75	13	21	38.1	3600	13	21	38.1	3600	13	21	38.1	3600	13	76	82.9	85	13	44	70.5	3601	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-75-1	75	13	17	23.5	3601	13	17	23.5	3600	13	17	23.5	3602	13	46	71.7	3685	13	28	53.6	3600	13	28	53.6	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-75-2	75	13	19	31.6	3601	13	19	31.6	3600	13	19	31.6	3602	13	74	82.4	3689	13	28	53.6	3600	13	28	53.6	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-75-3	75	13	18	27.8	3604	13	18	27.8	3600	13	18	27.8	3631	13	25	48.0	3718	13	28	53.6	3600	13	28	53.6	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-110-1	75	13	19	31.6	3600	13	19	31.6	3600	13	18	27.8	3603	13	47	72.3	3701	13	44	70.5	3601	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-110-2	75	14	18	22.2	3600	14	19	26.3	3600	14	19	26.3	3602	13	76	82.9	87	13	28	53.6	3600	13	28	53.6	3602	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-110-3	75	13	19	31.6	3612	13	19	31.6	3600	13	19	31.6	3617	13	76	82.9	1	13	28	53.6	3600	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-75-200-1	75	13	20	35.0	3600	13	20	35.0	3600	13	20	35.0	3605	13	76	82.9	1	13	76	82.9	3608	13	28	53.6	3600	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-100-110-2	100	15	22	31.8	3601	15	21	28.6	3600	15	21	28.6	3601	15	101	85.1	101	15	36	58.3	3600	15	36	58.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-100-110-3	100	15	21	28.6	3601	15	21	28.6	3601	15	21	28.6	1803	15	101	85.1	185	15	36	58.3	3601	15	36	58.3	3601	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-100-200-1	100	15	23	34.8	3602	15	23	34.8	3601	15	23	34.8	3600	15	101	85.1	181	15	58	74.1	3603	15	58	74.1	3607	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-100-200-2	100	15	25	40.0	3603	15	24	37.5	3601	15	24	37.5	3602	15	101	85.1	148	15	58	74.1	3605	15	58	74.1	3607	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-100-200-3	100	15	24	37.5	3602	15	24	37.5	3601	15	24	37.5	3601	15	101	85.1	178	15	58	74.1	3605	15	58	74.1	3605	-	-	-	-	-	-	-	-	-	-	-	-	-
drand-100-75-1	100	15																																				

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time
drand-112-75-1	112	16	21	23.8	3601	16	22	27.3	3600	16	22	27.3	3605	16	113	85.8	128	16	39	59.0	3601	16	27	40.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
drand-112-75-2	112	16	22	27.3	3601	16	22	27.3	3600	16	22	27.3	3603	16	113	85.8	123	16	64	75.0	3603	16	39	59.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-112-75-3	112	16	21	23.8	3601	16	21	23.8	3600	16	21	23.8	3600	16	113	85.8	136	16	39	59.0	3601	16	39	59.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
drand-112-110-1	112	16	22	27.3	3601	16	22	27.3	3600	16	22	27.3	3600	16	113	85.8	158	16	39	59.0	3601	16	27	40.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
drand-112-110-2	112	16	23	30.4	3602	16	23	30.4	3600	16	23	30.4	3601	16	113	85.8	84	16	39	59.0	3601	16	39	59.0	3601	-	-	-	-	-	-	-	-	-	-	-	-

Table 9: Results of 9 methods on luci instances

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time
luciferase-removed-15-1	13	7	7	0.0	18	7	7	0.0	127	7	7	0.0	125	7	7	0.0	594	7	7	0.0	29	7	7	0.0	159	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-15-3	13	8	8	0.0	1187	8	8	0.0	578	8	8	0.0	3600	7	8	12.5	3600	8	8	0.0	2518	6	10	40.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-15-4	13	8	8	0.0	857	8	8	0.0	382	8	8	0.0	3600	7	8	12.5	3600	8	8	0.0	1391	6	10	40.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-14-0	14	7	7	0.0	7	7	7	0.0	33	7	7	0.0	71	7	7	0.0	698	7	7	0.0	162	7	7	0.0	218	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-14-1	14	8	8	0.0	673	8	8	0.0	660	8	8	0.0	3522	7	8	12.5	3600	8	8	0.0	2287	6	10	40.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-14-2	14	6	8	25.0	3600	8	8	0.0	1174	6	8	25.0	3602	6	9	33.3	3601	6	15	60.0	3600	6	10	40.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-14-3	14	8	8	0.0	691	8	8	0.0	1016	8	8	0.0	1953	7	8	12.5	3601	8	8	0.0	896	8	8	0.0	2004	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-14-4	14	8	8	0.0	990	8	8	0.0	1183	8	8	0.0	3600	7	9	22.2	3600	8	10	20.0	3600	6	10	40.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-13-3	15	8	8	0.0	512	8	8	0.0	3600	8	8	0.0	3600	7	10	30.0	3600	6	11	45.5	3600	6	8	25.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-13-0	15	8	8	0.0	1987	8	8	0.0	3599	8	8	0.0	1944	6	10	40.0	3613	8	8	0.0	2874	7	8	12.5	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-13-1	15	7	8	12.5	3600	8	8	0.0	644	7	8	12.5	3601	7	10	30.0	3600	6	11	45.5	3600	6	11	45.5	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-12-0	16	7	8	12.5	3599	8	8	0.0	354	7	8	12.5	3600	7	10	30.0	3600	8	8	0.0	2076	7	9	22.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-12-2	16	7	8	12.5	3600	8	8	0.0	3599	8	8	0.0	2651	7	10	30.0	3600	7	12	41.7	3600	7	9	22.2	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-12-3	16	8	8	0.0	1637	8	8	0.0	657	8	8	0.0	3600	7	9	22.2	3600	8	9	11.1	3600	7	9	22.2	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-12-4	16	7	8	12.5	3600	8	8	0.0	3599	8	8	0.0	3600	7	9	22.2	3607	8	8	0.0	1256	7	9	22.2	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-11-0	17	8	8	0.0	2226	8	8	0.0	3226	8	8	0.0	3600	7	10	30.0	3604	8	9	11.1	3600	7	9	22.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-11-1	17	7	8	12.5	3600	8	8	0.0	1900	7	8	12.5	3600	7	13	46.2	3615	7	12	41.7	3600	7	12	41.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-11-2	17	7	8	12.5	3600	8	8	0.0	3600	7	8	12.5	3601	7	11	36.4	3600	7	12	41.7	3600	7	12	41.7	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-11-3	17	8	8	0.0	785	8	8	0.0	3452	7	8	12.5	3601	7	9	22.2	3600	7	12	41.7	3600	7	9	22.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-10-2	18	8	8	0.0	3002	8	8	0.0	3599	7	8	12.5	3601	7	8	12.5	3600	7	13	46.2	3600	7	9	22.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-10-4	18	7	8	12.5	3600	8	8	0.0	1178	8	8	0.0	1557	7	9	22.2	3600	7	13	46.2	3600	7	9	22.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-10-0	18	7	8	12.5	3600	8	8	0.0	410	7	8	12.5	3600	7	12	41.7	3600	8	9	11.1	3600	7	9	22.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-9-1	19	7	8	12.5	3600	8	8	0.0	3600	7	8	12.5	3601	7	14	50.0	3622	7	13	46.2	3600	7	9	22.2	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-9-3	19	7	8	12.5	3600	8	8	0.0	3600	8	8	0.0	3436	7	8	12.5	3600	7	13	46.2	3600	7	13	46.2	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-9-4	19	7	8	12.5	3614	8	8	0.0	3600	7	8	12.5	3602	7	11	36.4	3600	7	13	46.2	3600	7	13	46.2	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-8-0	20	7	8	12.5	3626	8	8	0.0	585	8	8	0.0	3600	7	13	46.2	3600	7	21	66.7	3600	7	10	30.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-8-1	20	8	8	0.0	1	8	8	0.0	1	8	8	0.0	1	7	12	41.7	3605	7	10	30.0	3600	7	14	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-8-2	20	8	8	0.0	3600	8	8	0.0	3600	8	8	0.0	3601	7	14	50.0	3601	7	14	50.0	3600	7	10	30.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-8-3	20	7	8	12.5	3601	8	8	0.0	3599	7	8	12.5	3600	7	12	41.7	3600	7	14	50.0	3600	7	14	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-8-4	20	7	8	12.5	3601	8	8	0.0	3599	8	8	0.0	2269	7	13	46.2	3603	7	14	50.0	3600	7	8	12.5	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-7-3	21	8	8	0.0	2	8	8	0.0	2	8	8	0.0	2	7	13	46.2	3600	7	14	50.0	3600	7	10	30.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-7-0	21	7	8	12.5	3602	8	8	0.0	3600	7	8	12.5	3601	7	13	46.2	3600	7	14	50.0	3600	7	14	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-7-1	21	8	8	0.0	7	8	8	0.0	7	8	8	0.0	7	7	13	46.2	3620	7	14	50.0	3600	7	14	50.0	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-6-0	22	8	8	0.0	3	8	8	0.0	2	8	8	0.0	2	8	13	38.5	3601	8	11	27.3	3600	8	11	27.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-6-2	22	8	8	0.0	2	8	8	0.0	2	8	8	0.0	2	8	14	42.9	3600	8	15	46.7	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-6-3	22	8	8	0.0	10	8	8	0.0	10	8	8	0.0	10	8	12	33.3	3600	8	15	46.7	3600	8	11	27.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-6-4	22	8	8	0.0	1	8	8	0.0	1	8	8	0.0	1	8	12	33.3	3600	8	11	27.3	3600	8	11	27.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-5-0	23	8	8	0.0	2	8	8	0.0	2	8	8	0.0	2	8	15	46.7	3609	8	24	66.7	3600	8	16	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-5-1	23	8	8	0.0	3	8	8	0.0	3	8	8	0.0	3	8	13	38.5	3601	8	16	50.0	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-5-2	23	8	8	0.0	12	8	8	0.0	12	8	8	0.0	12	8	12	33.3	3600	8	11	27.3	3600	8	16	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-5-3	23	8	8	0.0	2	8	8	0.0	2	8	8	0.0	2	8	16	50.0	3606	8	16	50.0	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-4-2	24	8	8	0.0	2	8	8	0.0	2	8	8	0.0	2	8	13	38.5	3601	8	16	50.0	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-

Continue on next page

Instances	D	nIP				nFEAS				nMAX				IP				FEAS				MAX				tIP				tFEAS				tMAX			
		LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time	LB	UB	gap	time
luciferase-removed-4-4	24	8	8	0.0	14	8	8	0.0	13	8	8	0.0	14	8	16	50.0	3600	8	16	50.0	3600	8	11	27.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-4-0	24	8	8	0.0	2	8	8	0.0	2	8	8	0.0	2	8	11	27.3	3600	8	16	50.0	3600	8	16	50.0	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-3-1	25	8	8	0.0	16	8	8	0.0	16	8	8	0.0	16	8	11	27.3	3600	8	26	69.2	3600	8	8	0.0	3586	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-3-2	25	8	8	0.0	12	8	8	0.0	12	8	8	0.0	12	8	13	38.5	3601	8	17	52.9	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-3-3	25	8	8	0.0	13	8	8	0.0	13	8	8	0.0	13	8	19	57.9	3601	8	12	33.3	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-3-4	25	8	8	0.0	5	8	8	0.0	4	8	8	0.0	5	8	15	46.7	3601	8	26	69.2	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-2-4	26	8	8	0.0	14	8	8	0.0	14	8	8	0.0	14	8	15	46.7	3611	8	17	52.9	3600	8	12	33.3	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-2-0	26	8	8	0.0	4	8	8	0.0	4	8	8	0.0	4	8	17	52.9	3608	8	17	52.9	3600	8	17	52.9	3601	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-2-1	26	8	8	0.0	5	8	8	0.0	5	8	8	0.0	5	8	14	42.9	3602	8	27	70.4	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-2-2	26	8	8	0.0	5	8	8	0.0	5	8	8	0.0	5	8	18	55.6	3601	8	17	52.9	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-1-1	27	8	8	0.0	14	8	8	0.0	14	8	8	0.0	14	8	15	46.7	3601	8	18	55.6	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-1-2	27	8	8	0.0	4	8	8	0.0	4	8	8	0.0	4	8	18	55.6	3601	8	12	33.3	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-1-3	27	8	8	0.0	14	8	8	0.0	14	8	8	0.0	14	8	18	55.6	3601	8	18	55.6	3600	8	9	11.1	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-removed-1-4	27	8	8	0.0	15	8	8	0.0	14	8	8	0.0	14	8	9	11.1	3601	8	28	71.4	3600	8	12	33.3	3600	-	-	-	-	-	-	-	-	-	-	-	-
luciferase-full	28	8	8	0.0	14	8	8	0.0	14	8	8	0.0	14	8	21	61.9	3601	8	18	55.6	3600	8	18	55.6	3600	-	-	-	-	-	-	-	-	-	-	-	-

References

- [1] Abbas, M. M., Bahig, H. M., *A fast exact sequential algorithm for the partial digest problem*, BMC Bioinformatics, 2016.
- [2] Ahrabian, H., Ganjtabesh, M., Nowzari-Dalini, A., Razaghi-Moghadam-Kashani, Z., *Genetic algorithm solution for partial digest problem*, Int j bioinform res appl. 9 (6), 584–94, 2013.
- [3] Alizadeh, F., Gonçalves, D., Krislock, N., Liberti, L., *Preface: Special issue dedicated to Distance Geometry*, Distance Geometry and Applications DGTA16, 1-162, 2019.
- [4] Benny Chor, Paul Lemke, Ziv Mador, *On the number of ordered factorizations of natural numbers*, Discrete Mathematics 214, 123-133, 2000.
- [5] Cieliebak, M., Eidenbenz, S., Penna, P., *Noisy data make the partial digest problem NP-hard*, Springer, 2003.
- [6] Cieliebak, M., Eidenbenz, S., Penna, P., *Partial Digest is hard to solve for erroneous input data*, Theoretical Computer Science 349 (3), 361-381, 2005.
- [7] Dakic, T., *On the turnpike problem*, Ph.D. thesis, Simon Fraser University, 2000.
- [8] Daurat, A., Gérard, Y., Nivat, M., *The chords' problem*, Theoretical Computer Science 282, 319-336, 2002.
- [9] Daurat, A., Gerard, Y., Nivat, M., *Some necessary clarifications about the chords problem and the partial digest problem*, Theoretical Computer Science 347, 432–436, 2005.
- [10] Fontoura, L., Martinelli, R., Poggi, M., Vidal, T. *The minimum distance superset problem: formulations and algorithms*, Journal of Global Optimization, 72(1), 27–53, 2018.
- [11] Jacek, B., Edmund, K.B., Marta, K., Alexandr, K., Mikhail, Y.K., *The Simplified Partial Digest Problem: approximation and a graph theoretic model*, European Journal of Operational Research 208 (2), 142-152, 2011.
- [12] Lemke, P., Skiena, S. S., Smith, W. D., *Reconstructing sets from interpoint distances*. In: Aronov, B., Basu, S., Pach, J., Sharir, M. (Eds.), Discrete and Computational Geometry: The Goodman-Pollack Festschrift. Springer Berlin Heidelberg, Berlin, Heidelberg, 597-631, 2003.
- [13] Karp, R. M., Newberg, L.A., *An algorithm for analysing probed partial digestion experiments*, Computer Applications in Biosciences 11 (3), 229-35, 1995.
- [14] Nadimi, R., Fathabadi, H.S., Ganjtabesh, M., *A fast algorithm for the partial digest problem*. Japan Journal of Industrial and Applied Mathematics (28), 315–25, 2011.
- [15] Newberg, L.A. , Naor, D. , *A lower bound on the number of solutions to the probed partial digest problem*, Advanced in Applied Mathematics (14), 172–183, 1993.
- [16] Skiena, S. S., Smith, W. D., Lemke, P., 1990, *Reconstructing sets from interpoint distances (extended abstract)*, Proceedings of the Sixth Annual Symposium on Computational Geometry, SCG'90, ACM, New York, NY, USA, 332-339, 1990.
- [17] Skiena, S. S., Sundaram, G., *A Partial Digest Approach to Restriction Site Mapping*, Bulletin of Mathematical Biology 56 (2), 275-294, 1994.
- [18] Tomohiro, S., Akihiro, F., *Solving subset problem using EN P system with active membranes*, Joint 8th International Conference on Soft Computing and Intelligent Systems and 17th International Symposium on Advanced Intelligent Systems, 2016.
- [19] Pandurangan, G., Ramesh, H., *The restriction mapping problem revisited*, Journal of Computer and System Sciences 65(3), 526-544, 2002.
- [20] Patterson, A. L., *A Direct Method for the Determination of the Components of Interatomic Distances in Crystals*, Zeitschrift für Kristallographie - Crystalline Materials 90 (1-6), 517-542, 1935.
- [21] Patterson, A. L., *Ambiguities in the X-Ray Analysis of Crystal Structures*, Physical Review 65 (1935), 195, 1944.
- [22] Zhang, Z., *An exponential example for a partial digest mapping algorithm*, Journal of Computational Biology 1 (3), 235-239, 1994.